

ZombieVerter VCU

Training Series

Complete reference guide for DIY EV conversion builders
30 modules - 6 tracks - Firmware V2.40A (August 2025)

github.com/robertwa1974/zombieverter-training
openinverter.org - evbmw.com

Track	Name	Modules
F	Foundation	F01-F04
H	Hardware	H01
W	Wiring	W01-W08
C	Configuration	C00-C06
I	Integration	I01-I04
A	Advanced	A01-A06

Table of Contents

FOUNDATION

- F01 - What Is a VCU and Why Do You Need One
- F02 - ZombieVerter Ecosystem Overview
- F03 - EV Drivetrain Fundamentals
- F04 - High Voltage Safety

HARDWARE

- H01 - VCU Hardware Walkthrough

WIRING

- W01 - Wiring Philosophy and Best Practices
- W02 - Throttle and Brake Wiring
- W03 - Contactor and Precharge Circuit
- W04 - HVIL High Voltage Interlock Loop
- W05 - Cooling System Control
- W06 - 12V Auxiliary and Ignition Wiring
- W07 - CAN Bus Wiring
- W08 - ISA IVT-S Shunt: Complete Reference

CONFIGURATION

- C00 - Firmware Version History
- C01 - Flashing Firmware
- C02 - Web Interface Walkthrough
- C03 - Essential Parameters: First Start
- C04 - Throttle Calibration
- C06 - Fault Codes and Status Flags

INTEGRATION

- I01 - Nissan Leaf Inverter
- I02 - Tesla Drive Units SDU and LDU
- I03 - GS450h Transaxle
- I03-X - GS450h Wiring and Oil Pump Deep Dive
- I04 - Mitsubishi Outlander Drive Unit

ADVANCED

- A01 - Regen Tuning
- A02 - Charge Control and Popular Chargers
- A03 - BMS Integration
- A04 - Gear Selectors and Shifters
- A05 - CAN Gateway and Multi-Node Systems
- A06 - Firmware Customisation and Contributing

What Is a VCU and Why Do You Need One

Track: Foundation

Prerequisites: None — start here

Audience: Complete beginners

Estimated reading time: 8 minutes

The Problem With Donor EV Components

When you salvage an inverter and motor from a wrecked Nissan Leaf, you have powerful, proven hardware — but that hardware was designed to talk to a Leaf BMS, a Leaf dashboard, a Leaf brake controller, and a Leaf battery management system. None of which you have.

Every donor EV component speaks its own proprietary CAN protocol. The inverter expects specific CAN messages on specific message IDs at specific baud rates. It won't spin without them. Building a custom controller to reverse-engineer and replicate those messages for every possible inverter type would take months per build.

A Vehicle Control Unit solves this problem. It's a purpose-built translator that sits between your donor drivetrain components and your conversion vehicle, speaking the language each component needs.

What a VCU Does

In a ZombieVerter-based conversion, the VCU is responsible for:

Contactor sequencing — closing the negative contactor, running precharge through a resistor, monitoring HV bus voltage, then closing the main contactor in the correct sequence. This protects the inverter capacitors from inrush current damage on every startup.

Torque control — reading the throttle pedal, mapping it to a 0–100% demand, and sending the appropriate CAN torque command to the inverter. Without this, the inverter has no way to know how much torque the driver is requesting.

Direction control — handling forward/reverse selection and communicating motor direction to the inverter.

Regen control — managing regenerative braking via throttle lift and brake pedal, sending negative torque commands to the inverter (V2.15A onwards).

Charge control — managing the onboard charger and DC-DC converter via CAN when plugged in.

BMS integration — reading current limits from a cell BMS and enforcing them as caps on motor torque and charge current.

Vehicle integration — sending spoofed CAN messages to keep the OEM instrument cluster functional (speedometer, tachometer, warning lights).

Safety — opening contactors immediately on fault conditions, HVIL loop monitoring, brake interlock enforcement.

What a VCU Is Not

A VCU is not a motor controller or inverter. It does not switch the high-voltage phases to the motor. The inverter does that. The VCU tells the inverter what torque to produce — the inverter does the actual power switching.

A VCU is not a BMS. It does not monitor individual cell voltages or temperatures, balance cells, or protect against cell-level faults. That is the job of a separate BMS (SimpBMS, Orion, etc.). The VCU reads the output of the BMS — current limits — and enforces them.

What Makes ZombieVerter Different

Before the ZombieVerter, DIY EV converters had two options: write custom Arduino/microcontroller code to replicate inverter CAN messages (high skill barrier, per-inverter effort), or use the openinverter framework on a custom PCB (powerful but complex to set up).

The ZombieVerter packages the openinverter framework in a purpose-built VCU board with:

- **No programming required** — all configuration through a browser-based web interface over Wi-Fi
 - **Multi-inverter support** — select your inverter type from a dropdown; the VCU handles the rest
 - **Multi-charger support** — Leaf PDM, VW OBC, Tesla Gen2/3, Outlander OBC, CHAdeMO, CCS
 - **IO Matrix** — configurable input/output pin functions without code changes
 - **Active community** — openinverter.org forum with hundreds of documented builds
-

The Hardware

The ZombieVerter board is built around two processors:

STM32F107 — the main MCU. Runs the VCU firmware, manages all I/O, handles CAN communication with all connected devices, runs the contactor sequencing logic, calculates torque demands.

ESP8266 — the Wi-Fi module. Hosts the web configuration interface. Broadcasts a Wi-Fi access point you connect to from any browser. Does not run VCU functions — it's purely the configuration layer.

The board connects to your vehicle through a 56-pin Aptiv automotive connector. All vehicle wiring — power, signals, CAN buses, contactor outputs — goes through this one connector.

Warning: Available from evbmw.com as a complete pre-programmed kit, partial kit, or PCB-only.

The Community

The ZombieVerter is designed and maintained by Damien Maguire (EVBMW) and built on Johannes Huebner's openinverter firmware framework. It is genuinely open source — schematics, firmware, and board files are on GitHub.

The community is at openinverter.org/forum. Hundreds of completed builds are documented there, covering BMW E-series, Volkswagen, Mitsubishi, classic vehicles, and many others.

Damien's YouTube channel (@Evbmw) documents everything he builds and every firmware release with walkthrough videos — an invaluable resource that directly informed this training series.

Next Steps

- **F02** — ZombieVerter Ecosystem Overview: supported hardware, community resources
 - **H01** — VCU Hardware Walkthrough: the board in detail
 - **C03** — Essential Parameters: First Start: get to run mode on your bench
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Source: Damien Maguire @Evbmw — "New VCU Project Introduction" (February 2021)

Last verified against firmware: V2.30A (August 2025)

ZombieVerter Ecosystem Overview

Track: Foundation

Prerequisites: F01 — What is a VCU and Why Do You Need One

Audience: All levels

Estimated reading time: 10 minutes

What Is the ZombieVerter?

The ZombieVerter is an open-source Vehicle Control Unit (VCU) designed specifically for DIY electric vehicle conversions. It was created by Damien Maguire (EVBMW) with contributions from the broader openinverter.org community.

Its core purpose is to solve a real problem: modern EV conversion projects often reuse salvaged components — motors, inverters, chargers, battery packs — from wrecked or retired EVs. Each of these components uses different control signals and communication protocols. Building a custom controller for every possible combination would be impractical. The ZombieVerter instead provides a general-purpose VCU with a wide variety of inputs, outputs, and pre-built support for the most common donor vehicle components.

In short: **the ZombieVerter is the brain that makes mismatched salvaged EV parts work together.**

The People Behind It

Person / Group	Role
Damien Maguire (EVBMW)	Primary hardware designer, firmware lead, project creator
Johannes Huebner (johu)	Creator of the underlying openinverter firmware framework the VCU builds on
Tom DeBree	GitHub repository management, release gating, and ZombieVerter VCU firmware coordination — the person who controls what gets merged and when releases go out
openinverter.org community	Ongoing firmware contributions, inverter support classes, testing, documentation

The project is genuinely open-source. Schematics and firmware are available on GitHub. PCB design files for the current production revision (V1.a) are available through Damien's Patreon at the Design Files tier.

Where the Community Lives

Resource	URL	What's There
Main wiki	openinverter.org/wiki	Hardware docs, wiring guides, parameter references
Forum	openinverter.org/forum	Support threads, development discussion, build logs
GitHub (firmware)	github.com/damienmaguire/Stm32-vcu	Source code, firmware releases
GitHub (hardware V1)	github.com/damienmaguire/Stm32-vcu/tree/master/Hardware/Zombie	PCB files (V1 only)
EVBMW Webshop	evbmw.com	Boards for purchase
Damien's YouTube	youtube.com/@Evbmw	Build videos, firmware walkthroughs
Patreon	patreon.com/evbmw	PCB design files (V1.a), early access content

Note: The forum is the single best source for troubleshooting. Most problems have been encountered and solved by someone before. Search before posting.

Hardware Features

The ZombieVerter VCU is built around an STM32F107 microcontroller (or GD32F107 on some early pandemic-era boards — see Module C01 for the distinction). An ESP8266 Wi-Fi module provides the wireless interface.

On-board interfaces:

- On-board Wi-Fi (ESP8266)
- 3× CAN bus interfaces
- LIN bus
- Synchronous serial interface (used for GS450h, GS300h, Prius)
- OBD-II interface
- 3× high-side PWM driver outputs
- 5× low-side outputs
- 3× ground-pull input pins

Connector: 56-pin Aptiv (formerly Delphi) connector. Part numbers:

- Connector (cable side): Aptiv 56-pin connector
- Header (board side): Aptiv 56-pin header
- Terminal removal tool: Aptiv/Delphi Part No. 210S048

Software Features

All configuration is done through a browser-based web interface — no laptop software to install, no USB cables required for normal operation. You connect to the VCU's Wi-Fi access point and navigate to its IP address in any browser.

What the firmware manages:

- Contactor sequencing (negative, positive, precharge)
- Charger control and charge timers
- Motor/inverter control (torque requests via CAN or serial)
- Heater control
- Water pump and coolant fan control
- Throttle mapping and regen
- BMS limits (charge/discharge current limiting)
- ISA IVT shunt initialization and monitoring
- Data logging and live graphing

Currently Supported Hardware

Warning: *This list grows with each firmware release. Always check the wiki for the latest.*

Motors / Drive Units

Component	Interface
Nissan Leaf Gen1/2/3 inverter/motor	CAN (180V minimum)
Lexus GS450h inverter / L110 gearbox	Sync serial
Lexus GS300h inverter / L210 gearbox	Sync serial
Toyota Prius/Yaris/Auris Gen3	Sync serial
Mitsubishi Outlander front/rear motors	CAN
OpenInverter controller board	CAN

Chargers / DC-DC Converters

Component	Notes
Nissan Leaf PDM (Gen1, 2, 3)	Charger + DC-DC combined unit
Mitsubishi Outlander OBC	Charger + DC-DC combined unit
Tesla Model S DC-DC converter	
BMW i3 LIM module	Enables CCS DC fast charging (Type 1 & Type 2)
CHAdeMO DC fast charging	Via internal CAN
Foccci CCS controller	
Elcon charger	

Heaters

Component	Interface
VAG/VW PTC water heater	LIN bus

Component	Interface
VAG/VW cabin heater	LIN bus
Opel Ampera / Chevy Volt 6.5kW heater	
Mitsubishi Outlander water heater	

BMS / Battery Monitoring

Component	Notes
ISA IVT shunt	Primary current/voltage monitoring method
Nissan Leaf ZE1 BMS / battery pack	
Renault Kangoo 36 BMS	
Orion BMS	
SimpBMS	
BMW S-BOX	
VW E-BOX	

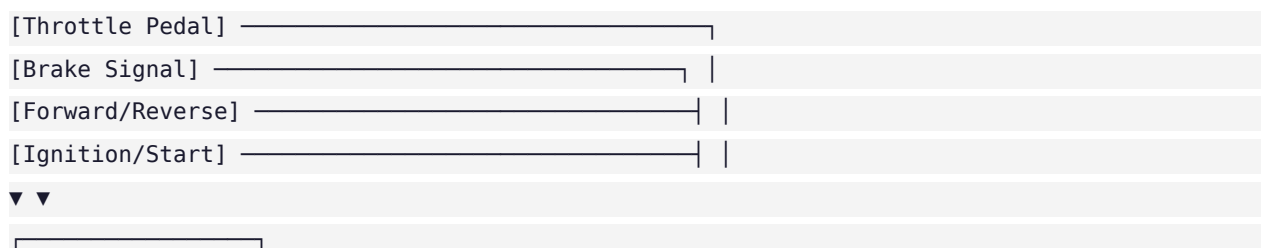
Warning: **Important:** The ZombieVerter does **not** implement direct BMS cell-level communication. It uses the ISA shunt for pack-level monitoring (voltage, current, state of charge). Cell-level BMS functions (balancing, per-cell voltage/temp) are handled by a separate BMS device such as SimpBMS, which communicates with the VCU via CAN.

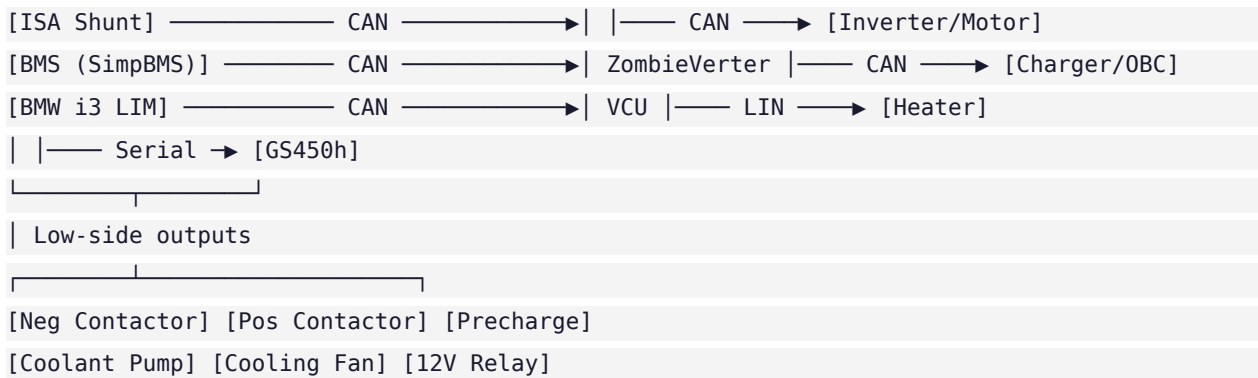
Vehicle CAN Integration (Dashboard / Instrument Cluster)

Vehicle	Notes
BMW E46 (1998–2005 3-series)	CAN + digital IO
BMW E39 (1996–2003 5-series)	CAN + digital IO
BMW E65 (2001–2008 7-series)	CAN
BMW E9x	CAN
Mid-2000s VAG vehicles	CAN
Subaru	CAN

How the VCU Fits Into a Conversion

A typical ZombieVerter-based conversion looks like this:





The VCU is the central coordinator. It reads driver inputs, monitors pack state via the shunt, sequences the contactors, and sends torque commands to the inverter — all configurable through the web interface without writing a line of code.

Getting a ZombieVerter Board

There are two ways to obtain a board:

Buy one from the EVBMW webshop:

- Fully-built (tested, programmed, no soldering required) — recommended for most users
- Partially-built kit (some assembly required)
- Lead time: up to 4 weeks from order

Build one yourself:

- PCB design files for V1 are on GitHub (free)
- PCB design files for the current V1.a revision require Patreon membership
- You will need to source and solder components yourself

WARNING: Note: Boards from the webshop come **pre-programmed** and ready to use. You do not need to flash firmware before first use.

What's Next

- **H01** — VCU Hardware Walkthrough: a physical tour of the board, connectors, and indicators
- **C01** — Flashing Firmware: only needed if you have a blank board or need to recover a bricked unit
- **C02** — The Web Interface Walkthrough: connecting to the VCU and navigating the UI

Source: openinverter.org/wiki/ZombieVerter_VCU | [openinverter.org community](https://openinverter.org/community)

Last verified against firmware: V2.30A (August 2025)

EV Drivetrain Fundamentals

Track: Foundation

Prerequisites: F01 — What Is a VCU

Audience: Complete beginners

Estimated reading time: 12 minutes

Two Voltage Domains

Every EV conversion has two completely separate electrical systems that must never be directly connected to each other:

High Voltage (HV) domain — the traction system. Battery pack, inverter, motor, onboard charger, DC-DC converter input. Typically 200–450V DC in ZombieVerter-based conversions. Orange cables by convention. Fatal at the current levels available.

Low Voltage (LV) domain — the vehicle electronics. 12V battery, VCU, lights, accessories, sensors, CAN buses, instrument cluster. Conventional automotive 12V. The VCU lives entirely in the LV domain — it controls the HV domain through contactors and CAN commands, but never carries HV itself.

The DC-DC converter bridges these domains in one direction only: it takes HV input and produces 12V output to charge the LV battery. Current never flows the other way.

The Traction Chain

From driver input to wheel torque, the path is:

Throttle pedal → VCU (torque demand) → Inverter (AC phase switching) → Motor → Gearbox/wheels

Throttle pedal — generates a 0–5V analog signal (Hall effect or potentiometer). The VCU reads this and maps it to a 0–100% torque demand.

VCU — translates driver demand to a CAN torque command in the inverter's protocol. Also enforces limits from the BMS, manages regen, handles direction.

Inverter — takes DC from the HV battery and switches it as three-phase AC to the motor. Switches at high frequency (typically 10–15 kHz) to produce sinusoidal current. Controls motor torque precisely by controlling the phase currents.

Motor — converts electrical energy to mechanical torque. In hybrid-derived systems (Leaf, GS450h, Prius, Outlander) the motor is a permanent magnet AC machine — efficient, high power density, well-suited for EV use.

Resolver or encoder — the motor position sensor. Tells the inverter exactly where the rotor is so it can switch phases at the right time. Without resolver feedback, the inverter cannot control the motor. The resolver cables run from the motor back to the inverter — these are the fragile signal cables that must be treated carefully.

Motor Types You'll Encounter

Permanent Magnet AC (PMAC / IPMSM) — used in Nissan Leaf (EM57, EM61), Mitsubishi Outlander, most modern EV traction motors. High efficiency, high torque density. Requires resolver feedback. The ZombieVerter supports several of these via CAN.

Induction (AC) — used in older Tesla Model S/X. Lower peak efficiency than PMAC but simpler magnetics. Supported via the openinverter board.

Dual motor (Power split) — the GS450h and Prius use two motor-generators (MG1 and MG2) with a planetary gearset between them. In EV conversion use MG2 provides primary traction, MG1 provides additional torque when the input shaft is locked. See Module I03 for detail.

Contactor Logic and Precharge

The HV battery cannot be connected directly to the inverter. The inverter contains large DC bus capacitors — connecting HV to uncharged capacitors causes an instantaneous high-current surge that can weld the contactor contacts permanently closed or destroy the capacitors.

The solution is precharge: a resistor in series with the positive HV rail that limits inrush current while the capacitors charge slowly.

The sequence:

Negative contactor closes (HV– connected to inverter)

Precharge contactor closes (HV+ through resistor to inverter)

Capacitors charge slowly — bus voltage rises over ~2-5 seconds

When bus voltage reaches threshold (udcsw), main positive contactor closes

Precharge contactor opens (resistor bypassed)

System enters run mode — full HV available

The VCU manages this entire sequence. The ISA shunt or equivalent sensor provides the bus voltage reading that tells the VCU when precharge is complete.

Warning: Why this matters for first-time builders: udcsw defaults to 330V. With no HV battery connected on the bench, the bus voltage never reaches 330V. Precharge appears to fail repeatedly. The fix is to set udcsw to 0 or 15V for bench testing. See Module C03.

Regenerative Braking

Regen reverses the energy flow. When the driver lifts the throttle or presses the brake, the VCU sends a negative torque command to the inverter. The inverter runs the motor as a generator, converting kinetic energy back to DC electricity and pushing it back into the battery.

The amount of regen is controlled by:

- throtmin — negative value = throttle-lift regen (e.g. -5 = light regen when foot off throttle)
- Brake input — dinbrake can trigger regen in addition to friction braking
- BMS current limits — the BMS caps how much current can flow back into the battery

Warning: Regen was first reliably implemented in V2.15A (April 2024). Content about regen from before this date refers to a non-functional or experimental implementation.

The ISA Shunt — Pack Monitoring

The VCU cannot directly measure HV bus voltage — it operates entirely in the 12V domain. An external sensor bridges this gap.

The ISA IVT-S shunt sits in series with the HV bus. It measures voltage, current, and calculates state of charge, then transmits these readings over CAN to the VCU. The VCU uses this data to:

- Monitor precharge progress (is the bus voltage rising?)
- Know when precharge is complete (has it reached `udcsw`?)
- Enforce discharge and charge current limits
- Report pack state to the driver

The ISA shunt must have permanent 12V power — the same always-on feed as the VCU. If powered from ignition-switched supply it loses its coulomb count state when the key is off, and the next startup may read stale or zero UDC.

Isolation and Grounding

The HV system must be electrically isolated from the vehicle chassis (LV ground). This isolation is safety-critical — if HV+ or HV− contact the chassis, the chassis becomes energised at HV potential.

The ISA shunt measures the voltage difference between HV+ and HV− without either being referenced to chassis ground. The inverter is similarly isolated. The battery pack enclosure may be grounded to chassis for EMC reasons but the battery terminals must not be.

HVIL (High Voltage Interlock Loop) is a low-current 12V loop that passes through every HV connector in the system. If any HV connector is opened or a cable is disconnected, the loop breaks, the VCU detects it and immediately opens the contactors. See Module W04.

Source: Damien Maguire @Evbmw — "Lexus GS450h EV Drivetrain Deep Dive" (June 2020)

Last verified against firmware: V2.30A (August 2025)

High Voltage Safety

Track: Foundation

Prerequisites: F03 — EV Drivetrain Fundamentals

Audience: All levels — read before working on any HV system

Estimated reading time: 10 minutes

WARNING: This module is not optional. High voltage EV systems can kill instantly, silently, and without warning. Read this before touching any orange cable, any contactor, or any HV terminal.

The Risk

A typical EV conversion pack runs at 200–450V DC. At these voltages, as little as 100mA of current through the body is sufficient to cause fatal ventricular fibrillation. The current available from a traction pack is measured in hundreds of amps — far beyond what causes death.

The danger is compounded because:

- **DC does not let go.** AC at 50/60 Hz causes muscle contractions that may allow you to pull away. DC causes sustained muscle contraction — if you grip a live conductor your hand locks closed.
- **The capacitors retain charge.** After contactors open, the inverter bus capacitors remain charged at near-pack voltage for minutes. The pack is isolated but the inverter is still dangerous.
- **Contactors can fail closed.** A welded contactor leaves the HV system live even when the VCU believes it is off.

The Two-Person Rule

Never work on a live HV system alone.

If something goes wrong — a shock, a fall, a fault — you need someone present who can kill the power and call for help. This is not a guideline. It is the single most important safety practice in this entire module.

Personal Protective Equipment

Minimum for any HV work:

- **Insulated gloves rated for the voltage** — Class 0 (1000V rated) or higher. Not rubber washing-up gloves, not latex medical gloves. Proper electrical insulating gloves from a safety supplier. Inspect before every use — a pinhole is invisible and fatal.
- **Safety glasses** — arc flash and battery acid protection.
- **No jewellery** — rings, watches, and chains are conductors. Remove them entirely before working near any electrical system, HV or LV.

Additional for battery work:

- Face shield

- Non-conductive tools
 - Insulated floor mat
-

The Discharge Procedure

Before working on any HV component — inverter, charger, battery, contactor box — verify the system is de-energised:

Key off, key removed from vehicle

Disconnect the 12V auxiliary battery — this removes power from the VCU and prevents contactor actuation

Open the HV service disconnect — every EV pack should have a manual service disconnect (MSD) that physically breaks the HV circuit. Pull it, pocket it.

Wait minimum 5 minutes — allows inverter capacitors to discharge through bleed resistors. Some inverters take longer. Check your inverter specification.

Verify with a multimeter — set to DC voltage, highest range. Measure across HV+ and HV– at the inverter DC bus terminals. Should read near zero. If it reads anything above 50V, wait longer and measure again.

Only then touch any HV component

Warning: **Never trust the contactors alone.** Contactors can fail closed (welded). The 12V interlock can fail. The VCU can malfunction. The only safe assumption is that the HV system is live until you have personally measured otherwise with a calibrated meter.

Fusing Philosophy

Every HV conductor must be protected by a fuse rated for the maximum fault current. For a ZombieVerter-based build:

Pack-level fusing: One main fuse as close to the battery positive terminal as possible. This protects the entire HV circuit. Typical rating: 200–400A depending on pack and inverter capability. Use a fuse rated for the DC voltage of your pack — standard AC fuses are not suitable.

Branch fusing: Each major HV branch (inverter, charger, DC-DC) should have its own fuse. This allows a fault in one branch to be isolated without killing the entire system.

Fuse ratings: The fuse rating protects the wiring, not the load. Size fuses to the cable rating, not to the maximum current of the device. An undersized cable is a fire hazard even if the fuse doesn't blow.

Service disconnect (MSD): A manually operated HV disconnect that physically breaks the HV circuit. Required for safe servicing. Should be accessible without tools, in a known location, and clearly labelled. Some builders use the BMW safety box or similar contactor box as the MSD — this is acceptable only if the manual override is clearly accessible.

HV Isolation vs HVIL — Two Different Concepts

This section introduces two distinct safety concepts that are frequently confused. They are not the same thing.

HV Isolation

What it is: A fundamental design property of the entire HV system.

The HV bus (battery positive and negative) must be **electrically floating** — isolated from the vehicle chassis (LV ground). Neither HV+ nor HV– should have a low-resistance path to chassis.

Why it matters: If HV+ or HV– contacts the chassis, the chassis becomes energised at HV potential. Anyone touching the vehicle while also contacting ground completes a circuit through their body.

How it's maintained:

- Battery terminals are never directly connected to chassis
- Inverter and charger HV ports are internally isolated by design
- The ISA shunt measures HV differentially — neither terminal is referenced to chassis ground

What breaks it: A chafed HV cable touching a body panel, a failed component creating an HV-to-chassis path, or incorrect wiring (e.g. connecting HV– to chassis as a "ground" — never correct).

This is a design property, not a circuit you wire.

HVIL — High Voltage Interlock Loop

What it is: A low-current 12V detection circuit that monitors connector integrity.

A simple series loop of 12V passes through every HV connector, service disconnect, and access panel in the system. When all connectors are fully mated, the loop is complete. When any connector is pulled, the loop opens and the VCU immediately opens the HV contactors.

What it does: Detects physical connector disconnection and triggers a contactor open. It does **not** detect isolation faults (HV contacting chassis).

This is an active 12V circuit you physically wire. See Module W04.

Summary

	HV Isolation	HVIL
What it is	A system design property	A 12V detection circuit
What it detects	Ground faults — HV touching chassis	Connector disconnection
How it works	Floating HV bus design	Series 12V loop through all HV connectors
What happens on failure	HV chassis energisation — silent danger	VCU opens contactors immediately
Do you wire it?	No — it's a design requirement	Yes — see Module W04

Warning: **Neither HVIL nor HV Isolation replaces the discharge procedure.** Contactors can weld closed. HVIL circuits can fail. Isolation can be compromised. The only safe assumption before touching any HV component is that the system is live until you have personally measured otherwise.

The BMW Safety Box (S-BOX) integrates HVIL with pack-level CAN monitoring in one unit. See Module W04 for HVIL wiring.

Cable Colour Conventions

Colour	Meaning
Orange	HV cables — always treat as potentially live
Red	12V positive
Black	12V negative / chassis ground
Blue	Neutral (some charger AC wiring)
Brown	Line (some charger AC wiring)

Warning: When working on an unfamiliar vehicle, treat all orange cables as live regardless of the system state you believe it to be in.

Bench Testing Safety

Most first-start testing is done on 12V only — no HV battery connected. This is intentionally the safest way to verify wiring and parameters before introducing HV risk.

When you do connect HV for the first time:

- Use a current-limited bench supply set to a low voltage (60V, 2A) before using full pack voltage
- Start with a bench precharge verification — watch udc rise in Spot Values before attempting run mode
- Have the 12V auxiliary battery disconnect within reach before and during the first HV test
- Verify contactor sequencing on 12V before connecting any HV cables

If Someone Is Shocked

Do not touch them — if they are still in contact with a live conductor, you will be shocked too

Kill the power — pull the service disconnect, disconnect the 12V battery, use a non-conductive object to break the contact if you cannot reach the disconnect

Call emergency services immediately — even if the person appears conscious and uninjured, cardiac arrest can occur up to 24 hours after an electrical injury

CPR if necessary — electrical shock can cause cardiac arrest without obvious external injury

Sources: *openinverter.org* community practice · *BMW Safety Box series* — *Good Enuff Garage* (April-May 2024) · *general electrical safety standards* Last verified: August 2025

VCU Hardware Walkthrough

Track: Hardware

Prerequisites: F02 — ZombieVerter Ecosystem Overview

Audience: All levels

Estimated reading time: 12 minutes

Board Overview

The ZombieVerter VCU is a compact single-board controller built around two main chips:

- **STM32F107** — the main microcontroller, running the VCU firmware
- **ESP8266** — a plug-in Wi-Fi module that hosts the web interface

WARNING: GD32F107 Warning: A small number of boards produced during the 2021-2022 semiconductor shortage used a GigaDevices GD32F107 chip instead of the STM32. These "GD boards" require a separate firmware branch that has since been abandoned and diverged significantly from the main codebase. If you have an older board, check your chip markings before flashing. See Module C01 for details.

The current production hardware revision is **V1.a**. An earlier V1 (pre-production) revision exists — the main wiki image labels it V1.2 to distinguish it. V1.a is the board you will receive from the webshop today.

Physical Layout

The board is housed in an IP-rated enclosure with a 56-pin Aptiv connector on one end. Key physical features:

The 56-Pin Main Connector

This is the single point of connection for all vehicle wiring. All power, ground, signal inputs, and outputs pass through this connector. It uses Aptiv (formerly Delphi) automotive-grade terminals.

Connector part numbers:

Part	Aptiv Part Number
Cable-side connector housing	Aptiv 56-pin connector
Board-side header	Aptiv 56-pin header
Terminal removal tool	210S048 (Aptiv/Delphi)

Alternative sources (AliExpress):

- Full enclosure kit with header, connector, and pins
- Connector and pins only
- Pre-wired connector harness (most convenient for initial bench testing)

The Wi-Fi Module

The ESP8266 plugs into a socket on the main board. It can be removed — this is sometimes recommended when doing an ST-Link firmware recovery, to avoid any interference during the flash process.

SSID: inverter or zomvcu (varies by firmware version) **Password:** inverter123 **Default IP:** 192.168.4.1

Warning: **Known issue with newer boards:** Some recent units have a new Wi-Fi module variant that does not automatically assign an IP address via DHCP. If you connect to the Wi-Fi but cannot reach 192.168.4.1, manually set your computer's IP address to 192.168.4.2 (or any address in the 192.168.4.x range other than .4.1). See the forum thread linked in the wiki for ongoing status.

Status LEDs

The board has several indicator LEDs:

LED	Label	Meaning
Activity	acty	Should be flashing when the board has 12V power. A steady (non-flashing) light or no light indicates a problem.
ESP8266 LEDs	on Wi-Fi module	Indicate Wi-Fi module status

If acty is not flashing after applying 12V power, the board is not running correctly. Common causes: insufficient power supply, bad ground connection, or a firmware issue requiring recovery.

SWD Test Points (C, G, D)

Three test points on the PCB are used for ST-Link firmware recovery:

- **C** — SWClk (clock)
- **G** — Ground
- **D** — SWDio (data)

These are only needed if the board is bricked and cannot be recovered via the web interface. See Module C01.

Power Requirements

The ZombieVerter requires a **permanent 12V supply** — not ignition-switched. This is intentional: the VCU needs to remain powered when the vehicle is parked to manage charge timers, monitor systems, and handle charging sessions.

Parameter	Value
Supply voltage	12V DC
Idle current draw	~150 mA
Power pin	Pin 56 (12V+)
Ground pin	Pin 55 (12V-)

150 mA at idle is modest — comparable to a small relay module. With a healthy 12V battery, this will not cause meaningful parasitic drain over days of parking.

WARNING: Do not use an ignition-switched supply for the main VCU power. If you do, the VCU will lose its UDC (pack voltage) history when the key is off, which can cause precharge failures on the next startup. See Module W03.

Complete Pin Reference

The following table covers all 56 pins on the main connector. Fixed-function pins are listed first, followed by the configurable I/O section.

Fixed Power & Signal Pins

Pin	Function	Direction	Notes
55	12V Ground	PWR	Permanent ground
56	12V Positive	PWR	Permanent 12V — NOT ignition switched
32	Ignition relay control	Output (low-side)	Pulls ground pin of 12V relay to ground — relay switches 12V "ignition" to inverter/charger/DCDC
15	Ignition ON input	Input	Pull HIGH (to 12V+) with ignition switch in ON position
52	Start input	Input	Pull HIGH momentarily with ignition switch in START position
53	Reverse input	Input	Pull HIGH (to 12V+) when reverse selected
54	Forward input	Input	Pull HIGH (to 12V+) when forward selected
49	Brake input	Input	Pull HIGH (to 12V+) when brake pedal pressed
45	Throttle ground	PWR	Ground reference for throttle pot(s)
46	Throttle channel 2	Analog input	Second pot wiper (dual-channel throttle)
47	Throttle channel 1	Analog input	First pot wiper
48	Throttle 5V supply	PWR	5V supply for throttle pot(s)
31	Negative contactor	Output (low-side)	Pulls negative contactor coil ground
33	Positive contactor	Output (low-side)	Pulls positive contactor coil ground

Pin	Function	Direction	Notes
34	Precharge contactor	Output (low-side)	Pulls precharge contactor coil ground

WARNING: Contactor wiring rule: All contactor output pins are **low-side switching** — they pull the ground leg of the contactor coil to ground. The positive leg of each contactor coil must be connected to 12V+. Do NOT connect 12V+ to these output pins.

Configurable I/O Pins

These pins can be assigned to different functions through the web interface. However, the hardware type of each pin is fixed — software cannot change a low-side output into a high-side output, for example.

Low-Side Outputs

These pull to ground when activated. Use them to switch relay coils (coolant pumps, fans, etc.).

Not suitable for directly driving loads.

Pin assignment name	Default use
GP Out 3	General purpose
GP Out 2	General purpose
Neg Contactor switch / GP Out 1	Negative contactor (or general purpose)
Trans SL1–	GS450h transmission solenoid 1 (if not using GS450h, available as GP)
Trans SL2–	GS450h transmission solenoid 2 (if not using GS450h, available as GP)

High-Side PWM Outputs

These supply 12V+ when activated, with optional PWM modulation. Use for gauges, PWM-controlled fans, or indicator signals. **Not suitable for controlling relay coils** — use low-side outputs for relays.

Pin assignment name	Default use
PWM 1	General purpose PWM
PWM 2	General purpose PWM
PWM 3	General purpose PWM
Pump PWM	GS450h oil pump PWM, or tachometer PWM output

Ground-Pull Digital Inputs

These pins detect when pulled to ground. **Do not apply any voltage to these pins** — they are ground-detect only, not 12V-tolerant inputs.

Pin assignment name
PB1

Pin assignment name
PB2
PB3

Configurable Pin Functions (Software-Assigned)

The following functions can be assigned to the configurable I/O pins via the web interface. Note the type constraint — a function expecting a low-side output cannot be assigned to a PWM pin.

Function Name	Type	Description
ChaDemoAlw	OUTPUT	Activates when CHAdeMO charger handshake initiates
OBCEnable	OUTPUT	Activates as part of the ExtCharger module
HeaterEnable	OUTPUT	Activates in run mode when coolant pump is on
RunIndication	OUTPUT	Activates when VCU is in run mode
WarnIndication	OUTPUT	Activates when an error occurs
CoolantPump	OUTPUT	Activates during precharge and run
NegContactor	OUTPUT	Activates when negative contactor should close
BrakeLight	OUTPUT	Activates when brake light threshold met
ReverseLight	OUTPUT	Activates when reverse is selected
CoolingFan	OUTPUT	Activates when FanTemp setpoint is reached
HVActive	OUTPUT	Activates when contactors are closed in run or charge mode
BrakeVacPump	DIGITAL OUTPUT	Activates when BrakeVacSensor threshold is met
CpSpoof	PWM OUTPUT	Spoofs CP signal to OBC (used with FOCCI or i3 LIM)
GS450Hpump	PWM OUTPUT	Controls GS450h oil pump speed
HeatReq	DIGITAL INPUT	Heater request signal
HVRequest	DIGITAL INPUT	Currently non-functioning

Function Name	Type	Description
DCFCRequest	DIGITAL INPUT	CHAdEMO charge interface enable
ProxPilot	ANALOGUE INPUT	Detects J1772 charge cable insertion
BrakeVacSensor	ANALOGUE INPUT	Vacuum sensor for brake booster pump control
Switch_NoRegen	DIGITAL INPUT	Inhibits regen (useful with manual transmission) — check firmware release notes for availability

WARNING: Critical warning: The web interface will allow you to assign a function to a pin type it is not compatible with (e.g. assigning an output function to an input pin). The software will accept the assignment but the hardware will not respond. Always check that your assignment matches the physical pin type. A common mistake is wiring an input switch but assigning it to a function that requires an output.

The Proximity Pilot (ProxPilot) Circuit

The ProxPilot analogue input is used to detect when a J1772 charge cable is plugged in, which triggers the VCU to enter charge mode. This is required for chargers like the Mitsubishi Outlander OBC that don't have their own cable detection logic. (The Nissan Leaf PDM handles this internally, so ProxPilot is not needed for Leaf stacks.)

The detection circuit requires two external resistors:

```

5V (Pin 48)
|
|
330Ω ← R_pullup
|
|————— ProxPilot analogue input pin
|
R5 ← value depends on connector type:
| Type 1 (J1772): 2.7kΩ
| Type 2 (Mennekes): 4.7kΩ
|
GND

```

Note: Note: The charge port itself may already contain this resistor — check before adding an external one.

Configuration steps:

Assign ProxPilot function to the chosen analogue input pin in the web UI

Monitor the PPVal spot value while plugging in a cable

Set PPthreshold to a value just above the reading seen when the cable is inserted

Be aware that the exact resistance (and therefore PPVal) will vary depending on the cable's rated current capacity

What's Next

- **H02** — Inputs & Outputs Deep Dive: electrical characteristics, voltage limits, current ratings
- **H03** — CAN Bus on the VCU: which CAN bus is which, termination, and baud rates
- **W01** — Wiring Philosophy & Best Practices: before you touch a wire

Source: openinverter.org/wiki/ZombieVerter_VCU | openinverter.org/wiki/ZombieVerter_IO

Last verified against firmware: V2.30A (August 2025)

Wiring Philosophy and Best Practices

Track: Wiring

Prerequisites: H01 — VCU Hardware Walkthrough

Audience: All levels

Estimated reading time: 15 minutes

The Philosophy

A ZombieVerter wiring harness is not a single entity — it is three logically separate fuse blocks, each with a distinct purpose, feeding different groups of components. Understanding this structure before crimping a single terminal will save hours of debugging later.

The three groups are:

Permanent power — always live, even when the vehicle is parked

Ignition ON — live when the key is in the ON position

Start position — momentary signal only, like turning a key to START

Each group has its own fuse block. Each fuse block has its own feed from the 12V battery. Mixing these groups — feeding an ignition-switched device from permanent power, or running too much current through the ignition switch — is the most common source of mysterious electrical gremlins in EV conversions.

Fuse Block Strategy

Permanent Power Block

What belongs here: VCU (Pin 56), ISA shunt, any device that needs to run when the vehicle is parked (charge timer management, telematics).

Fuse sizing: 1-3A for VCU alone. Add appropriately for other permanent loads. Each device on this block should have its own individual fuse — don't share.

Why separate: The VCU and ISA shunt must have always-on power. If you feed them from ignition-switched supply, the shunt loses UDC state on key-off and precharge fails on every startup. See Module C03.

Practical tip: Use modular automotive fuse blocks (e.g. Dorman 4-position blocks). They connect together like Lego — start with 4 positions, add another block as devices accumulate. The bus bar makes adding feeds trivial.

Ignition ON Block

What belongs here: Devices that should wake up when the key is turned on but should not draw permanent power. In a ZombieVerter build this block is typically fed from the inverter enable relay (Pin 32) rather than the ignition switch directly — the VCU controls when these devices get power.

Why use the relay: The inverter, charger, and DC-DC converter should not power up until the VCU is ready to manage them. Feeding them from a direct ignition feed bypasses this protection.

Start Position Block

What belongs here: The start signal (Pin 52) only. This is a momentary 12V pulse — 500ms minimum — that triggers the contactor sequence. Not a sustained feed.

Key principle: Never run high current through the ignition switch. The ignition switch should only carry enough current to activate relays and signal pins. Fuse the ignition switch output at 5A or less.

Wire Gauging

Wire gauge is specified in AWG in North America, metric cross-section (mm²) in Europe. Higher AWG number = thinner wire — the opposite of the metric system.

Application	Suggested gauge
HV main traction cables	35-70mm ² (0 AWG or larger)
HV charger / DC-DC	10-16mm ² (6-4 AWG)
12V main battery feed	6-10mm ² (10-8 AWG)
12V fuse block feeds	4mm ² (12 AWG)
Signal wires (VCU pins)	0.5-1mm ² (20-18 AWG)
CAN bus	0.5mm ² twisted pair (22 AWG)

Warning: Rule: Rule: Size the wire for the load current. Then size the fuse just below the wire's current rating. The fuse protects the wire — the wire must already be sized to handle the device.

Connector Selection

For the VCU 56-pin header: Use the correct Aptiv (formerly Delphi) connector with the matching terminal crimps. Available as a complete kit on AliExpress. Use a proper ratcheting crimp tool — a generic crimp tool produces unreliable crimps on these terminals.

For signal connections: Use automotive-grade waterproof connectors (Deutsch DT or equivalent) where connections may be exposed to moisture or vibration. Avoid bare blade connectors in the engine bay.

For CAN bus: Twisted pair. Either purpose-made CAN cable or twisted 22 AWG signal cable. Termination at both ends with 120Ω resistors.

Joining wires: Solder and heat-shrink, or proper crimp splice connectors. Never twist-and-tape. Wire nuts are not suitable for automotive use.

Labelling

Label every wire at both ends before installation. Once a harness is routed and cable-tied, tracing an unlabelled wire is extremely time-consuming.

A simple label convention:

P56-12VPERM ← Pin 56, permanent 12V

P15-IGNON ← Pin 15, ignition ON signal

CAN1H-INV ← CAN1 High, to inverter

Use heat-shrink label sleeves or a label maker with vinyl tape. Marker pen on tape fades in heat and oil.

Dupont Cables — For Bench Only

Dupont (breadboard) jumper cables are useful for initial bench testing — they let you power up the board without committing to a permanent header. However they are loose, unreliable, and not rated for automotive vibration. Remove and replace with proper crimped terminals before the vehicle goes on the road.

Building the Harness — The Sequence

Good Enuff Garage's wire harness series (April-June 2024) documents the complete ZombieVerter harness build sequence. The recommended order:

Permanent power fuse block — build first, verify VCU powers up and Wi-Fi appears

Ignition switch to VCU — add ignition ON (Pin 15) and momentary start (Pin 52)

Verify run mode on bench — 12V only, no HV — before adding anything else

Forward/reverse switch — add direction control (Pins 53/54), verify `din_forward/din_reverse` in Spot Values

Brake signal — add brake switch (Pin 49), verify `din_brake` goes on/off correctly

Throttle wiring — add pot signals, calibrate

Inverter relay — add Pin 32 output relay, test switched supply

Contactor wiring — add precharge, negative, main contactor coil wiring

CAN bus — add inverter CAN, shunt CAN

HV connections — final step, after all 12V functionality is verified

Build incrementally and verify each step before adding the next. A fault in step 3 is easy to find. The same fault buried under 8 more steps takes hours.

The Safety Box Harness

The BMW Safety Box (S-BOX) is often used in BMW-based conversions to provide pack monitoring and HVIL in one unit. The S-BOX harness is covered in the wire harness series (May 2024, Parts 1 and 2). Key point: the S-BOX communicates over CAN2 and provides the `udc` reading that the VCU uses for precharge. Set `shuntType` = SBOX and `shuntCan` = CAN2.

Common Wiring Mistakes

Mistake	Consequence	Prevention
VCU on ignition-switched power	Shunt loses UDC, precharge fails every startup	Permanent power to Pin 56 always
Brake switch on normally-closed terminal	dinbrake stuck on, no torque ever	Use normally-open (NO) terminal
CAN H and CAN L swapped	No communication, no fault message	Use twisted pair, check colour code
Missing CAN termination	Intermittent communication faults under load	120Ω at each end of each bus

Mistake	Consequence	Prevention
throtmax left at 0	Throttle does nothing	Set to 100 — see C04
Contactors coil+ wired to VCU output pin	VCU output damaged — outputs are low-side only	Coil+ to 12V, coil– to VCU pin

Source: Good Enuff Garage "DIY Wire Harness" series (April-June 2024, 10 videos) · Damien Maguire @EVBmw

Last verified against firmware: V2.30A (August 2025)

Throttle and Brake Wiring

Track: Wiring

Prerequisites: H01 — VCU Hardware Walkthrough

Audience: All levels

Estimated reading time: 12 minutes

Overview

The throttle and brake inputs are the primary driver interface signals. Getting them right — especially the throttle calibration — is essential for safe, predictable vehicle behaviour. This module covers the wiring; calibration parameters are covered in Module C04.

Throttle Pedal Wiring

Dual vs Single Channel

The ZombieVerter supports both single and dual-channel throttle pedals. **Dual-channel is strongly recommended** for safety — if one channel fails (open circuit, short circuit, or mechanical linkage failure), the VCU can detect the discrepancy between channels and cut torque rather than applying full throttle.

Single-channel is available but provides no redundancy. If the single pot fails in the high direction, the result is unintended full throttle. Use dual-channel wherever possible.

Pin Assignments

VCU Pin	Function	Connect to
Pin 48	Throttle 5V supply	Throttle pot(s) positive (Vcc)
Pin 47	Throttle channel 1	Channel 1 wiper
Pin 46	Throttle channel 2	Channel 2 wiper
Pin 45	Throttle ground	Throttle pot(s) negative (GND)

For a dual-channel pedal, both pots share the 5V supply (Pin 48) and ground (Pin 45), with their wipers going to Pins 47 and 46 respectively.

For a single-channel pedal, only Pins 48, 47, and 45 are used. Set potmode to single in the web interface.

Wiring Diagram

Dual-channel throttle:

Pin 48 (5V) ——— Pot 1 Vcc

————— Pot 2 Vcc

Pin 47 (Ch1) ——— Pot 1 Wiper

Pin 46 (Ch2) ————— Pot 2 Wiper
 Pin 45 (GND) ————— Pot 1 GND
 ————— Pot 2 GND

Single-channel throttle:

Pin 48 (5V) ————— Pot Vcc
 Pin 47 (Ch1) ————— Pot Wiper
 Pin 45 (GND) ————— Pot GND

Understanding How the ZombieVerter Reads Throttle

This is one of the most important things to understand before calibrating — **the ZombieVerter handles potmin/potmax differently from most inverters and motor controllers.**

The Standard Approach (Most Controllers)

Most motor controllers set potmin and potmax *outside* the mechanical travel range of the pedal. For example, if the pedal physically travels from an ADC value of 400 to 2600, you might set potmin to 300 and potmax to 2700. This way, if a value outside those limits is detected (due to an open wire, short, or mechanical failure), the controller faults safely.

The ZombieVerter Approach (Opposite!)

The ZombieVerter sets potmin and potmax *inside* the mechanical travel range:

- potmin should be set **just above** the value seen at the resting (off-throttle) position
- potmax should be set **just below** the value seen at full throttle

This means the pedal physically travels slightly beyond both limits. The VCU uses these values to calculate a normalised 0-100 potnom value.

Warning: This trips up almost everyone coming from other systems. If you set potmin below your actual resting position, the VCU will interpret the pedal as always slightly depressed. If you set potmax above your actual WOT value, you'll never reach 100% torque.

Calibration procedure (covered in detail in Module C04):

Watch the pot spot value in the web interface while moving the pedal

Note the value at rest — set potmin just above this

Note the value at full travel — set potmax just below this

Do the same for pot2min / pot2max if using dual-channel

Throttle Parameters Quick Reference

Parameter	Description
potmin	ADC value just above off-throttle resting position
potmax	ADC value just below wide-open throttle
pot2min	Same as potmin but for channel 2
pot2max	Same as potmax but for channel 2
potmode	single or dual — must match your wiring

Parameter	Description
potnom	Resulting normalised 0-100 throttle value (spot value, read-only)
throtmin	Minimum allowed potnom — can be negative (for throttle-lift regen)
throtmax	Maximum allowed potnom in forward direction
throtramp	Rate of potnom increase (potnom change per %/10ms) — limits rate of torque rise
throtramprpm	Motor RPM above which throtramp no longer applies
revlim	Rev limiter (motor RPM)

Critical Gotcha: throtmax Defaults to Zero

throtmax defaults to **0** in a fresh configuration. This means the VCU will allow zero forward torque regardless of throttle position. **This is the most common reason for a "throttle does nothing" problem on first start.** Set throtmax to 100 (or your desired maximum) before expecting torque output.

Brake Signal Wiring

Purpose

The brake signal serves two functions in the ZombieVerter:

Safety interlock — dinbreak must read "off" before the VCU will allow torque output. If the brake is stuck "on", the VCU will not produce torque even if throttle is applied.

Regen trigger — applying the brake can trigger regenerative braking (configured separately — see Module A01).

Wiring

The brake input is a simple digital input:

VCU Pin	Function	Connect to
Pin 49	Brake input	12V+ when brake pedal is pressed

When the brake pedal is pressed, Pin 49 is pulled to 12V+. When released, Pin 49 should return to 0V (or float).

Do not leave Pin 49 floating — a floating input can read randomly as "on" due to electrical noise, blocking torque output unpredictably. If you are not wiring a brake signal for now (bench testing), either connect Pin 49 to ground through a resistor or confirm dinbreak reads "off" in spot values before troubleshooting torque issues.

BrakeLight Output

The VCU can drive a brake light output from a configurable low-side output pin. Assign the BrakeLight function in the web interface to activate this output when the brake is pressed above

a configurable threshold.

Forward / Reverse Wiring

VCU Pin	Function	Connect to
Pin 54	Forward	12V+ when forward selected
Pin 53	Reverse	12V+ when reverse selected

Both are digital inputs pulled high to 12V+ to select direction. The direction signal can come from:

- A physical gear selector lever with detents (park/reverse/neutral/drive)
- A toggle switch
- A momentary button (set `dirmode` to `button` or `buttonReversed` in the web interface)

`dirmode` parameter options:

- `switch` — pin high = direction active
- `button` — momentary press toggles direction
- `switchReversed` — inverted logic
- `buttonReversed` — inverted momentary logic

Common Wiring Mistakes

Problem	Cause
dinbreak always shows "on"	Brake pin wired to constant 12V+, or brake switch wired NC instead of NO
dinbreak always shows "on"	Floating pin picking up noise — add pull-down resistor
potnom never reaches 100	potmax set higher than actual WOT ADC value
potnom shows value at rest	potmin set lower than actual resting ADC value
No torque output	throtmax set to zero (default)
Erratic torque	Poor ground connection between throttle pot GND and Pin 45
Channel 2 fault	potmode set to dual but <code>pot2min</code> / <code>pot2max</code> not calibrated

What's Next

- **C04** — Throttle Calibration: step-by-step parameter tuning in the web interface
- **W03** — Contactor & Precharge Circuit
- **A01** — Regen Tuning: brake and throttle-lift regen configuration

Source: openinverter.org/wiki/ZombieVerter_VCU | [openinverter.org forum threads](https://openinverter.org/forum/threads)

Last verified against firmware: V2.30A (August 2025)

Contactor and Precharge Circuit

Track: Wiring

Prerequisites: H01 — VCU Hardware Walkthrough, F03 — EV Drivetrain Fundamentals

Audience: All levels

Estimated reading time: 18 minutes

Why This Module Matters

Contactor and precharge wiring is the most safety-critical part of a ZombieVerter installation. Getting it wrong can result in:

- Violent inrush current that welds contactors closed permanently
- Failure to enter run mode (the most common first-start problem)
- Unexpected HV live conditions on the vehicle chassis

Read this module completely before wiring. Do not skip the UDC section.

What the Contactor System Does

The ZombieVerter manages three contactors in a standard EV conversion:

Contactor	Purpose
Negative	Connects the battery negative to the HV system ground
Precharge	Connects HV positive through a resistor, to slowly charge inverter capacitors
Positive (Main)	Connects battery positive directly to the HV bus once precharge is complete

The sequencing logic is:

```

START signal received
|
▼
Negative contactor CLOSES ← always first, immediate on MOD_PRECHARGE entry
|
▼
Inverter 12V enable fires ← inv_out SET (Pin 32 relay), inverter begins booting
|
[250ms stagger]
|
▼
Precharge contactor CLOSES ← current limited by precharge resistor

```

HV bus begins rising
▼
1 second minimum precharge time enforced regardless of UDC
▼
UDC ≥ udcsw AND UDC < udclim AND throttle released?
YES — [250ms stagger] — Main (positive) contactor CLOSES
▼
VCU enters RUN mode — opmode = "run"
NO — wait (up to 5 seconds total from start)
PRECHARGE FAULT — prec_out cleared, ERR_PRECHARGE posted
Warning: Important: The precharge contactor does not open when the main contactor closes. It remains energised throughout the entire drive session and is only de-energised when the VCU returns to MODOFF. This is the actual firmware behaviour — the precharge resistor runs in parallel with the main contactor during normal driving. See [Precharge Contactor Behaviour](#precharge-contactor-behaviour) below.

UDC — The Most Important Concept in This Module

UDC is the VCU's measurement of HV bus voltage. It is the single most critical input to the contactor sequencing logic.

Without a valid UDC reading, the ZombieVerter **will fail precharge and never enter run mode**. This is the most common cause of first-start failures.

The VCU can receive UDC from several sources, selected via the shuntType parameter:

shuntType value	Source
ISA	ISA IVT current/voltage shunt (most common)
SBOX	BMW S-BOX battery management unit
VAG	Volkswagen/Audi battery system
LEAF	Nissan Leaf inverter (reports pack voltage via CAN)

You must configure shuntType to match your actual hardware. If it is set to a source that is not present or not communicating, UDC will read zero and precharge will fail immediately.

ISA Shunt — Permanent Power Requirement

If you are using an ISA IVT shunt (the most common choice), it must have **permanent 12V power** — the same always-on supply as the VCU itself. The ISA shunt continuously monitors and reports UDC.

If the shunt is powered from an ignition-switched supply, it will lose its UDC reading when the key is off. When you next turn the key and start precharge, the VCU may see a stale or zero UDC value

and behave unpredictably.

Warning: Rule: ISA shunt power = always on. Same feed as the VCU.

Contactor Wiring

Electrical Concept: Low-Side Switching

All contactor outputs on the ZombieVerter are **low-side switches**. This means:

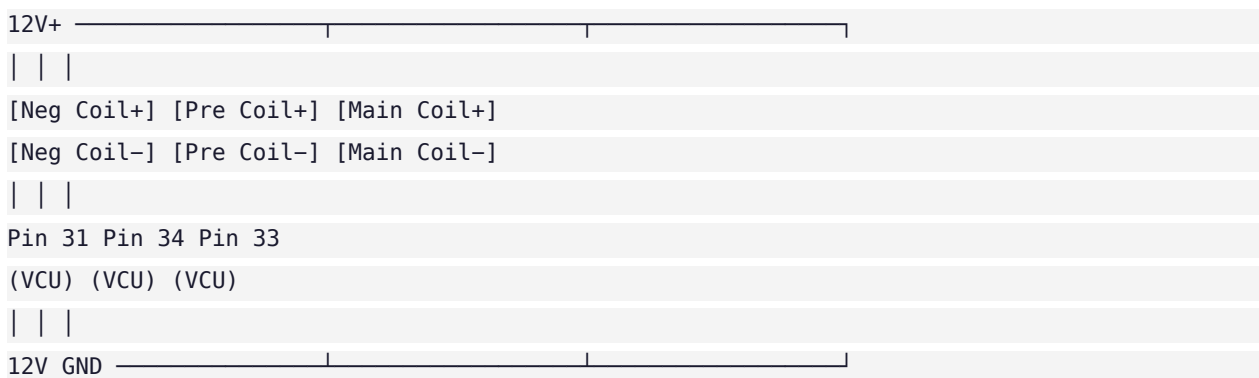
- The VCU output pin connects to the **ground leg** of the contactor coil
- When the VCU activates the output, it pulls that pin to ground, completing the circuit
- The **positive leg** of every contactor coil must be connected to 12V+

Never connect 12V+ to the VCU contactor output pins. These are ground-side switching outputs only.

Pin Assignments

VCU Pin	Function	Connects to
Pin 31	Negative contactor	Ground leg of negative contactor coil
Pin 33	Positive (main) contactor	Ground leg of positive contactor coil
Pin 34	Precharge contactor	Ground leg of precharge contactor coil

Wiring Diagram (Conceptual)



Contactor Polarity

Some contactors are polarity-sensitive — their coil has a defined positive and negative terminal. If wired backwards, they will not close (or may close but with reduced hold force and reduced life). Check your specific contactor datasheet.

Precharge Resistor

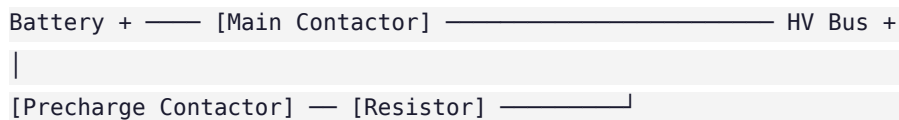
The precharge resistor limits current flow during the precharge phase, protecting both the contactor and the inverter capacitors from inrush damage.

Sizing guidelines:

- Typical values: 40Ω–100Ω at 50W–100W rating

- Higher resistance = slower precharge (more protection, but must complete within 5 seconds)
- Lower resistance = faster precharge (less protection, may stress contactors)
- 100Ω / 100W is a common and safe starting point for most conversions up to ~400V

Placement: The precharge resistor connects in series with the precharge contactor, between the positive battery terminal and the positive HV bus (inverter side).



Precharge Contactor Behaviour

Warning: ☐ **This differs from standard EV practice — read carefully.**

In most EV systems the precharge contactor opens as soon as the main contactor closes, removing the resistor from the circuit. **The ZombieVerter firmware does not do this.**

Looking at the state machine in `stm32vcu.cpp`: `precout` is set during `MODPRECHARGE` and is never cleared on transition to `MODRUN` or during normal driving. The only places `precout` is explicitly cleared are:

- `MODOFF` — after the shutdown `rlyDly` countdown completes
- `MODPCHFAIL` — on a precharge timeout fault

This means **the precharge contactor remains closed in parallel with the main contactor for the entire drive session**. The precharge resistor is in the circuit throughout normal operation. At typical drive currents flowing through the main contactor, the resistor carries negligible current (the main contactor provides the low-impedance path), so this has no practical effect on performance.

The implication for hardware selection: your precharge contactor and resistor must be rated for sustained energisation, not just the brief precharge interval. Most standard precharge contactors are already rated for continuous coil energisation, but verify your specific part.

UDC Measurement Point — ISA Shunt

The ISA shunt must be positioned to measure the voltage at the correct point. The shunt's U1 terminal reads voltage relative to HV ground.

During precharge, U1 should be connected to the HV bus on the inverter side of the main contactor. As the precharge resistor charges the inverter capacitors, U1 rises toward pack voltage. When it crosses `udcsw`, the VCU closes the main contactor.

A common mistake is connecting U1 to the battery side of the main contactor. In this position, U1 reads full pack voltage immediately (before precharge even starts), and the VCU will try to close the main contactor without actually precharging the inverter capacitors — potentially welding the main contactor on first use.

Warning: Check: During initial power-up with no HV connected, U1 should read 0V. Apply HV with only precharge closed — U1 should rise slowly from 0V toward pack voltage. Only then will the VCU close the main contactor.

Startup Timing Detail

The firmware enforces deliberate stagger delays between each contactor event. Understanding these helps diagnose first-start timing issues.

Full contactor sequence with timing:

MOD_PRECHARGE entry:

t=0ms NEG contactor closes (immediate)

t=0ms inv_out SET – inverter 12V enable fires (immediate)

t=250ms PREC contactor closes (after 25-tick rlyDly)

t=250ms–1250ms HV bus rising, 1-second minimum precharge enforced

t=≥1250ms UDC check passes

MOD_RUN entry:

t+0ms rlyDly=25 begins

t+250ms MAIN contactor closes (after 25-tick rlyDly)

t+250ms Drive enabled

MOD_OFF (shutdown):

t=0ms rlyDly=250 begins (2.5-second grace period)

t=2500ms All contactors open simultaneously

The **250ms gap between NEG and PREC** is intentional — it prevents multiple contactors pulling current simultaneously and gives the inverter time to begin its own boot sequence before HV arrives on the bus.

The **2.5-second shutdown delay** (10× longer than the startup stagger) gives the inverter time to ramp torque down to zero and safely discharge gate drivers before contactors open.

Inverter 12V Enable — Timing and Exceptions

The VCU fires invout (Pin 32 relay) at the very start of MODPRECHARGE, **before** the precharge contactor closes. This is deliberate: the inverter needs time to boot its own firmware and establish CAN communication before the VCU expects to talk to it. The 250ms stagger before precout provides that window.

Exception — OpenInverter-based inverters: When the selected inverter type is set to OpenInverter (Inverter = 0I), invout is **not** fired during precharge. OpenInverter manages its own startup independently. If you are using an OI-based inverter and the enable relay is not firing, this is expected behaviour, not a fault.

Key Parameters

These parameters control contactor and precharge behaviour:

Parameter	Description	Typical Value
udcmin	Minimum pack voltage — below this, VCU will not enter run mode	Set to ~80% of your minimum expected pack voltage (e.g. 20V for bench testing)
udclim	Maximum pack voltage — above this, VCU derates	Set just above your fully charged pack voltage
udcsw	Voltage threshold that triggers main contactor close	Set to ~95% of your nominal pack voltage

Parameter	Description	Typical Value
shuntType	Source of UDC measurement	Match to your hardware: ISA, SBOX, VAG, or LEAF

Common First-Start Mistakes

Symptom	Likely Cause
UDC reads 0, precharge never completes	shuntType wrong, shunt not powered, shunt not initialised, shunt CAN bus not connected
UDC reads full pack voltage immediately, main contactor fires instantly	ISA U1 connected to battery side instead of inverter side
Precharge times out (stays in PRECHARGE state)	udcsw set too high, precharge resistor too large, shunt not reporting correctly
VCU never attempts precharge	udcmin set too high (pack voltage is below the minimum threshold)
Precharge completes but VCU won't enter run mode	dinbreak is showing "on" — brake signal is active or stuck, blocking run mode
Inverter enable relay not firing, OI inverter	Expected — invout is intentionally suppressed for OpenInverter type
Fault code appears briefly then clears on entering run mode	Normal — ErrorMessage::UnpostAll() is called every 10ms cycle while in MODRUN

ISA Shunt Initialisation

A new ISA shunt must be initialised before it will communicate on CAN. This is a one-time procedure:

Wire the ISA shunt to permanent 12V and the correct CAN bus

In the web interface under **Comms**, set shuntCan to the CAN bus the shunt is connected to

Save parameters to flash

Under **Comms**, set ISAMode to **Init**

Save parameters to flash

Power cycle the VCU and shunt simultaneously (they should be on the same 12V feed)

The shunt initialises

Set ISAMode back to **Normal**

Save parameters to flash

Reboot the VCU

If initialisation fails: Separate the shunt and VCU power supplies. Power on the shunt first, wait 2–3 seconds, then power on the VCU. This delay allows the shunt to fully boot before the VCU attempts to communicate with it. This has resolved initialisation failures for a number of users.

Contactor Firing During Precharge — Inverter Enable Relay

One subtle but important wiring detail: the inverter (and charger, DCDC) should not receive its 12V "ignition/enable" signal until the VCU is ready to energise it. Premature inverter power-up can cause:

- CAN bus conflicts before the VCU has configured the inverter
- Inverter self-initialisation that conflicts with VCU sequencing

The ZombieVerter controls an external relay (via Pin 32, low-side switching) to provide a switched 12V "ignition" signal to these devices. This relay fires at the start of MODPRECHARGE, giving the inverter a 250ms head-start before the precharge contactor closes and HV arrives on the bus.

Relay wiring:

```
12V+ — [Relay coil +]  
[Relay coil -] — Pin 32 (VCU) — GND  
[Relay switch NO] — 12V+  
[Relay switch COM] — Inverter/charger enable/ignition pin
```

Check that this relay is firing during precharge by watching the inverter's power indicator or CAN traffic when you apply the Start signal.

Troubleshooting Checklist

Before assuming a hardware fault, verify all of the following:

- [] shuntType matches actual hardware
- [] ISA shunt has permanent 12V power (not ignition-switched)
- [] ISA shunt has been initialised (ISAMode Init → Normal cycle)
- [] ISA shunt CAN bus matches shuntCan parameter setting
- [] CAN bus has correct termination (120Ω at each end of the bus)
- [] ISA shunt U1 is connected to inverter side of main contactor (not battery side)
- [] All contactor coil positive leads are connected to 12V+
- [] All contactor coil negative leads are connected to the correct VCU pins (31, 33, 34)
- [] udcmin is set to a sane value (try 20V for bench testing)
- [] udcsw is set below actual pack voltage
- [] dinbreak reads "off" in spot values (brake signal not stuck on)
- [] Inverter enable relay is firing during precharge (Pin 32) — except OpenInverter type

What's Next

- **W04** — HVIL: High Voltage Interlock Loop
- **W07** — CAN Bus Wiring: termination and topology
- **C03** — Essential Parameters: First Start

Source: openinverter.org/wiki/ZombieVerter_VCU | [openinverter.org forum threads](https://openinverter.org/forum/threads) | `stm32_vcu.cpp` V2.40A-RW

Last verified against firmware: V2.30A (August 2025)

HVIL High Voltage Interlock Loop

Track: Wiring

Prerequisites: F04 — High Voltage Safety, W03 — Contactor & Precharge Circuit

Audience: Intermediate

Estimated reading time: 10 minutes

What Is HVIL?

The High Voltage Interlock Loop (HVIL) is a low-current 12V detection circuit that passes through every HV connector, service disconnect, and access panel in the system. If any connector is unplugged, any panel is removed, or any HV cable is broken, the loop opens — and the VCU immediately opens the HV contactors.

HVIL transforms accidental HV contact from a likely fatal event into a contactor-open event. Instead of unplugging an HV cable while the system is live, you unplug the HVIL pin first (or simultaneously), the VCU sees the broken loop, and the HV bus drops before you can touch a live terminal.

Warning: HVIL vs HV Isolation — not the same thing:

Warning: HVIL is a 12V circuit that detects connector disconnection and triggers a contactor open.

Warning: HV Isolation is the design property that HV+ and HV– are electrically floating from chassis ground. A ground fault (HV touching chassis) is an isolation failure — HVIL cannot detect it.

Warning: See Module F04 for the full explanation of both concepts.

How It Works

The HVIL circuit is a simple series loop:

VCU HVIL out → Connector 1 loop → Connector 2 loop → Battery disconnect → VCU HVIL in

Each HV connector in the system has two small signal pins (the HVIL pins) that are internally shorted together within the connector. When all connectors are fully mated, the loop is complete and current flows. When any connector is pulled, the loop opens, current stops, and the VCU opens the contactors.

The VCU monitors the loop return signal. A break in the loop triggers an immediate contactor open regardless of opmode.

The BMW Safety Box (S-BOX)

A commonly used solution in ZombieVerter BMW builds is the BMW Safety Box. The S-BOX is an OEM battery isolation module from early E-series BMW/Mini hybrid vehicles that combines:

- Negative and positive contactors with precharge
- HVIL monitoring

- Pack voltage monitoring (CAN)
- Manual service disconnect

For BMW-based conversions it provides all of these functions in a single tested unit. Set `shuntType = SBOX` in VCU parameters.

The S-BOX communicates over CAN2 by default. Set `shuntCan = CAN2`. The S-BOX provides `udc` to the VCU — this is what drives precharge completion detection.

Wiring HVIL From Scratch

For builds not using the S-BOX, HVIL can be implemented with:

HV connectors with HVIL pins: Use HV connectors that have dedicated interlock pins (e.g. Delphi/Aptiv HV connectors used on Leaf, many OEM components). These have two small signal pins alongside the power pins. Wire all HVIL pins in series.

Manual service disconnect: A physical plug or lever that disconnects both the HV circuit AND the HVIL loop when pulled. The HVIL break triggers a contactor open, then the HV disconnect is physical.

VCU integration: The HVIL signal connects to a VCU digital input pin. Assign the HVIL function to this pin in the Din section. Configure as active-low (pin pulled to ground through the loop, open = fault) or active-high depending on your implementation.

S-BOX Integration

VCU Wiring

S-BOX terminal	Connect to
12V+	Permanent 12V (same feed as VCU)
GND	Chassis ground
CAN H	VCU CAN2H
CAN L	VCU CAN2L
Contactor coil– (neg)	VCU Pin 31
Contactor coil– (pos)	VCU Pin 33
Precharge coil–	VCU Pin 34
HV+	Battery positive (via fuse)
HV–	Battery negative

VCU Parameters

`shuntType = SBOX`

`shuntCan = CAN2`

The VCU will send CAN commands to the S-BOX to open and close its internal contactors as part of the normal precharge sequence.

Connectivity Check

With S-BOX powered and CAN connected:

- Check udc in Spot Values — should show a non-zero value if the S-BOX is communicating and the HV battery is connected
 - Check status — should not show a CAN communication error
-

HVIL on OEM Drive Unit Connectors

Many OEM inverters and drive units have built-in HVIL connectors:

Nissan Leaf: The large HV connector on the Leaf PDM/inverter has an integral interlock. In OEM use this connects to the vehicle HVIL loop. In a conversion it should be either looped (jumped) or integrated into your HVIL circuit.

Tesla drive units: Have an HVIL connector that must be looped or connected to the vehicle HVIL circuit. An unlooped HVIL connector prevents the inverter from entering run mode.

GS450h: The HV connectors at the brass pillar connection points do not have integral HVIL pins — the interlock is handled at the battery/contact box level.

Without HVIL

Builds without HVIL are possible — many simple conversions omit it. Without HVIL:

- Disconnecting any HV connector while the system is live is potentially fatal
- The VCU has no automatic protection against accidental HV exposure during service
- You rely entirely on the manual service disconnect and your own procedural discipline

If omitting HVIL, the discharge procedure (Module F04) must be followed rigorously every single time before touching any HV component. There are no shortcuts.

Source: Good Enuff Garage BMW Safety Box series (April-May 2024) · openinverter.org/wiki

Last verified against firmware: V2.30A (August 2025)

Cooling System Control

Track: Wiring

Prerequisites: H01 — VCU Hardware Walkthrough, C02 — Web Interface Walkthrough

Audience: Intermediate

Estimated reading time: 10 minutes

Three Pumps — Three Different Jobs

The VCU DOUT section lists several pump-related output functions in the same place. Before wiring anything, understand that these control **completely different systems** with **different purposes**:

VCU Function	What it controls	Purpose
CoolantPump	Electric water/glycol pump	Cools the inverter
GS450Hpump	GS450h transmission oil pump	Enables traction — not cooling
BrakeVacPump	Electric vacuum pump	Brake servo assistance

Warning: ⚠ **GS450h builders:** The GS450h oil pump **pressurises the transmission** to engage its internal clutch plates so the gearset can transmit torque to the wheels. It is **not a cooling pump** and has nothing to do with inverter cooling. The GS450h inverter uses a separate water/glycol coolant loop — that is what CoolantPump controls. Without oil pressure the transmission cannot transmit torque, regardless of inverter state. See Module I03-X for full oil pump wiring and commissioning.

These three functions are independent. A build may need all three, some, or just one.

Why Cooling Matters More in a Conversion

In an OEM EV, the cooling system is engineered alongside the drivetrain with precise flow rates, thermal mass, and thermostat control tuned for that specific motor and inverter. In a conversion, you are building a cooling system from scratch around salvaged components that were designed with different assumptions.

Under-cooling in a conversion manifests as:

- Thermal derating — the inverter reduces available torque as temperature rises
- Thermal shutdown — the inverter or motor stops completely to protect itself
- Accelerated component degradation over time

Over-cooling wastes energy running pumps and fans at full speed continuously. The VCU can manage both.

What the VCU Can Control

The VCU has dedicated output functions for cooling components, assignable via the Dout section of the web interface:

Function	What it controls	How it activates
CoolantPump	Electric coolant pump relay	On during pre-charge, charge and run mode
GS450Hpump	GS450h transmission oil pump PWM	PWM speed control via PumpPWM parameter
CoolingFan	Cooling fan relay or speed controller	Activates when temp exceeds FanTemp
BrakeVacPump	Brake vacuum pump relay	Activates when brake vacuum is low

Coolant Pump

An electric coolant pump must run whenever the inverter is active — from precharge through to run mode shutdown. Assign CoolantPump to a low-side digital output pin.

Circuit: 12V+ → relay coil → relay coil– → VCU Dout pin (low-side). When the VCU activates the output, the relay closes and supplies 12V to the pump motor.

Pump selection: Match the pump to the inverter's cooling system requirements. Most donor inverters (Leaf, Tesla) are designed for water/glycol coolant with flow rates in the 2–8 L/min range. An automotive heater core pump (e.g. Davies Craig EWP series) is commonly used in conversions.

GS450h oil pump: Uses PWM speed control, not a simple relay. See Module I03-X for full oil pump controller wiring. PumpPWM parameter (added V2.20A) controls duty cycle. Starting point: 30%.

Cooling Fan — FanTemp (V2.20A+)

FanTemp sets the inverter heatsink temperature (temphs) at which the VCU activates the cooling fan output.

FanTemp = 40 → fan on when heatsink reaches 40°C

Set this based on your cooling system design. A well-designed cooling loop should keep the inverter heatsink below 60°C under sustained load. If temphs regularly reaches 80°C+, the cooling system needs more capacity.

Monitoring: Watch temphs and tempm in Spot Values during sustained driving. These are your primary thermal health indicators.

GS450h Oil Pump — Traction, Not Cooling

Warning: This is not a coolant pump. It is documented here because it appears in the same Dout section of the web interface alongside cooling outputs.

The GS450h transmission uses hydraulic oil pressure to engage its internal clutch plates. Without oil pressure, the transmission cannot transmit torque — the wheels will not turn regardless of inverter state. The pump must run from precharge throughout run mode.

- Current draw: ~25A at full speed — **must** have its own dedicated 12V feed with a 30A minimum fuse
 - Do not power from a VCU output pin
 - Start at 30% PWM duty cycle
 - From V2.20A, the PumpPWM output function is freely assignable (not limited to GS450h builds)
- See **Module I03-X** for full GS450h oil pump wiring, specifications, and commissioning procedure.

Brake Vacuum Pump

Many non-hybrid donor vehicles use engine vacuum for brake servo assistance. In an EV conversion with no engine, this vacuum disappears. An electric vacuum pump is required for brake servo operation.

The VCU can monitor a vacuum sensor and activate a vacuum pump as needed:

Assign BrakeVacSensor to an analogue input pin

Assign BrakeVacPump to a digital output pin

The VCU activates the pump when vacuum drops below threshold

Alternatively, use a standalone electric vacuum pump controller — these are available as off-the-shelf units that self-regulate without VCU integration.

Note: Fix note — V2.05A: Before V2.05A, the brake vacuum pump output stayed active after ignition off. This was corrected in V2.05A (March 2024). If your pump runs continuously after key-off, update firmware.

Temperature Monitoring

The VCU receives temperature data from several sources:

Spot value	Source
temphs	Inverter heatsink temperature (from inverter CAN)
tempm	Motor temperature (from inverter CAN, where supported)
tempcool	Coolant temperature (if thermistor connected to VCU analogue input)

For vehicle integration, temphs and tempm can be used to drive the OEM temperature gauge (where the vehicle class supports it) — giving the driver a real indication of drivetrain thermal state.

Thermal Derating

Most supported inverters implement their own thermal derating — as heatsink temperature rises above a threshold, the inverter reduces available torque. This is done inside the inverter firmware, not by the VCU. The VCU sees reduced torque output but cannot override inverter-internal derating.

Signs of thermal derating:

- Throttle at 100% but torque spot value below maximum

- temps above 60–70°C
- Performance gradually reduces after sustained high-load driving

Solution: more cooling. Larger radiator, higher flow pump, or additional cooling surface.

Common Cooling Mistakes

Not running the pump during precharge: The inverter capacitors charge and the inverter initialises during precharge. Thermal load begins immediately. Assign CoolantPump to activate from the start of precharge, not just when the run mode is reached.

Pump on ignition-switched power: If the cooling pump is powered from ignition-switched supply and the VCU is on permanent power, the pump stops when the key is off but the inverter may still be thermally hot. Wire the pump to the VCU-controlled relay output so the VCU can run the pump after shutdown if needed.

No fan control in summer: A cooling loop without fan control may be adequate in winter ambient temperatures and completely inadequate in summer. Design for worst-case ambient temperature.

Source: Damien Maguire @Evbmw — V2.20A firmware walkthrough (December 2024) · openinverter.org community builds

PumpPWM and FanTemp added V2.20A · Vacuum pump fix: V2.05A (March 2024) Last verified against firmware: V2.30A (August 2025)

12V Auxiliary and Ignition Wiring

Track: Wiring

Prerequisites: H01 — VCU Hardware Walkthrough

Audience: All levels

Video sources: "EV Conversion DIY Wire Harness" series (Apr-May 2024, Good Enough Garage); "ZombieVerter VCU Forward & Reverse" (Apr 2024) **Estimated reading time:** 12 minutes

Overview

The ZombieVerter's 12V wiring is divided into three logical groups:

Permanent power — always on, powers the VCU itself

Ignition ON — switched when the key is turned to "on" position

Start — momentary signal when the key is turned to "start" position

Understanding why these are separate — and keeping them separate — is important for correct contactor sequencing and reliable operation.

Permanent Power

The VCU must remain powered at all times, even when the vehicle is "off." This is required for:

- Charge timers
- Charging session management
- Monitoring systems during parking
- ISA shunt continuous UDC reporting

VCU Pin	Function
Pin 56	12V+ permanent positive
Pin 55	12V– ground

Fusing: Fuse the positive lead. A 1–3A fuse is appropriate for the VCU alone at ~150mA idle current. If you are powering other devices from the same feed, size accordingly.

Power supply options for bench testing:

- Bench power supply (recommended — set to 13V, limit to 1–2A)
- Car battery (add a fuse)
- 9V battery (works for basic power-up verification only — not for any functional testing)

WARNING: Do not use an ignition-switched supply for permanent power. The VCU and ISA shunt must stay on when the key is off. See Module W03 for why this matters for precharge.

Fuse Block Strategy

For a clean installation, use separate fuse blocks for:

Fuse block	Powers	Notes
Permanent power block	VCU (Pin 56), ISA shunt, any always-on devices	Direct from battery through main fuse
Ignition ON block	Devices that should power on with key	Fed from VCU-controlled relay (Pin 32 output)
Start position block	Start signal only	Momentary feed

Using separate fuse blocks makes troubleshooting far easier — you can isolate each logical group.

Ignition ON Signal

When the driver turns the key to the "on" position (not start), Pin 15 must see 12V.

VCU Pin	Function	Signal type
Pin 15	Ignition ON	12V switched — stays on while key is in ON position

This is not a momentary signal. It stays high for the entire time the vehicle is in use.

Simple bench test: Connect a toggle switch between 12V+ and Pin 15. Toggle = key to ON position.

Start Signal

When the driver turns the key to the "start" position (momentary crank position), Pin 52 must see a brief 12V pulse.

VCU Pin	Function	Signal type
Pin 52	Start	Momentary 12V — brief pulse, like turning a key to start

The start signal triggers the precharge sequence. It does not need to stay high — a pulse of at least ~500ms is sufficient. The VCU latches into precharge mode once triggered.

Simple bench test: Connect a momentary pushbutton between 12V+ and Pin 52. Press = key to start.

Common mistake: Holding the start signal on too long or wiring it as a sustained signal rather than momentary. This can cause issues with re-triggering precharge after a fault. Wire it like an ignition start — momentary.

The Inverter/Charger Enable Relay (Pin 32)

The VCU controls an external relay through Pin 32 (low-side switching) to provide a managed "ignition" or "enable" signal to the inverter, charger, and DC-DC converter. These devices should not receive their enable signal until the VCU is ready for them.

How it works:

- Pin 32 is a low-side output — it pulls the relay coil ground to ground when activated
- The relay switches 12V to the enable pins of the inverter, charger, and/or DC-DC

- The VCU controls when this relay fires based on operating state

Wiring:

12V+ ————— [Relay coil +]
[Relay coil -] — Pin 32 (VCU) — GND

12V+ ————— [Relay switch COM]
[Relay switch NO] — Inverter enable / ignition input
└─ Charger enable input
└─ DC-DC enable input

Warning: This relay is sometimes called the "inverter relay" or "ignition relay" in forum discussions. It's the same thing — a VCU-controlled switch that powers up the drivetrain components when the VCU is ready.

Forward and Reverse Signals

VCU Pin	Function	Signal type
Pin 54	Forward	12V when forward direction selected
Pin 53	Reverse	12V when reverse direction selected

Direction Mode Options

Configure via `dirmode` parameter (under Throttle in the web interface):

dirmode setting	How it works
defaultFwd	No wiring needed for forward. Default direction is forward when in run mode. Add a button on Pin 53 to select reverse.
button	Use a momentary button. No pin active = default direction. Pin active = alternate direction.
switch	Use a 3-position switch. Pin 54 high = forward, Pin 53 high = reverse, neither high = neutral.
switchReversed	Same as switch but motor direction inverted. Use if drivetrain spins backwards.
buttonReversed	Same as button but inverted.

Behaviour in Each Mode

defaultFwd: The simplest setup. As soon as the VCU enters run mode, direction is automatically Drive. Add a single wire from a momentary button to Pin 53 for reverse. When the button is released, direction returns to Drive.

switch (recommended for most builds): Wire a 3-position switch:

- Position 1 → 12V to Pin 54 → Forward/Drive
- Position 2 → Centre (neither pin active) → Neutral

- Position 3 → 12V to Pin 53 → Reverse

The direction shown in spot values will track the switch position. Note: the switch wiring can be counterintuitive — the "Forward" label on the switch often corresponds to the opposite blade terminal. Check your specific switch's wiring diagram.

What happens when you turn off ignition with switch in Forward: Direction goes to Neutral when opmode goes to Off. When you next start the vehicle with the switch in Forward, it will immediately go to Drive — correct behaviour.

Torque Direction

Forward sends positive torque commands to the inverter. Reverse sends negative torque commands — the motor spins in the opposite direction. There is no physical reverse gear in most EV transaxles (Prius, Outlander, Leaf) — reverse is achieved by commanding negative torque.

Exception: The Lexus GS450h transmission does have a reverse gear and uses a physical gear selector with linkage. This is handled differently — see Module I03.

Swapping forward/reverse direction: If your motor spins the wrong way for a given direction signal (common when reorienting a drive unit from its OEM position), use `switchReversed` or `buttonReversed` in `dirmode` to invert the direction without rewiring.

Brake Signal

VCU Pin	Function	Signal type
Pin 49	Brake	12V when brake pedal pressed

The brake signal is a safety interlock. `dinbreak` must read "off" (brake not pressed) for the VCU to allow torque output.

Wire to the normally-open (NO) terminal of the brake light switch, so Pin 49 sees 12V only when the brake pedal is depressed.

If Pin 49 is left unwired (floating): It may pick up electrical noise and read as "on" intermittently or constantly, blocking torque. Add a 10kΩ pull-down resistor from Pin 49 to ground if you have noise issues.

The "Holy Trinity" — Minimum for Run Mode

From bench testing experience, the minimum wires required to reach run mode are:

Pin 56 — permanent 12V+

Pin 55 — permanent ground

Pin 15 — ignition ON (12V switched)

Pin 52 — start (momentary 12V)

Pin 54 or 53 — forward or reverse direction

Plus correct parameters (especially `udcsw`, `shuntType`, and `throtmax`).

What's Next

- **W07** — CAN Bus Wiring
- **W03** — Contactor & Precharge Circuit
- **C03** — Essential Parameters: First Start

Source: "EV Conversion DIY Wire Harness" series, "ZombieVerter VCU Forward & Reverse" — Good Enough Garage (Apr-May 2024)

Last verified against firmware: V2.30A (August 2025)

CAN Bus Wiring

Track: Wiring

Prerequisites: H01 — VCU Hardware Walkthrough, W03 — Contactor & Precharge

Audience: All levels

Estimated reading time: 10 minutes

Overview

The ZombieVerter communicates with the inverter, shunt, BMS, charger, and other devices over CAN bus. Getting the physical wiring, termination, and bus assignment parameters right is essential — a misconfigured CAN bus is one of the most common sources of "nothing works" at first startup.

The Three CAN Buses

The VCU has three CAN interfaces:

Bus	Physical interface	Speed	Notes
CAN1	STM32 native CAN1	500 kbps — hardcoded	Primary bus. CanSDO (web interface parameter access) is hardcoded here.
CAN2	STM32 native CAN2	500 kbps — hardcoded	Secondary bus. Remapped pins on the VCU PCB.
CAN3	MCP25625 SPI-CAN chip	Configurable via CAN3Speed	Used for CHAdeMO (fault-tolerant CAN).

Both CAN1 and CAN2 are hardcoded at 500 kbps in firmware — there is no parameter to change this. All devices on CAN1 or CAN2 must run at 500 kbps.

CAN3 Speed Options

CAN3Speed parameter values:

Value	Speed	Use case
0	33.3 kbps	CHAdeMO fault-tolerant CAN
1	500 kbps	Standard high-speed CAN
2	100 kbps	—

CAN1 vs CAN2 — Are They Interchangeable?

For most devices, yes. Both run at 500 kbps and carry the same callback handler. However there is one hardware asymmetry:

Warning: **CanSDO is hardcoded to CAN1**. The SDO communication used by the openinverter protocol (CAN ID 0x601) listens on both buses, but SDO replies are sent on CAN1 only. This only matters if you are using SDO-based tools directly — it does not affect normal VCU operation.

In practice, use whichever bus your wiring makes convenient. The conventional layout — inverter on CAN1, shunt/BMS on CAN2 — simply reduces the chance of CAN ID conflicts between devices and is a good habit, not a hardware requirement.

Baud Rates by Device

All devices used with CAN1 or CAN2 must run at **500 kbps**.

Device	Baud rate	Notes
Nissan Leaf inverter	500 kbps	
Mitsubishi Outlander	500 kbps	
ISA IVT-S shunt	500 kbps ¹	
SimpBMS / Orion BMS	500 kbps	
BMW i3 LIM	500 kbps	
CHAdemo socket	33.3 kbps fault-tolerant	CAN3 only — 6 PCB solder jumpers required
OBD2 / ELM327	500 kbps	

¹ **Applies to IVT-S units flashed with the ZombieVerter-specific firmware.** Stock IVT-S units from ISA default to 1 Mbps and will not communicate with the VCU without reflashing. The ZV firmware flash is available from the openinverter.org community. A correctly flashed unit can share CAN1 or CAN2 with other 500 kbps devices. See **W08** for full IVT-S reference.

Bus Assignment Parameters

Each device class has a parameter to select which physical bus it is on:

Parameter	Assigns	Default
InverterCan	Traction inverter	CAN1
VehicleCan	Vehicle CAN (instrument cluster)	CAN2
ShuntCan	ISA shunt / S-BOX / VAG box	CAN1
LimCan	i3 LIM charge interface	CAN1
ChargerCan	OBC / charger	CAN2
BMSCan	Cell BMS	CAN2
OBD2Can	OBD2 interface	CAN1
CanMapCan	Custom CAN map messages	CAN1

Parameter	Assigns	Default
DCDCCan	DC-DC converter	CAN2
HeaterCan	CAN-controlled heater	CAN2

All are configurable — the defaults shown are firmware defaults from `paramprj.h` and may not match your wiring. Set each parameter to match where you physically wired the device.

Typical Bus Layout

A common starting configuration:

CAN1 @ 500 kbps:

Traction inverter (InverterCan = CAN1)

Leaf PDM charger (ChargerCan = CAN1 – same bus as Leaf inverter)

BMW i3 LIM (LimCan = CAN1)

OBD2 dongle (OBD2Can = CAN1)

CAN2 @ 500 kbps:

ISA IVT-S shunt (ShuntCan = CAN2) – ZV flash, 500 kbps

SimpBMS / Orion (BMSCan = CAN2)

ESP32 CAN bridge – shares bus, 500 kbps transceiver

CAN3 @ 33.3 kbps fault-tolerant:

CHAdEMO socket – requires 6 PCB solder jumpers

This is a recommendation, not a requirement. Any device can go on either CAN1 or CAN2 as long as all devices on the same bus run at 500 kbps and there are no CAN ID conflicts between them.

Physical Wiring Rules

Use Twisted Pair

All CAN runs must use twisted pair wire. CAN H and CAN L must be twisted together. Untwisted runs in electrically noisy environments (near HV cables, near inverters) cause intermittent communication failures that are very difficult to diagnose.

Use shielded twisted pair for any run longer than ~500mm or in close proximity to HV wiring.

Bus Topology — Not Star

CAN is a **bus** topology. All devices connect to the same two wires (CAN H and CAN L) running from one end of the bus to the other. Avoid star topologies where wires branch out from a central point.

CORRECT – bus topology:

[VCU] ————— [Inverter]

| |

[Shunt] [BMS]

WRONG – star topology:

[VCU]

/ | \ \

/ | \ \

[Inv] [Shunt] [BMS] [Charger]

Keep stub branches (drops from the main bus to a device) as short as possible — under 100mm ideally.

Termination

CAN bus requires **120Ω termination at each end of the bus** — two resistors total, one at each physical endpoint. No more, no less.

Measure across CAN H and CAN L with all devices powered off. You should read ~**60Ω** (two 120Ω resistors in parallel).

- Less than 60Ω: too many terminators — find and remove the extras
- More than 60Ω (or open circuit): missing a terminator, or terminator not in place

Some inverters and OBCs have built-in 120Ω termination. Check before adding external resistors.

Assigning Devices to Buses — Step by Step

Physically wire all CAN devices — CAN H to CAN H, CAN L to CAN L, twisted pair throughout

Place 120Ω terminators at the two physical ends of each bus

In the VCU web interface, set each Can parameter to match where you wired that device

Save to flash

Reboot the VCU

Verify communication by checking relevant spot values (invstat, udc, BMS values)

Warning: Known issue: CAN1 and CAN2 are labelled in reverse in some older wiring diagrams. If a device won't communicate despite correct physical wiring, try swapping the Can parameter between 0 (CAN1) and 1 (CAN2) before rewiring anything.

Diagnosing CAN Communication Failures

Work through in order:

Is the correct bus assignment parameter set? Check InverterCan, ShuntCan, etc.

Is baud rate consistent? All devices on CAN1/CAN2 must be 500 kbps.

Is termination correct? Measure ~60Ω across H/L with power off.

Try swapping the Can parameter between 0 (CAN1) and 1 (CAN2) — CAN1/CAN2 label swap issue.

Are CAN H and CAN L wires the right way around? Try swapping them.

Is the cable twisted pair? Long untwisted runs in noisy environments cause intermittent failures.

Does the device need initialisation? (ISA shunt requires IsaInit — see W08)

Related Modules

- **W08** — ISA IVT-S Shunt: Complete Reference (CAN IDs, init procedure, spot values)
- **C02** — Web Interface Walkthrough (where to find the Can parameters)
- **C03** — Essential Parameters: First Start
- **A03** — BMS Integration

Source: *ZombieVerter firmware* `stm32\_vcu.cpp`, `param\_prj.h` · openinverter.org/wiki

Last verified against firmware: V2.40A

ISA IVT-S Shunt: Complete Reference

Track: Wiring · **Module:** W08

□ Current — verified against ZombieVerter firmware V2.40A source *Source files reviewed:* isashunt.cpp, isashunt.h, stm32vcu.cpp, paramprj.h, hwinit.cpp

Overview

The ISA IVT-S is a combined current/voltage/temperature sensor that communicates over CAN. It is the primary source of pack voltage (udc) and current (idc) for the ZombieVerter VCU. Without it — or an equivalent shunt type — the VCU cannot perform precharge sequencing, SOC calculation, or current limiting.

This module documents the shunt's CAN protocol, initialisation procedure, firmware integration, and spot value mapping, derived directly from the ZombieVerter firmware source code.

CAN Bus — Critical Fact

Warning: The ZombieVerter-specific IVT-S flash runs at 500 kbps.

Both CAN1 and CAN2 on the ZombieVerter are hardcoded at 500 kbps in firmware:

```
Stm32Can c (CAN1, CanHardware::Baud500);
Stm32Can c2(CAN2, CanHardware::Baud500, true);
```

The IVT-S with ZV firmware can share CAN1 or CAN2 with other 500 kbps devices — inverters, BMS modules, ESP32 interfaces, etc. **No dedicated bus is required.** Use whichever bus your wiring makes convenient; set ShuntCan to match.

Warning: ⚠ **Stock (unflashed) IVT-S units default to 1 Mbps** and will not communicate with the ZombieVerter. The unit must be flashed with the ZombieVerter-specific firmware before use. This flash is available from the openinverter.org community.

ShuntType Parameter

Set in the web UI under General Setup:

ShuntType value	Meaning
0 — None	No shunt. SOC calculations disabled.
1 — ISA	IVT-S shunt. Standard mode.
2 — SBOX	BMW S-Box pack monitor
3 — VAG	VW/Audi e-Box
4 — ISA_udcsw	IVT-S shunt, alternative precharge logic variant

For the IVT-S, set ShuntType = ISA (value 1).

CAN Message IDs

The IVT-S broadcasts 8 messages continuously once running. The VCU registers all 8 in `ISA::RegisterCanMessages()`:

CAN ID	Measurement	Units (raw)	Spot value
0x521	Current (Amperes)	mA × 1 (int32)	idc
0x522	Voltage U1	mV × 1 (int32)	udc
0x523	Voltage U2	mV × 1 (int32)	udc2
0x524	Voltage U3	mV × 1 (int32)	udc3
0x525	Temperature	°C × 10 (int32)	tmpaux
0x526	Power	W (int32)	power
0x527	Amp-hours	Ah (int32)	AMPh
0x528	Kilowatt-hours	kWh (int32)	KWh

All values are big-endian int32, bytes 2–5 of the 8-byte frame (bytes 0–1 are a status/type header).

The VCU sends commands to the shunt on 0x411.

U1, U2, U3 Voltage Inputs

The IVT-S has three independent voltage sense inputs:

- **U1** — Primary voltage sense. This is the input the VCU uses for udc (pack voltage for precharge logic). **Must be connected to the inverter/load side of the main contactor.**
- **U2** — Secondary voltage sense. Optional. Can monitor battery-side voltage. Reported as udc2.
- **U3** — Third voltage sense. Optional. Reported as udc3. $\Delta V (= U1 - U2)$ is also computed and shown in Spot Values.

Warning: ⚠ **U1 placement is safety-critical.** U1 must sense voltage on the load side of the main contactor — after precharge has charged the bus capacitors. If U1 is connected to the battery side, the VCU sees full pack voltage before precharge begins and will close the main contactor immediately into uncharged capacitors.

Warning: ⚠ **ΔV is a useful diagnostic.** During a healthy precharge sequence, ΔV (U1 – U2, with U2 on battery side) starts at pack voltage and drops toward zero as the bus caps charge. A precharge that stalls partway shows as a stable non-zero ΔV .

12V Power Supply

The IVT-S requires a **permanent 12V supply** — the same feed as the VCU itself.

- The VCU and shunt must power up together on the same permanent 12V rail
- This is required for the initialisation sequence to complete correctly
- If the shunt powers up after the VCU, init timing issues can occur (see Troubleshooting)
- **Never connect to an ignition-switched feed** — the shunt must remain powered to maintain coulomb counting during charging when ignition may be off

Initialisation — How It Works in Firmware

From stm32vcu.cpp main():

```
if (Param::GetInt(Param::IsaInit) == 1)
```

```
ISA::initialize(shunt_can);
```

IsaInit is checked **once at boot** — not polled continuously. Setting IsaInit = 1 and saving to flash means the next power cycle will call ISA::initialize().

ISA::initialize() performs this sequence:

Sends **STOP** to the shunt — halts current measurement

Sends 9 channel configuration messages

Sends **STORE** after each — writes config to shunt flash

Sends **START** — resumes measurement

Hardware delays (~110ms) are inserted between each step via a busy-wait loop on the STM32.

Step-by-step procedure

Wire IVT-S: CAN H/L to VCU CAN1 or CAN2, permanent 12V+, GND

Set ShuntType = ISA, ShuntCan to match your wiring — save to flash

Set IsaInit = 1 — save to flash

Power cycle VCU and shunt simultaneously (shared 12V feed means this happens automatically)

The shunt initialises — udc in Spot Values should now show a reading

Set IsaInit = 0 — save to flash

Reboot the VCU

Warning: ⚠ IsaInit **is not automatically cleared by the firmware**. If you leave it at 1, the full init sequence — including coulomb counter reset — runs on every boot. Always set it back to 0 and save to flash after a successful init.

Additional Commands

These commands are available in the firmware but not exposed as web UI parameters:

Function	Command ID	byte[0]	Effect
STOP	0x411	0x34	Halts measurement
START	0x411	0x34 + byte[1]=0x01	Resumes measurement
STORE	0x411	0x32	Saves config to shunt flash
RESTART	0x411	0x3F	Zeros Ah and kWh counters — does not re-init
deFAULT	0x411	0x3D	Returns shunt to factory defaults — wipes ZV flash

WARNING: ⚠ **Do not send deFAULT to a ZV-flashed shunt**. It erases the ZV firmware configuration and reverts the shunt to 1 Mbps default baud rate, requiring reflashing.

Warning: **RESTART is useful for bench sessions.** It zeroes the coulomb counter (Ah, kWh) without requiring a full re-init. Use it at the start of each test session for a clean SOC baseline.

Spot Values — Complete Mapping

From `paramprj.h`:

Spot Value	Source	Description
udc	ISA Voltage (0x522 / U1)	Pack voltage — primary input for precharge logic
udc2	ISA Voltage2 (0x523 / U2)	Secondary voltage sense
udc3	ISA Voltage3 (0x524 / U3)	Tertiary voltage sense
deltaV	U1 – U2	Voltage difference — useful precharge diagnostic
idc	ISA Amperes (0x521)	Pack current (positive = discharge, negative = regen/charge)
power	ISA KW (0x526)	Instantaneous power in kW
AMPh	ISA Ah (0x527)	Cumulative amp-hours since last RESTART or init
KWh	ISA KWh (0x528)	Cumulative kWh since last RESTART or init
tmpaux	ISA Temperature (0x525)	Shunt temperature
SOC	Calculated from Ah + BattCap	State of charge %

SOC calculation runs every 100ms only when `ShuntType != 0`.

Precharge Integration

The VCU's precharge state machine uses `udc` directly. Precharge exits to RUN or CHARGE mode only when:

- `udc >= udcsw` (bus voltage has reached the switchover threshold), AND
- A minimum 1-second precharge time has elapsed, AND
- No throttle pressed, no voltage faults

If `udc < udcsw` after **5 seconds** (hardcoded `PRECHARGETIMEOUT` in firmware — not configurable), the VCU posts `ERRPRECHARGE` and enters `MODPCHFAIL`.

Troubleshooting

`udc` **reads zero after init**:

- Verify `ShuntType = ISA` and `ShuntCan` is correct — save to flash and reboot
- Check CAN H/L connections and 120Ω termination at both bus ends
- Confirm the shunt has been flashed with ZV firmware — stock units at 1 Mbps will not respond

- Re-run the init procedure

Init timing issue — shunt doesn't respond to init:

- Separate the 12V supplies temporarily — power the shunt first, wait 2–3 seconds, then power the VCU
- Fixes a race condition seen on some units where the shunt isn't ready when the VCU fires the init sequence at boot

udc reads full pack voltage during precharge — precharge completes instantly:

- U1 is connected to the battery side instead of the load side — move U1 to the inverter/load side of the main contactor

Ah/kWh not resetting between sessions:

- Coulomb counters reset on power loss to the shunt — this is expected behaviour if the shunt shares the permanent 12V feed
- To manually zero them mid-session, send a RESTART command (0x3F on 0x411)

IsaInit keeps re-running on every boot:

- IsaInit was not set back to 0 after initialisation — set to 0, save to flash

Related Modules

- **W03** — Contactor and precharge circuit (physical wiring)
- **W07** — CAN bus wiring (topology, termination, bus assignment parameters)
- **C03** — Essential parameters: first start (udcmin, udclim, udcsw)
- **A03** — BMS integration (shunt + cell BMS together)

Source: ZombieVerter firmware — ``isa_shunt.cpp`` (Jack Rickard / EVtv, adapted by Damien Maguire), ``stm32_vcu.cpp`` (Tom de Bree, Johannes Hübner, Damien Maguire), ``param_prj.h``

Last verified against firmware: V2.40A

Firmware Version History

Track: Firmware & Configuration

Prerequisites: None — read this before any other C-track module

Audience: All levels

Purpose: Version reference and staleness guide for the entire training series **Last updated against:** V2.30A (August 2025 — latest as of this writing)

Why This Module Exists

The ZombieVerter firmware has evolved significantly since first release. YouTube videos, forum posts, and even parts of this training series may reference older behaviour, renamed parameters, or features that didn't yet exist at the time of recording. This module is the **ground truth reference** for what changed when.

How to use it:

- Before following instructions from any video or forum post, check the video date against this changelog
- If a parameter name doesn't match what you see in your firmware, look here for renames
- If a feature described in a module isn't visible in your web interface, check the version it was introduced

Firmware Version Summary Table

Version	Release date	Era	Key theme
1.00.A	Jun 2022	V1	First official release
1.06A	Nov 2022	V1	Throttle mode bug fix, throtdead introduced
1.07A	Nov 2022	V1	Leaf inverter bug fix
1.10A	Nov 2022	V1	CAN3, Leaf CAN improvements
1.11A	Feb 2023	V1	DC current limit fix
2.00A	May 2023	V2 — major	IO matrix, virtual functions, module decoupling
2.01A	Sep 2023	V2	Outlander charger, VW EBox shunt, BMS module

Version	Release date	Era	Key theme
2.04A	Jan 2024	V2	IOMatrix analog inputs, gear shifter class, Tesla DCDC, OBD2
2.05A	Mar 2024	V2	Throttle UDC limit fix, LIN heater, Park gear, 250ms contactor delay
2.15A	Apr 2024	V2	Regen, Outlander rear inverter, GS450h auto shifting, brake light regen
2.17A	May 2024	V2	Outlander rear motor bug fix
2.20A	Dec 2024	V2	Foccci integration, PWM functions, E90 CAN, revRegen, Kangoo BMS
2.22A	Mar 2025	V2	IS300h fix, LIN fix, Leaf Gen3 PDM improvement
2.30A	Aug 2025	V2 — current	Throttle smoothing, derate simplification, Leaf charging fix

The V1 → V2 Boundary (May 2023) — The Most Important Break

V2.00A was a major architectural rewrite. If you see content from before mid-2023, treat it with caution. Key changes:

- **IO Matrix introduced** — configurable input/output pin assignments via web interface. Before V2, many pins had fixed functions. The Din/Dout parameter sections in the web UI are a V2 feature.
- **Module decoupling** — Leaf stack components (inverter, PDM, charger) can now be used independently. Before V2, Leaf stack required all components together.
- **Charge interface decoupling** — Any OBC can now work with any charge interface (LIM, CHAdeMO). Previously coupled.
- **Subaru vehicle module added**
- **IMPORTANT:** Upgrading from V1.xx to V2.xx requires a power cycle after update (different RTC settings)

Detailed Release Notes

2.30A — August 2025 (Current)

☐ **Current — this training series targets this version**

- Leaf reverse torque limiter removed
- Leaf charging fix for low current PP detection

- Throttle smoothing — potnom float fix
- Derate logic simplified — ramps removed
- Throttle smoothing — throttle ramping after derating added
- Tested in 3 vehicles before release: BMW E39 (GS450h/Tesla PCS/CHAdemo), BMW E46 (Leaf Gen1/Outlander charger/i3 LIM CCS/Ampera heater), BMW E31 (Tesla LDU/Tesla Gen2 charger/Tesla DCDC/VW coolant heater/CHAdemo)

⚠ **Upgrade note:** No specific upgrade warnings for this release.

2.22A — March 2025

- Lexus IS300h inverter fix
- LIN bus fix for VW coolant heater
- Nissan Leaf Gen3 PDM operation improvement

⚠ **Upgrade note:** When upgrading from earlier versions, you **must reselect your inverter type and save params** — inverter sequence numbering was updated. If running a Toyota inverter, **reboot the VCU after upgrade** or the inverter will not initialise.

2.20A — December 2024

New parameters introduced (relevant to training modules):

New parameter	Function	Module
revRegen	Toggle regen on/off in reverse	A01
PumpPWM	Output pin function now selectable (was GS450h only)	H01, H05
FanTemp	Inverter or charger temp threshold for cooling fan	W05
TachoPPR	Pulses per revolution for tachometer output	H05
TorqueDerate	Spot value — set any time potnom derates	C06

Other notable changes:

- Foccci one-click setup and ZombieVerter integration
- CpSpoof PWM output for CP signal spoofing
- GS450h oil pump PWM now a selectable output function (PumpPWM)
- Outlander charge CP spoofing no longer hard-coded
- E65 shifting now uses shifter class (no longer hard-coded)
- BMW E90 lock state over CAN, warning lights, vehicle speed
- Classic Vehicle option added — has tacho output, uses 12V inputs
- Speedo output option on GS450h oil pump pin
- Voltage and current limiting now actually tapering (was broken before)
- Kangoo BMS integration added
- Heater fix — checks for HeatReq assignment
- Outlander heater + heartbeat message

- GS450h/IS300h: option for blending MG1/MG2 until 50% potnom

⚠ **Upgrade note:** Must reselect inverter type and save params. Must reboot after upgrade. Toyota inverter users must reboot or inverter won't initialise.

2.17A — May 2024

- Fix Outlander rear motor bug (minor point release)
-

2.15A — April 2024

Regen introduced for the first time as a working feature.

Key additions:

- **Regenerative braking** — working implementation
- Outlander rear inverter support added
- Prius Gen3 temperature reading fix (was causing uneven throttle)
- Brake light output option triggered by regen
- **GS450h auto shifting** introduced
- JLR gear shifter support added
- No Vehicle CAN option added
- BMW CAN sleep and wake
- Throttle ramping fixed

❏ **VERIFY for regen content:** Any video or forum content about regen from before April 2024 refers to a non-working or experimental implementation. Regen only became reliable in V2.15A.

2.05A — March 2024

- Throttle UDC limit bug fixed (was broken)
- Brake vacuum pump control fix (now turns off when ignition off)
- 250ms delay between HV contactor activations (needed for parallel packs)
- LIN-controlled VW heater added — **hardware note: R50 must be changed to 1kΩ and C45 to 1nF on boards ordered before 14/03/24**
- Park gear added to gear selection
- GS300h battery discharge limit fixed

⚠ **Upgrade note:** Toyota drivetrain users must reset VCU after firmware load.

2.04A — January 2024

Major feature additions:

- IO Matrix extended to analog inputs — brake vacuum pump control via vacuum sensor, charge port PP detection
 - Gear shifter class added — supports 12V inputs and BMW F30 CAN-based shifter
 - Tesla Gen2 DCDC converter class added
 - OBD2 class for Torque Pro via CAN/ELM327
 - BMW E31 vehicle class added
 - OpenInverter CAN class updated
-

2.01A — September 2023

- Mitsubishi Outlander PHEV charger/DCDC support added
- VW PHEV EBox shunt option added
- BMS module merged (bench tested only at release)
- Heater bug fix

⚠ **Upgrade note:** If upgrading from V1.xx, power cycle after update.

2.00A — May 2023 — MAJOR VERSION

See "V1 → V2 Boundary" section above for full impact.

1.06A — November 2022 — Important V1 note

Throttle potmin/potmax behaviour clarified in release notes:

Warning: "Please check your potmin, pot2min, potmax, pot2max values: They are the ABSOLUTE minimum and maximum. Please set them in a way that the potval can never reach them."

This confirms the inside-the-range calibration method has been present since at least V1.06A.

New in V1.06A:

- throtdead parameter introduced (throttle deadzone)
- Dual-channel throttle limp mode: if channels deviate >10%, throttle limited to 50% and THROTTLE12DIFF error stored
- Regen disabled entirely (re-enabled properly in V2.15A)

Content Staleness Reference

Use this table when consuming training content from external sources (YouTube, forum posts):

Content date	Firmware era	Trust level	Notes
Before Jun 2022	Pre-release	❑ Outdated	Pre-release firmware, significant changes since
Jun 2022 – Apr 2023	V1.00–V1.11	❑ Outdated	V1 era — IO Matrix, regen, many features missing
May 2023 – Dec 2023	V2.00–V2.01	❑ Verify	V2 architecture but many features not yet present
Jan 2024 – Mar 2024	V2.04–V2.05	❑ Verify	Most wiring/config content valid, regen not working
Apr 2024 – Nov 2024	V2.15–V2.17	❑ Mostly current	Regen works, most features present, minor gaps
Dec 2024 onwards	V2.20+	❑ Current	Full feature set, matches this training series

Parameter Renames / Removals Between Versions

Parameter	Status	Notes
ISAMode	☐ Stable	Present since V1, still current
udcsw	☐ Stable	Present since V1, still current
throtdead	☐ Stable	Introduced V1.06A
revRegen	☐ Added V2.20A	Not present in older firmware
PumpPWM	☐ Added V2.20A	Was GS450h-only before this
FanTemp	☐ Added V2.20A	
regenRamp	☐ Check	Listed as "new but not used yet" in V1.06A release notes — verify current status
HVRequest	☐ Non-functioning	Listed in IO wiki as "NOT FUNCTIONING" — do not rely on
Outlander (inverter=6)	☐ Deprecated	Replaced by RearOutlander (inverter=8)

The "Inverter Sequence Numbering" Warning

This affects anyone upgrading from V2.05A or earlier to V2.15A or later, and from any version to V2.22A.

The numeric values of the inverter parameter enum changed between versions. If you upgraded without reselecting your inverter type, the VCU may be trying to control the wrong inverter protocol.

After any major firmware upgrade, always:

Go to Parameters → Comms

Reselect your inverter type from the dropdown

Save to flash

Reboot the VCU

Staying Current

Firmware releases are managed by **Tom DeBree**, who coordinates what gets merged into the main codebase and controls when official releases are tagged on GitHub. If you see activity on the repository or a new release appear, Tom is typically the person who made it happen.

- **GitHub releases:** github.com/damienmaguire/Stm32-vcu/releases — always check here before updating
- **Forum thread:** openinverter.org/forum — development thread has release announcements
- **Damien's YouTube:** New firmware releases are typically accompanied by a walkthrough video

Source: github.com/damienmaguire/Stm32-vcu/releases — full release history

This module should be updated with each new firmware release. Current latest firmware: V2.30A (August 2025)

Flashing Firmware

Track: Firmware & Configuration

Prerequisites: None (standalone reference module)

Audience: All levels

Estimated reading time: 10 minutes

Do You Need This Module?

Probably not. Boards purchased from the EVBMW webshop come pre-programmed and ready to use. You do not need to flash firmware before first use on a new board.

You need this module if:

- You have a blank/unprogrammed board
- The web interface shows `firmware: null`
- A firmware update failed partway through and the board is unresponsive
- You want to run a specific firmware version

Know Your Chip

Before doing anything with firmware, identify which microcontroller is on your board.

Chip	Markings	Notes
STM32F107	ST logo + STM32F107	Standard chip — use main firmware branch
GD32F107	GD logo + GD32F107	Pandemic-era substitute — requires separate firmware branch

The GD32F107 ("GD chip") was used on some boards produced during the 2021–2022 semiconductor shortage. A separate firmware branch (GDZombie) exists on GitHub but development was abandoned shortly after release. As of the current main firmware (V2.20A), the GD branch has substantially diverged and is not kept up to date. **If you have a GD board, you are limited to an older firmware version and will miss recent features.**

If you are unsure which chip you have, look at the main IC on the board under good lighting. The manufacturer logo and chip number are printed on the chip.

Normal Firmware Update (Web Interface)

For routine firmware updates on a working board, use the web interface. No external tools required.

Steps:

Download the latest firmware .bin file from the [GitHub releases page](https://github.com/damienmaguire/Stm32-vcu/releases)

Connect to the VCU via Wi-Fi

Navigate to 192.168.4.1 in your browser

Find the **UART Update** field in the interface

Select your .bin file

Upload and wait for the update to complete

WARNING: Known issue — recent units: Some newer VCU boards do not automatically assign an IP address to your device via DHCP. If 192.168.4.1 does not load, manually set your computer's Wi-Fi adapter to a static IP of 192.168.4.2 (subnet 255.255.255.0), then try again. Any address in the 192.168.4.x range other than .1 will work.

Important notes:

- The file must be named stm32vcu.bin — rename it if it has a version suffix
- Windows 10 has known reliability issues with the UART update method. If you are on Windows 10, consider using Ubuntu (WSL2 is fine) or switching to the ST-Link method if the UART update fails
- Do not power-cycle the board mid-update

Verifying success: After reboot, connect to the web interface and check the firmware version displayed at the top of the page. If it shows firmware: null, the update failed and you will need to use the ST-Link recovery method below.

Full Recovery via ST-Link (Bricked Board)

Use this method if:

- The web interface shows firmware: null
- The board is completely unresponsive (no Wi-Fi, acty LED not flashing)
- UART update failed and board will not boot

What You Need

- ST-Link V2 dongle (cheap clones from Amazon/AliExpress work, though results vary)
- STM32CubeProgrammer software (free, from ST Microelectronics)
- Bootloader .hex file (from [Johannes Huebner's bootloader repo](https://github.com/jsphuebner/tumanako-inverter-fw-bootloader/releases))
- Latest ZombieVerter firmware .hex file (from [Damien's GitHub releases](https://github.com/damienmaguire/Stm32-vcu/releases/))
- 12V power supply for the VCU

Warning: Use .hex files for ST-Link flashing (not .bin). The .hex format includes address information so STM32CubeProgrammer places the data correctly without you needing to manually specify flash addresses.

Procedure

Download STM32CubeProgrammer from the ST website and install it

Upgrade your ST-Link firmware using STM32CubeProgrammer — not strictly required but recommended

Disconnect the Wi-Fi module from the VCU board (gently pull the ESP8266 module from its socket) — this eliminates one potential source of interference during flashing

Connect ST-Link to the VCU SWD test points:

ST-Link pin → VCU test point

SWCLK → C

GND → G

SWDIO → D

Apply 12V power to the VCU (Pins 55 and 56) — the ST-Link alone does not power the board

Open STM32CubeProgrammer and connect to the device via the ST-Link interface — you should see the chip detected

Perform a full chip erase — this wipes all existing flash content including any corrupted firmware

Flash the bootloader .hex file first

Flash the firmware .hex file second

Disconnect the ST-Link

Reconnect the Wi-Fi module

Power cycle the VCU and verify the Wi-Fi access point appears and the web interface loads

If STM32CubeProgrammer Cannot Connect

- Check your SWD wiring — SWCLK and SWDIO are often swapped
- Try a different ST-Link clone — quality varies widely
- Ensure 12V is present on the VCU before attempting to connect
- Try with the Wi-Fi module disconnected if you haven't already

Serial Connection Parameters

If you use a serial terminal to communicate with the VCU (for diagnostics or the serial parameter interface):

Baud rate: 115200

Data bits: 8

Parity: None

Stop bits: 2 ← This is unusual! Most tools default to 1 stop bit.

Flow ctrl: None

The 2 stop bits is a common gotcha — if your serial terminal is not receiving coherent data, check the stop bit setting first.

Firmware Sources

What	Where
Latest stable firmware (.bin and .hex)	github.com/damienmaguire/Stm32-vcu/releases
Bootloader (.hex)	github.com/jsphuebner/tumanako-inverter-fw-bootloader/releases
GD32F107 firmware (older, unsupported)	github.com/damienmaguire/Stm32-vcu/tree/GD_Zombie
STM32CubeProgrammer	st.com/en/development-tools/stm32cubeprog.html

Always use a released version unless you have a specific reason to build from source. The GitHub releases page lists changelogs so you can see what changed between versions.

Building From Source (Advanced)

If you want to compile your own firmware from the repository:

```
git clone --recurse-submodules git@github.com:damienmaguire/Stm32-vcu.git
```

The `--recurse-submodules` flag is critical — it pulls in `libopeninv` and other dependencies. Without it, the build will fail with missing includes.

GD32F107 boards require specific code modifications — see the forum thread linked in the wiki before attempting to compile for a GD board.

What's Next

- **C02** — The Web Interface Walkthrough: navigating the UI once the board is running
 - **C03** — Essential Parameters: First Start
-

Source: openinverter.org/wiki/ZombieVerter_VCU | openinverter.org/wiki/Zombieverter_programing

Last verified against firmware: V2.30A (August 2025)

Web Interface Walkthrough

Track: Firmware & Configuration

Prerequisites: H01 — VCU Hardware Walkthrough

Audience: All levels

Estimated reading time: 10 minutes

Overview

All ZombieVerter configuration is done through a browser-based web interface served by the on-board ESP8266 Wi-Fi module. There is no desktop software to install, no USB cable required for normal operation, and no command line interaction needed for day-to-day use.

Connecting to the Web Interface

Minimum Requirements for First Power-Up

- A fully built ZombieVerter VCU
- Two wires connecting Pin 55 (ground) and Pin 56 (12V+) to a stable 12V supply
- A computer, tablet, or phone with Wi-Fi

Step-by-Step

Apply stable 12V power to Pins 55 and 56

The acty LED on the board should begin **flashing** — this confirms the MCU is running

On your device, scan for Wi-Fi networks and connect to the ZombieVerter access point:

- **SSID:** inverter or zomvcu (varies by firmware version)
- **Password:** inverter123

Open a browser and navigate to: 192.168.4.1

The openinverter web interface should load

If You Cannot Connect

Symptom: Wi-Fi network appears but browser cannot reach 192.168.4.1

Some recent boards use a new Wi-Fi module variant that does not automatically assign your device an IP address via DHCP. If this happens:

- Manually set your device's IP address to 192.168.4.2 (or any address in the 192.168.4.x range other than .4.1)
- Set the subnet mask to 255.255.255.0
- Set the gateway to 192.168.4.1
- Try the browser again

Symptom: acty LED is not flashing

The MCU is not running. Check:

- 12V is present on Pin 56

- Ground is present on Pin 55
- If the board previously had a failed firmware update, it may need ST-Link recovery (see Module C01)

Interface Layout

The web interface is divided into several sections. The exact layout varies slightly between firmware versions, but the core structure is consistent.

Top Bar

Displays:

- Firmware version (if this shows null, the firmware is corrupted — see Module C01)
- Current VCU status / opmode
- Live spot values for key parameters (UDC, motor RPM, etc.)

Parameters Tab

The main configuration area. Parameters are grouped into categories:

Category	What's in it
Motor	Throttle limits, ramp rates, rev limiter, direction mode
Charger	Charger type, charge current limits, charge timer settings
Comms	CAN bus assignments, inverter type, shunt type, ISA mode, CANID settings
Throttle	potmin, potmax, pot2min, pot2max, potmode
Din	Configurable digital input pin assignments
Dout	Configurable digital output pin assignments
Temps	Temperature setpoints for fan, pump, heater enable
Contactors	udcmin, udclim, udcsw, shunt type
BMS	BMS current limits, cell voltage limits

Spot Values Tab

Live read-only values from the VCU. Essential for diagnostics and calibration. Key spot values:

Spot Value	What it shows
pot	Raw ADC reading from throttle channel 1
pot2	Raw ADC reading from throttle channel 2
potnom	Normalised throttle position (0-100)
udc	HV bus voltage from shunt

Spot Value	What it shows
invudc	HV voltage reported by the inverter
idc	DC current from shunt
opmode	Current VCU operating mode
invstat	Inverter status
dinbrake	Brake input state (must be "off" for torque)
dinforward	Forward input state
dinreverse	Reverse input state
dinstart	Start input state
speed	Motor speed (RPM)
torque	Requested torque
temphs	Heatsink temperature
tempm	Motor temperature
PPVal	Proximity pilot ADC value (if used)

Logging / Graphing

The interface includes a real-time graphing capability. Select which spot values to plot, set the update interval, and watch live traces. Useful for:

- Verifying throttle linearity
- Watching precharge voltage rise
- Diagnosing temperature issues
- Tuning regen behaviour

Save / Load

Parameters can be saved to flash (persistent across power cycles) or exported/imported as JSON files. Always **save to flash** after changing parameters — changes held only in RAM are lost on power cycle.

Parameter Workflow

The typical workflow for any configuration change:

Make the change in the Parameters tab

Observe the effect in Spot Values (some changes take effect immediately)

When satisfied, click **Save to Flash**

Reboot if required (some parameters only take effect after reboot)

Warning: Always save to flash before testing with HV. A power interruption during testing will otherwise revert your configuration.

Configuring Input/Output Pin Functions

Digital input and output pin assignments are made through the Din and Dout parameter categories. This is where you tell the VCU what function each configurable pin serves.

Important: The web interface does not prevent you from making invalid assignments — for example, assigning an output function to an input-only pin. The VCU will accept the assignment but the hardware will not respond. Always verify that your assignment matches the physical pin type (see Module H01 for the pin type table).

Inverter Type Selection

One of the first and most important settings is the inverter parameter under **Comms**. This tells the VCU which motor control protocol to use:

Value	Inverter
0	None — no inverter connected
1	Nissan Leaf Gen1/2/3 (CAN)
2	Lexus GS450h (sync serial)
3	UserCAN — not currently used
4	OpenInverter board (CAN)
5	Toyota Prius Gen3 (sync serial)
6	Outlander — deprecated
7	Lexus GS300h (sync serial)
8	Mitsubishi Outlander rear (CAN)

Set this before attempting to communicate with your inverter.

Vehicle Integration Type

The vehicle parameter selects the CAN profile for dashboard/instrument cluster integration:

Value	Vehicle
0	BMW E46
1	BMW E6x+ (E60, E90, etc.)
2	Classic — digital IO only, no vehicle CAN
3	None
5	BMW E39
6	VAG
7	Subaru
8	BMW E31

If you are not integrating with an OEM instrument cluster, set this to None or Classic.

ESP32 CAN Bus Web Interface (Alternative)

The VCU also supports an ESP32-based CAN bus web interface as an alternative to the built-in ESP8266. If you use this:

- The Node ID is **hard-coded to 3**
 - Note that the FOCCCI (CCS charge controller) has a default Node ID of 22 — no conflict, but be aware if you are also using FOCCCI
-

What's Next

- **C03** — Essential Parameters: First Start
 - **C04** — Throttle Calibration
-

Source: openinverter.org/wiki/ZombieVerter_VCU

Last verified against firmware: V2.30A (August 2025)

Essential Parameters: First Start

Track: Firmware & Configuration

Prerequisites: C02 — Web Interface Walkthrough, W02, W03

Audience: All levels

Estimated reading time: 15 minutes

Overview

This module walks you through the minimum parameter set required to bring a ZombieVerter system to life for the first time. Follow these steps in order. Do not skip ahead to HV testing until the LV checks are complete.

Before You Start: LV Bench Test

You should be able to complete the entire first-start sequence with **12V only** — no HV battery connected. This lets you verify wiring, parameters, and sequencing logic before introducing any HV risk.

Required for LV bench test:

- ZombieVerter VCU with 12V power
- Wi-Fi connected device with web interface open
- Throttle pedal (or a potentiometer simulating one)
- Brake signal (a simple switch to 12V)
- Forward/Reverse signal (switches to 12V)
- Ignition ON and START signals (switches to 12V)
- Contactor coils connected (they will click — no HV needed to verify sequencing)
- ISA shunt connected and initialised (see Module W08)

Step 1: Set the Inverter Type

Parameter: Inverter (under General Setup)

Set this to match your actual inverter. If you set it wrong, the VCU will send CAN messages on the wrong IDs at the wrong baud rate and nothing will communicate.

Your inverter	Set Inverter to
Nissan Leaf (any gen)	1
Lexus GS450h	2
Toyota Prius Gen3	5
Mitsubishi Outlander (rear)	8
OpenInverter board	4

Your inverter	Set Inverter to
Nothing yet	0

Save to flash.

Step 2: Set the Shunt Type

Parameter: ShuntType (under General Setup)

Your UDC source	Set ShuntType to
ISA IVT-S shunt	ISA
BMW S-BOX	SBOX
VW/Audi system	VAG
Nissan Leaf inverter	LEAF
None	None

Save to flash.

ISA Shunt Initialisation Procedure

If you are using an ISA shunt it must be initialised before it will report UDC. The VCU cannot precharge without a valid UDC reading.

Warning: The ISA shunt and VCU must share the same permanent 12V feed — this is required for the init sequence to work and for accurate readings during charging when ignition is off.

Wire the ISA shunt to permanent 12V+ and the CAN bus

Set ShuntType = ISA and ShuntCan to the correct CAN bus — save to flash

Set IsaInit = 1 — save to flash

Power cycle the VCU and shunt **at the same time** (shared 12V feed means this is automatic)

The shunt will initialise — udc in Spot Values should now show a reading

Set IsaInit = 0 — save to flash

Reboot the VCU

Warning: ⚠ IsaInit is not cleared automatically by the firmware. If left at 1, the full init sequence (including coulomb counter reset) runs on every boot. Always set it back to 0 after a successful init.

Warning: If the shunt doesn't initialise: Separate the shunt and VCU power supplies temporarily, power the shunt first, then power the VCU 2-3 seconds later. This fixes a timing issue seen with some units.

For the complete IVT-S reference including CAN IDs, spot value mapping, and troubleshooting, see **W08**.

Step 3: Set the CAN Bus Assignments

Parameters: InverterCan, ShuntCan (under General Setup)

The ZombieVerter has three CAN interfaces. You need to tell the VCU which physical bus each device is connected to.

Note: There is a known documentation issue where CAN1 and CAN2 are labelled in reverse in some older wiring diagrams. If your inverter is not communicating despite correct CAN wiring, try swapping the bus assignment parameter value before rewiring.

Common starting configuration:

- InverterCan = CAN1 (or whichever bus your inverter is on)
- ShuntCan = match your shunt wiring (CAN1 or CAN2)

Save to flash.

Step 4: Set Voltage Limits

Parameters: udcmin, udclim, udcsw

Parameter	Description	Firmware default	What to set
udcmin	Below this, VCU will not enter run mode	450V	~80% of minimum pack voltage
udclim	Above this, VCU derates	520V	Just above fully charged pack voltage
udcsw	Precharge complete — main contactor closes	330V	~95% of nominal pack voltage

Warning: ⚠ udcmin **defaults to 450V** and udclim **to 520V**. Both must be lowered for any build using a sub-400V pack or the VCU will never enter run mode.

For initial LV bench testing with no HV connected, set udcsw to a low value (15–20V) so precharge "completes" quickly and lets you verify the contactor sequencing.

Warning: **Precharge timeout is 5 seconds** — hardcoded in firmware. If udc has not reached udcsw within 5 seconds, the VCU posts ERRPRECHARGE and enters MODPCHFAIL. Ensure your precharge resistor is sized to charge the bus capacitors within this window.

Save to flash.

Step 5: Configure Throttle (Minimum Settings)

Parameters: potmin, potmax, potmode, throtmin, throtmax

First, set potmode:

- SingleChannel — one throttle pot
- DualChannel — two throttle pots (recommended for safety)

Then watch the pot spot value while moving the throttle pedal and set:

- potmin = value just **above** the resting (off-throttle) ADC reading
- potmax = value just **below** the full-travel ADC reading

Warning: ⚠ **ZombieVerter** potmin/potmax **are set inside the mechanical travel range** — opposite to most systems. This is intentional: if the sensor fails or the connector pulls out, the VCU sees a value outside the calibrated range and cuts throttle safely.

Then set:

- throtmin = 0 (or a small negative value for throttle-lift regen)
- throtmax = 100

Warning: ⚠ throtmax **defaults to 100**. If your throttle does nothing, check this parameter first — it is the most common first-start gotcha.

If using dual throttle, also calibrate pot2min and pot2max the same way while watching pot2.

Save to flash.

Step 6: Verify Brake Signal

In Spot Values, check dinbrake.

- It should show **"off"** when the brake pedal is not pressed
- It should show **"on"** when the brake pedal is pressed

Warning: dinbrake must be **"off"** for the VCU to allow torque output. If it is stuck "on" with no brake pressed, check the brake switch wiring (normally open vs normally closed), check for a floating pin, or add a pull-down resistor to ground.

Step 7: Verify Direction Inputs

Apply 12V to Pin 54 (forward) and check dinforward in Spot Values — should show "on". Apply 12V to Pin 53 (reverse) and check dinreverse — should show "on".

Step 8: First Precharge Test (LV Only)

With everything configured and saved to flash:

Apply ignition ON signal to Pin 15

Briefly apply START signal to Pin 52 (momentary 12V pulse)

Watch Spot Values:

- Contactor outputs should fire in sequence (you'll hear them click)
- udc should update
- If udc rises above udcsw, the main contactor closes
- opmode should reach run

If the system stays in PRECHARGE:

- Is udc reading a valid value? If zero, the shunt is not communicating — check W08
- Is udcsw reachable? For LV bench testing with no HV, lower udcsw to 5V
- Remember: precharge has a 5-second timeout — if udc doesn't reach udcsw in time, the VCU faults to MODPCHFAIL

If opmode never reaches "run":

- Check dinbrake is "off"
- Check dinforward or dinreverse is "on" — VCU requires a direction selected
- Check udc is above udcmin

Step 9: First Torque Test (LV Only)

With opmode showing run:

Apply forward direction (Pin 54 to 12V)

Slowly apply throttle

Watch potnom — should rise from 0 to 100 as you press the pedal

Watch torque — should rise proportionally

Without HV or a connected inverter, the VCU sends torque commands via CAN. Verify with a CAN analyser if available.

Complete First-Start Parameter Checklist

Parameter	Firmware default	What to set	Notes
Inverter	0	Match your hardware	General Setup
ShuntType	0 — None	Match your hardware	ISA, SBOX, VAG, or LEAF
InverterCan	CAN1	Match your wiring	
ShuntCan	CAN1	Match your wiring	
udcmin	450V	~80% of min pack voltage	⚠ Must lower for all sub-400V builds
udclim	520V	Just above fully charged pack voltage	
udcsw	330V	~95% of nominal pack voltage	Set 15-20V for LV bench test
potmode	SingleChannel	Match wiring	
potmin	0	Just above off-throttle ADC reading	Inside mechanical travel
potmax	4095	Just below WOT ADC reading	Inside mechanical travel
pot2min	4095	Same for channel 2 (dual only)	
pot2max	4095	Same for channel 2 (dual only)	
throtmin	-100	0 (or small negative for lift regen)	
throtmax	100	100	Default is correct — verify if throttle unresponsive
throtmaxRev	30	Raise if more reverse torque needed	Firmware defaults to 30% in reverse
revlim	6000 rpm	Set appropriate for your motor	

Parameter	Firmware default	What to set	Notes
IsaInit	0	Toggle 1→0 to init ISA shunt	See W08 — must reset to 0 after init
Vehicle	0 (BMW E46)	None or Classic	Unless integrating with OEM cluster

What's Next

- **C04** — Throttle Calibration: fine-tuning potmin/potmax, ramp rates, dual-pot plausibility
- **C05** — CAN Configuration: inverter CAN profiles, baud rates, message IDs
- **C06** — Fault Codes & Status Flags: reading and interpreting faults
- **W08** — ISA IVT-S Shunt: Complete Reference

Source: openinverter.org/wiki · ZombieVerter firmware `param\_prj.h`, `stm32\_vcu.cpp`

Last verified against firmware: V2.40A

Throttle Calibration

Track: Firmware & Configuration

Prerequisites: W02 — Throttle & Brake Wiring, C02 — Web Interface Walkthrough

Audience: All levels

Video source: "ZombieVerter VCU Throttle Calibration" (May 2024, Good Enuff Garage)

Estimated reading time: 15 minutes

Throttle Pedal Types

Before calibrating, understand what kind of throttle pedal you have.

Hall Effect vs. Potentiometer

Modern drive-by-wire throttle pedals from roughly 2005 onwards use **Hall effect sensors** — magnetic, no physical contact between moving parts. Older pedals used **potentiometers** (physical wiper contact). The ZombieVerter parameter names still use "pot" as a historical holdover, but the calibration procedure is identical regardless of sensor type. Hall effect pedals are generally more reliable and longer-lasting.

Single vs. Dual Channel

All Hall effect pedals contain **two independent sensors** (two channels) for redundancy. The ZombieVerter supports both:

- **Dual channel (recommended):** Both sensors are wired and monitored. If one channel fails or reads an implausible value, the VCU can detect the discrepancy and cut torque safely.
- **Single channel:** Only one sensor wired. No redundancy — if the single channel fails high, the result is unintended full throttle. Use dual channel if at all possible.

Mounting Styles

Throttle pedals from donor vehicles come in different mounting configurations:

- **Floor-mounted / pedestal mount** — Common in European cars (BMW, Mercedes). The pedal mounts to the floor on a bracket.
- **Firewall-mounted** — Common in American and Japanese cars. Mounts directly to the firewall, hanging position.
- **Bracket-mounted** — Some pedals (like certain Chevy/GMC units) use a separate bracket that can be cut, heated, and welded for custom fitment. Useful for classic car conversions.

Popular Donor Pedals

Pedal	Notes
BMW E46 (floor mount)	All plastic, comes with separate bracket — buy the bracket too
Honda Fit	Metal construction — can be cut and welded. Popular choice.

Pedal	Notes
Toyota Prius (Gen 3)	Hall effect, commonly available, pinout on openinverter wiki
Chevy Avalanche / Envoy	Metal body, firewall mount
Generic junkyard	Use Pick-n-Pull type yards; 50% off sales around holidays

Warning: **Tip:** Find a pedal with a known pinout diagram *before* buying. The openinverter wiki has pinout diagrams for BMW and Prius pedals. For others, search "[year] [model] throttle pedal pinout" or check alldata.com (subscription).

Wiring Six Wires to Four Pins

Every dual-channel Hall effect throttle pedal has **six wires** — three per channel:

- Ground (×2)
- 5V supply (×2)
- Signal/wiper (×2)

These six wires connect to **four VCU pins** by combining the shared grounds and shared 5V supplies:

Pedal wire → VCU pin

Channel 1 ground —

Channel 2 ground — Pin 45 (throttle ground)

Channel 1 5V —

Channel 2 5V — Pin 48 (throttle 5V supply)

Channel 1 signal — Pin 47 (throttle channel 1)

Channel 2 signal — Pin 46 (throttle channel 2)

Warning: **Important:** The throttle grounds and 5V supply connect to the VCU pins, not to vehicle chassis ground or the 12V battery. The VCU manages power to the throttle internally. Connecting throttle ground to chassis can cause instability.

BMW Pedal Color Guide (Version 1 — brown/yellow/white with green stripe variants)

Wire	Function
Brown	Channel 1 ground
Brown/green stripe	Channel 2 ground
Yellow	Channel 1 5V (or signal — verify with pinout)
Yellow/green stripe	Channel 2 5V (or signal)
White	Channel 1 signal
White/green stripe	Channel 2 signal

Always verify with the official pinout on the openinverter wiki before crimping.

Calibration Procedure

Step 1: Verify Wiring

With the VCU powered and connected to the web interface:

Navigate to **Spot Values**

Find pot and pot2

Move the throttle pedal slowly from rest to full travel

Both pot and pot2 should **change** as you move the pedal

If pot or pot2 does not change, the relevant channel is not wired correctly. Check your wiring before proceeding.

Step 2: Read the Resting Values

With the pedal **completely released** (foot off):

Read pot — note this value (e.g. 435)

Read pot2 — note this value (e.g. 221)

Step 3: Read the Full-Travel Values

Press and **hold** the pedal at full travel:

Click Refresh in spot values

Read pot at full travel (e.g. 2410)

Read pot2 at full travel (e.g. 1211)

Step 4: Set potmin and potmax

Navigate to **Parameters → Throttle**.

The ZombieVerter sets potmin/potmax **inside** the mechanical travel range (opposite to most inverters):

Parameter	Calculation	Example
potmin	Resting value + 10	$435 + 10 = 445$
potmax	Full-travel value – 10	$2410 - 10 = 2400$
pot2min	Channel 2 resting + 10	$221 + 10 = 231$
pot2max	Channel 2 full-travel – 10	$1211 - 10 = 1201$

Warning: **Why ±10?** This creates a small buffer inside the mechanical limits. The resulting potnom range (0–100) still covers 100% of usable throttle — you lose only a tiny fraction of ADC range in exchange for stability at the extremes.

Press **Enter** after each value, then **Save Parameters to Flash**.

Step 5: Set potmode

Your wiring	Set potmode to
Both channels wired	dual
One channel only	single

Step 6: Verify potnom

Go back to Spot Values. Watch potnom:

- At rest: should read **0** (or close to 0)
- At full travel: should read **100** (or close to 100)
- Movement should be smooth and proportional

If potnom reads a small non-zero value at rest, your potmin is set slightly too low — increase it by 5–10 counts.

Step 7: Set throtmin and throtmax

This is the most commonly missed step.

Parameter	Description	Typical value
throtmin	Minimum allowed potnom (can be negative for lift-off regen)	0 (or small negative for regen)
throtmax	Maximum allowed potnom in forward	100

WARNING: throtmax defaults to 0. A VCU with throtmax = 0 will never produce forward torque regardless of throttle position. If your throttle appears to do nothing, check this parameter first.

Additional Throttle Parameters

Parameter	Default	Description
throtramp	10 %/10ms	Rate of potnom increase per 10ms interval. Limits how fast torque rises when pedal is pressed quickly. Higher = more responsive, lower = smoother.
throtramprpm	20000 rpm	Motor RPM above which throtramp no longer applies. Default effectively disables the ramp limit — lower this if you only want smooth low-speed response.
revlim	6000 rpm	Motor RPM hard rev limiter. Set appropriate for your motor — 6000 rpm is conservative for most builds.
throtmaxRev	30%	Maximum torque allowed in reverse direction. Firmware default is deliberately conservative — raise if your reverse feels sluggish.
throtrpmfilt	15 rpm/10ms	Filters rapid speed changes fed into the throttle map. If you have low-speed judder, increase this value.

Parameter	Default	Description
throtdead	10%	Deadband around zero potnom before torque begins.

Using the Spot Values Visibility Filter

The web interface spot values list is long. For throttle calibration, you only need to watch a few values. Use the **Visible** column to hide everything except:

- pot
- pot2
- potnom

Check the Visible box for those three, set the rest to non-visible, then use Auto Refresh. This keeps the display clean during calibration.

Dual-Channel Plausibility

With potmode = dual, the VCU continuously compares pot and pot2. If the two channels disagree by more than a configured threshold, the VCU will flag a throttle fault and cut torque.

This means **both channels must be calibrated correctly**. A common mistake is calibrating only channel 1 and leaving pot2min/pot2max at defaults — this will cause immediate plausibility faults when potmode is switched to dual.

Troubleshooting Throttle Issues

Symptom	Cause	Fix
pot or pot2 doesn't change with pedal	Wrong pin, broken wire, or open circuit	Re-check wiring to pins 47, 46
potnom stuck at non-zero at rest	potmin set too low	Increase potmin by 5-10
potnom never reaches 100	potmax set too high	Decrease potmax by 5-10
No torque despite valid potnom	throtmax = 0	Set throtmax to 100
Throttle fault in dual mode	pot2min/pot2max not calibrated	Calibrate both channels
Erratic potnom	Poor ground between pedal and Pin 45	Check ground connection

What's Next

- **A01** — Regen Tuning: configuring throtmin negative values for lift-off regen
- **C03** — Essential Parameters: First Start (if not already completed)

Source: "ZombieVerter VCU Throttle Calibration" — Good Enuff Garage (May 2024) | openinverter.org/wiki/ZombieVerter_VCU

Last verified against firmware: V2.30A (August 2025)

Fault Codes and Status Flags

Track: Firmware & Configuration

Prerequisites: C02 — Web Interface Walkthrough, C03 — Essential Parameters: First Start

Audience: All levels

Video source: "ZombieVerter VCU Won't RUN" (May 2024, Good Enuff Garage)

Estimated reading time: 12 minutes

Overview

When the ZombieVerter doesn't behave as expected, the web interface gives you two key feedback mechanisms:

- `opmode` — the current operating state of the VCU
- `status` / `lastError` — what the VCU last tried to do and why it stopped

Understanding these is the difference between spending five minutes diagnosing a problem and spending five hours replacing things that weren't broken.

Operating Modes (`opmode`)

`opmode` is visible in Spot Values and shows the current state of the VCU state machine.

opmode value	Meaning
off	VCU is powered but ignition is not on, or system is in standby
precharge	Start signal received — negative and precharge contactors closed, waiting for UDC to rise
pch / PCH	Same as precharge (abbreviated display)
run	Precharge complete, main contactor closed, system ready for torque commands
charge	System is in charge mode (HV contactors closed for charging)

Normal startup sequence:

`off` → `precharge` → `run`

If the system goes `off` → `precharge` → `off` or `off` → `precharge` → `pch` fail, something prevented precharge from completing.

The `lastError` and `status` Fields

These two spot values tell you what the VCU was trying to do and why it stopped. They are often more useful than `opmode` alone.

Note: **Note on naming:** The field called `lastError` is not always actually an error — it often records the last significant status change. Think of it as "last known status" rather than "last fault." Similarly, `status` reports the current blocking condition.

Common status Messages

status message	Meaning	Common cause
UDC below udcsw	Pack voltage hasn't reached the precharge completion threshold	udcsw set too high, shunt not communicating, wrong shuntType
Throttle 1 2	Throttle channel fault	pot2min/pot2max not calibrated, potmode mismatch, wiring issue
dinbreak on	Brake signal is active	Brake wired normally-closed instead of normally-open, floating input

The Most Common Failure: `udcsw`

This is the single most frequently missed parameter for first-time users.

What it does: `udcsw` is the voltage threshold that tells the VCU "precharge is complete — close the main contactor and enter run mode." It defaults to **330V**.

The problem: On a workbench with no HV battery connected, the VCU will never see 330V. Precharge will start, time out, and fail — every time.

The fix: Set `udcsw` appropriately for your situation:

Situation	Set <code>udcsw</code> to
Bench testing, no HV connected	0-5V (just to prove the sequencing works)
Bench testing with low-voltage HV source	Just below your source voltage
Real vehicle, known pack voltage	~95% of nominal pack voltage

WARNING: Why does it default to 330V? Setting it too low is dangerous — the main contactor could close before the inverter capacitors are adequately charged, potentially damaging or destroying the inverter. The 330V default is conservative and safe for common Nissan Leaf/Prius systems. It is intentionally set to require you to make a conscious decision about your pack voltage.

Procedure to diagnose and fix:

Go to Spot Values, enable Auto Refresh

Turn ignition ON → press START

Watch `opmode` — does it go to precharge? Good. Does it then fail?

Check `status` — if it says `UDC below udcsw`, this is your issue

Go to Parameters → Contactor Control → `udcsw`

Set to 0 for bench testing (no HV) or appropriate value for your pack

Save Parameters to Flash

Try again

Precharge Failure Diagnosis Tree

System won't reach RUN mode

|

▼

Does opmode reach "precharge"?

| |

NO YES

| |

▼ ▼

Check: Does it then fail?

- Ignition |

signal on — YES

Pin 15 |

- Start signal ▼

on Pin 52 Check status field:

|

|

| | |

"UDC below "din_break Other

udcs" on"

| | |

▼ ▼ ▼

Check Brake Check shunt,

udcs" signal CAN wiring,

setting wiring udcmin

Throttle Status Errors

After precharge succeeds and opmode reaches run, the next common status message involves the throttle:

Throttle 1 2 (or similar)

This means the VCU has a problem with one or both throttle channels. Common causes:

potmode **set to dual but only one channel wired** — Channel 2 reads a fixed value that disagrees with channel 1's movement

pot2min/pot2max **not calibrated** — Channel 2 plausibility check fails immediately

Wiring fault — One channel is open or shorted

Diagnosis:

Go to Spot Values

Watch pot and pot2 while moving the pedal

Both should change. If one stays fixed, that channel has a wiring problem.

If both change but the system still faults, check that pot2min/pot2max are calibrated (not at defaults)

dinbreak — The Brake Signal Interlock

dinbreak is a critical safety interlock. **The VCU will not produce torque while dinbreak shows "on"**, regardless of throttle position or operating mode.

This causes significant confusion because the system appears to be in run mode, the throttle appears to work (potnom changes), but the motor does nothing.

Check in Spot Values:

- dinbreak = **off** → normal, torque is allowed
- dinbreak = **on** → brake is active, torque is blocked

Causes of dinbreak being stuck "on"

Cause	Fix
Brake wired to constant 12V	Check brake switch wiring — should be normally open
Brake switch wired normally-closed (NC)	Use the normally-open (NO) terminal
Pin 49 floating with electrical noise	Add 10kΩ pull-down resistor from Pin 49 to ground
Brake pedal held down during testing	Release the brake

Parameter Abbreviations Reference

The ZombieVerter interface uses abbreviated parameter names that can be confusing. Common ones:

Abbreviation	Full meaning
udc	Battery/pack voltage (U = voltage in German/Latin, DC = direct current)
udcsw	UDC switch — voltage at which precharge is considered complete
udcmin	UDC minimum — below this voltage, VCU will not enter run mode
udclim	UDC limit — maximum pack voltage before derating
invudc	Inverter-reported DC bus voltage
opmode	Operating mode
invstat	Inverter status
pot	Throttle channel 1 raw ADC value (pot = potentiometer, legacy name)
pot2	Throttle channel 2 raw ADC value
potnom	Normalised throttle demand (0–100)
dinbreak	Digital input — brake signal

Abbreviation	Full meaning
dinforward	Digital input — forward direction signal
dinreverse	Digital input — reverse direction signal
dinstart	Digital input — start signal
temphs	Temperature — heat sink (inside inverter)
tempm	Temperature — motor
idc	DC current from shunt
pch	Precharge (abbreviated in status messages)

Reading the Interface Efficiently

Use Ctrl+F

The web interface is a single long page. Use browser find (Ctrl+F) to locate specific parameters or spot values quickly. Search for `udcs` to jump directly to that parameter.

Use Auto Refresh Strategically

Auto Refresh updates all spot values continuously. For diagnosis, turn it on and watch specific values while performing actions (turning ignition on, pressing start, moving throttle). For calibration, it's easier to turn it off and use the manual Refresh button so values don't jump while you're trying to read them.

Use the Visible Filter

Check the Visible box for only the spot values you're currently watching, hide the rest. This dramatically reduces clutter, especially useful when diagnosing precharge issues (watch just `opmode`, `status`, `udc`) or throttle issues (watch just `pot`, `pot2`, `potnom`, `dinbreak`).

What's Next

- **C07** — Data Logging & Live Telemetry
- **W03** — Contactor & Precharge Circuit (for deeper precharge theory)

Source: "ZombieVerter VCU Won't RUN" — Good Enuff Garage (May 2024) | openinverter.org/wiki/ZombieVerter_VCU

Last verified against firmware: V2.30A (August 2025)

Nissan Leaf Inverter

Track: Integration

Prerequisites: C03 — Essential Parameters: First Start, W03 — Contactor & Precharge

Audience: All levels — Leaf is the most common ZV drivetrain

Estimated reading time: 15 minutes

Overview

The Nissan Leaf inverter is the most common drivetrain in ZombieVerter-based conversions. It uses a CAN-based interface — straightforward wiring, well-documented, and supported across all three Leaf generations. The Leaf motor (EM57 or EM61) is a permanent magnet AC machine with excellent power density and broad speed range.

VCU parameter: inverter = 1

Leaf Generations

Generation	Pack	Motor	Inverter type	Notes
Gen1 (ZE0)	24kWh	EM57 (80kW)	Single inverter	2011-2017. Most common in conversions.
Gen2 (AZE0)	30/40kWh	EM57 (80-110kW)	Single inverter	2016-2022. Minor CAN differences.
Gen3 (ZE1)	40/62kWh	EM57/EM90	Revised inverter	2018+. Improved in V2.22A.

For conversion use, Gen1 and Gen2 are nearly identical from the VCU's perspective. Gen3 has some CAN differences that were addressed in V2.22A firmware.

Vehicle Connections

The Leaf inverter needs very few external connections:

Signal	Connection
Permanent 12V	12V+ to inverter (two gray wires — join together, add fuse)
Ground	Chassis ground to inverter (two gray wires — join together)
Ignition ON	12V when key is in ON position (orange wire from OEM harness)
CAN H	CAN bus high to VCU CAN1H

Signal	Connection
CAN L	CAN bus low to VCU CAN1L

That's five connections. The Leaf inverter is genuinely simple to wire.

Warning: Using the OEM harness: If you have the Leaf wire harness, extract the inverter signal plug branch. If not, you can connect directly to the inverter connector pins — Damien's 2019 "Grey Goose" video documents exactly which pins carry which signals, including colour codes.

Resolver Wiring

The Leaf motor position is sensed by a resolver mounted in the motor. The resolver connects from the motor to the inverter via a separate harness plug (typically 8-pin).

The resolver carries: sine pair, cosine pair, exciter pair, and motor temperature sensor. All must be connected correctly or the inverter will report a resolver fault and refuse to run.

If you have the motor harness: Plug it in as-is. The connector is keyed and can only go in one way.

If you don't have the harness: You need to identify the pins from the resolver winding resistance (sine/cosine = $\sim 50\Omega$, exciter = $\sim 100\Omega$) and wire accordingly. This is documented in the openinverter wiki and in Damien's 2019 Leaf inverter video.

VCU Parameters

```
inverter = 1 (Leaf)
```

```
inverterCan = CAN1 (or CAN2 – try both if no communication)
```

The Leaf PDM (charger/DC-DC) can be added on the same CAN bus:

```
chargerType = LeafPDM
```

```
chargerCan = CAN1
```

Connectivity Check

With the Leaf inverter powered (12V applied, CAN connected, inverter = 1):

- Check invstat in Spot Values — should show a non-zero value when the inverter is communicating
- Check temphs — should show a plausible heatsink temperature (25–40°C at room temperature)
- If both read zero: check CAN assignment, check baud rate (should be 500 kbps), try swapping CAN1/CAN2 assignment

Minimum Pack Voltage

The Leaf inverter requires a minimum pack voltage to operate. Below approximately 200V the inverter will not run. This is hardcoded in the inverter firmware and cannot be changed from the VCU.

For bench testing at low voltage: use `udcs` = 0 and set `udcmin` low to allow the VCU to attempt precharge. The inverter itself will refuse to produce torque below 200V — you will see the contactors sequence correctly but `invstat` will indicate the inverter is not ready.

The Leaf PDM — Charger and DC-DC

The Leaf Power Delivery Module (PDM) combines the AC charger and DC-DC converter. It uses the same CAN bus as the inverter and communicates using the same 500 kbps baud rate.

See Module A02 for full Leaf PDM charging setup.

Gen3 Specific Notes

The Gen3 ZE1 Leaf (2018+) has some differences in CAN message content that caused issues with early V2.x firmware. V2.22A (March 2025) addressed the Gen3 PDM specifically. If using Gen3 hardware, ensure firmware is V2.22A or later.

Known Issues and Gotchas

"Inverter not communicating" despite correct wiring: Try swapping inverterCan between CAN1 and CAN2. There is a known documentation issue where these labels are reversed in some older wiring diagrams.

Resolver error on first run: Usually caused by incorrect resolver wiring. The motor must not be spinning when the VCU first initialises the resolver. Power up with the motor stationary.

Inverter runs but no torque: Check `throtmax` — defaults to 0. Also check `dinbrake` is showing "off" and a direction (`din_forward` or `din_reverse`) is selected.

Motor makes noise but doesn't produce useful torque: Check that the HV voltage is above 200V. The inverter needs minimum voltage to produce full torque output.

Common Build Context

The Leaf inverter/motor combination is physically an FWD transverse unit — compact and suited to front-wheel drive conversions. For RWD applications it requires a custom adapter to convert from the integrated FWD drive axle output to a propeller shaft or transfer case.

The motor + inverter typically weigh around 60kg combined. The motor alone is approximately 35kg.

Source: Damien Maguire @Evbmw — "The Grey Goose — Leaf Inverter Wiring" (July 2019) · "Nissan Leaf Motor Running" (2018)

openinverter.org wiki — Nissan Leaf inverter page Last verified against firmware: V2.30A (August 2025)

Tesla Drive Units SDU and LDU

Track: Integration

Prerequisites: C03 — Essential Parameters: First Start, W03 — Contactor & Precharge

Audience: Intermediate

Estimated reading time: 15 minutes

Overview

Tesla drive units are increasingly popular in EV conversions — powerful, well-engineered, and becoming more available as early Model S vehicles reach end-of-life. The ZombieVerter controls Tesla drive units via an openinverter replacement logic board that installs in the inverter in place of the OEM Tesla logic board.

Warning: **Critical distinction:** The openinverter board replaces the LOGIC board inside the inverter — not the entire inverter. The power electronics, motor, and housing remain OEM. Only the small control board that sits inside the inverter housing is replaced.

Small Drive Unit (SDU) vs Large Drive Unit (LDU)

	SDU	LDU
Model	Model S rear (pre-2016), Model 3 rear	Model S/X rear (2013+)
Peak power	~220kW	~375kW
Weight	~100kg	~145kg
Layout	Compact — suits smaller builds	Larger — needs more space
Availability	Good	Good and increasing
OI board version	V2.1	V2.1

Both use the same openinverter board and the same ZombieVerter integration approach.

The OpenInverter Board

The openinverter replacement board is available from evbmw.com or can be built from the GitHub files. It provides:

- CAN interface to the VCU
- Motor position encoding
- Gate drive signals to the IGBT modules in the power stage
- Replacement of all OEM Tesla logic functionality needed for EV conversion use

Set inverter = OIBoard (4) in VCU parameters.

Pre-Installation Checks — Do These Before Soldering Anything

Damien's 2023 video documents a systematic 8-step process. Inverter failures in Tesla conversions almost always result from skipping these steps. Follow them in order.

Step 1 — DC Bus Diode Test

With the inverter completely unpowered and disconnected, set multimeter to diode test mode. Connect positive probe to HV+ terminal, negative probe to HV− terminal. You should see: voltage rising from near zero as the capacitors charge, then climbing to open circuit. This confirms no dead short on the DC bus.

Reverse the probes (negative to HV+, positive to HV−). You should see approximately 0.6V — the forward voltage of the body diodes. This is correct.

WARNING: If you see a direct short (0V) in either direction: the inverter has failed IGBT modules. Do not proceed.

Step 2 — Phase Diode Tests

With multimeter negative on HV− bus, positive probe on each of the three motor phase terminals (U, V, W) in turn. Each should show increasing voltage then open circuit (capacitors charging). This confirms the upper IGBT body diodes are intact.

Reverse: negative probe on HV+, positive on each phase. Should see ~0.6V on each. This confirms the lower IGBT body diodes are intact.

Any phase showing a direct short = failed IGBT on that phase. Do not proceed.

Step 3 — 12V Logic Board Test (OEM board still installed)

Apply 12V power to the logic board connector only — do not connect HV. The board should draw approximately 200–300mA and show some activity (LEDs, CAN frames on a bus monitor). If the board draws no current or draws excessive current, investigate before proceeding.

Step 4 — Solder the OpenInverter Board

Remove the OEM logic board. Solder the openinverter board connectors carefully — cold joints on the gate driver connections cause intermittent or failed gate switching. Inspect under magnification before proceeding.

Step 5 — 12V Test With OI Board

Apply 12V to the logic board connector with the OI board installed. The board should power up, CAN should be active, no excessive current draw.

Step 6 — Install Board Into Inverter

Reinstall the cover. The board is now installed but HV has not been applied yet.

Step 7 — 12V Test With OI Board Installed in Inverter

Apply 12V again — verify normal operation with the board in its installed position.

Step 8 — Encoder Test Before Any HV

With the VCU connected, CAN active, and 12V applied: rotate the motor shaft slowly by hand. Watch encoder or resolver readings in the VCU Spot Values. The values should change smoothly as the shaft rotates. If they don't — investigate the encoder wiring before applying HV.

Warning: Only after completing all 8 steps should you apply HV to the drive unit.

Why These Steps Matter

Damien's 2023 video was specifically created in response to a pattern of inverter failures in the community. The failures were caused by:

Applying HV before the DC bus was verified — a pre-existing fault that would have been caught by the diode test

Applying HV before the encoder was verified — the inverter attempts motor commutation immediately on HV application; if encoder data is wrong it commands incorrect phase currents and destroys the power stage

The OI board itself is not expensive. The IGBT modules in the inverter power stage are expensive and difficult to source. The 8-step process is 30 minutes of work that prevents a catastrophic and expensive failure.

HVIL on Tesla Drive Units

Tesla drive units have an HVIL connector that must be looped for the inverter to operate. The HVIL connector carries a low-current loop through the HV connector housing — if the HV cover is removed, the loop breaks.

In a conversion, the HVIL connector must either be looped (a simple wire jumper connecting both pins) or integrated into the vehicle's HVIL circuit. Leaving it unconnected will prevent the inverter from entering run mode.

Cooling Requirements

Tesla drive units are liquid cooled. In the OEM vehicle, they use Tesla's proprietary coolant loop with a pump, radiator, and chiller. In a conversion, you need an aftermarket electric coolant pump, a radiator or heat exchanger, and thermostat control.

The VCU can control a coolant pump via the `CoolantPump` output function and a fan via `CoolingFan` with the `FanTemp` threshold parameter (added V2.20A).

Minimum coolant flow should be maintained whenever the inverter is active. Unlike the GS450h which uses oil, the Tesla drive unit uses water/glycol coolant.

VCU Parameters

```
inverter = 4 (OpenInverter board)
```

```
inverterCan = CAN1
```

Community Resources

The Volvo V50 Tesla Model 3 drive unit project (2025) is the most recent documented Tesla LDU conversion with ZombieVerter. Three build videos are available on Damien's channel documenting the integration process with current firmware.

Source: Damien Maguire @Evmw — "Using Tesla Drive Units with OpenInverter Board" (July 2023) · "Volvo V50 Tesla Model 3 Drive Unit" series (2025)

Last verified against firmware: V2.30A (August 2025)

GS450h Transaxle

Track: Inverter Integration

Prerequisites: I00 — Generic Inverter Integration Principles, W07 — CAN Bus Wiring

Audience: Intermediate

Video source: "ZombieVerter VCU Prius Inverter Connection" series (May-Jun 2024); "2013 Lexus GS450h Transmission Wire Harness DIY EV Conversion" (Jun 2025, Good Enough Garage)

Estimated reading time: 20 minutes

Why the GS450h is Different

The Lexus GS450h (and GS300h) transaxle is one of the most popular choices for rear-wheel-drive EV conversions — particularly for classic American muscle cars, trucks, and anything with a traditional longitudinal drivetrain layout. It is, however, the most mechanically complex integration the ZombieVerter supports.

Key differences from Leaf/Outlander/Prius integrations:

Feature	Leaf / Outlander	GS450h
Drive unit	Transverse FWD	Longitudinal RWD
Motor count	1 traction motor	2 motors (MG1 + MG2)
Reverse gear	Software (negative torque)	Physical gear in transmission
Physical shifter	Not needed	Required (Park/Reverse/Neutral/Drive)
Inverter comms	CAN bus	Synchronous serial (8-wire)
Gearbox	None (single fixed ratio)	Two-speed (high $\approx$ 2:1, low $\approx$ 4:1)

The Two Motors: MG1 and MG2

The GS450h contains two motor-generators:

Motor	Role
MG1	Generator / starter — primarily used for engine starting and power split in the OEM hybrid system
MG2	Traction motor — the primary drive motor in an EV conversion

In a ZombieVerter conversion, MG2 is used for traction. MG1 may or may not be utilised depending on the conversion.

HV Connection — Critical Warning

WARNING: ⚠ DO NOT connect HV to the OEM positive/negative terminals on the GS450h inverter. Doing so routes current through the boost inductor and boost IGBT, which will destroy the inverter under sustained load. At least two confirmed inverter failures from this mistake are documented in the community.

HV must be connected to the **brass pillars** — custom M6 brass standoffs installed onto the internal busbars, accessed by removing the boost converter cover. The brass pillars are **not supplied** with the inverter; you must fabricate or purchase them.

See **I03-X — GS450h Wiring & Oil Pump Deep Dive** for the full step-by-step procedure: cover removal, busbar access, pillar installation, and HV cable connection.

Synchronous Serial Interface

Unlike the Leaf (CAN) and Outlander (CAN), the GS450h inverter uses a **synchronous serial bus** — an 8-wire interface specific to Toyota/Lexus systems. This is also used for the Toyota Prius Gen3 and the Lexus GS300h.

Why this matters:

- The VCU's CAN buses are not used for inverter communication with GS450h
- The sync serial wires connect to specific pins on the ZombieVerter's 56-pin connector
- The wiring is more complex than CAN (8 wires vs 2)

Sync Serial Pin Assignments (ZombieVerter)

The four synchronous serial pairs connect to the following VCU pins:

Signal pair	VCU Pin (–/negative)	VCU Pin (+/positive)	Notes
REQ	16	17	Request
CLK	18	19	Clock
MTH	20	21	Motor to hybrid (inverter → VCU)
HTM	22	23	Hybrid to motor (VCU → inverter)

Warning: Pin assignments may vary slightly between firmware versions and board revisions — always verify against the current wiki pinout diagram.

Inverter Connector

The GS450h inverter connector is a large multi-pin connector (approximately 36 pins) that combines:

- The synchronous serial interface (8 pins used)
- Power (12V supply — 2 wires redundant positive, 2 wires redundant ground)
- Resolver connections - the resolver signals for MG1 and MG2, used for motor position sensing. This is **OEM harness territory** — use the original Toyota/Lexus wire harness wherever possible. Building resolver cables from scratch is possible but requires shielded twisted pairs and careful attention to wiring.

Note: **Note:** This connector carries 12V logic power and signals only. HV connects separately via the brass pillars — see the HV Connection section above.

Power Wiring to the Inverter Connector

Wire colour (OEM)	Function	Notes
Green (×2)	12V positive supply to inverter	Two wires, combined — add an inline fuse
White/black (×2)	Ground	Two wires, combined

The redundant positive wires run from the connector back to a common point and rejoin — they are Toyota's built-in redundancy. In an EV conversion, join them together and run to a fused 12V supply.

Start with a 1A fuse and work upwards if it blows (2A, 3A, 5A). The inverter draws more than the VCU at startup.

Wire Harness — OEM vs. DIY

Use OEM Harnesses Where Possible

The synchronous serial wiring is particularly important to get right. Toyota's harness uses:

- Twisted shielded pairs for the sync serial signals
- Specific loop-back connections within the connector (wires that exit and re-enter the same connector)
- Shielding for EMI protection

Building these from scratch is achievable but error-prone. Buying the OEM harness and extracting only what you need is strongly recommended.

What to Extract from the GS450h Harness

The GS450h "engine wire" harness (also called the transmission wire harness) contains everything you need but also a lot you don't need. A typical harness extraction yields:

What you extract	Used for
MG1 resolver cable (4-pin or 10-pin connector)	Motor position sensing
MG2 resolver cable	Motor position sensing
Shift solenoid harness (SL1, SL2)	Transmission gear control (high/low)
Neutral safety switch harness	Park/Reverse/Neutral/Drive detection

You'll only use approximately 12% of the total harness — the rest is OEM engine/hybrid control wiring that can be discarded.

Harness Identification Tips

When buying a used GS450h harness:

- Look for the "squiggly" plastic guide that runs along the transmission side
- The resolver connectors are at one end — MG1 and MG2 each have their own plug

- The resolver plugs are at the **opposite end** from the main inverter connection — if a harness is cut at the wrong place, you lose the resolver plugs
- The neutral safety switch plug is gray with plastic shielding, on a branch that includes many ground wires
- Gen 3 (2007–2011) and Gen 4 (2013+) harnesses differ — Gen 3 is the supported conversion platform as of current firmware

Resolver Cable Identification

On a 4th gen harness:

- MG2 resolver: gray/green wires + temp sensor wires (red/blue)
- MG1 resolver: separate plug, typically 4 or 10 pins
- Temp sensor wires must be cut (they don't have through-plugs, only connect on one end)

Shift Solenoids

The GS450h has two speeds — high gear (~2:1 ratio) and low gear (~4:1 ratio). The shift solenoids SL1 and SL2 control which gear the transmission is in. In ZombieVerter, these are controlled via:

VCU pin assignment	Function
Trans SL1–	Transmission solenoid 1 ground-side control
Trans SL2–	Transmission solenoid 2 ground-side control

For basic operation, many builders leave the transmission in high gear and don't connect the shift solenoids. The zombie does have parameters for managing gear shifts, but this requires more advanced configuration.

Gear Shifter Switch

The gear shifter switch tells the VCU which gear position the physical shifter is in (Park, Reverse, Neutral, Drive). The switch physically activates the parking pawl when in PARK, all other positions are simply an electrical switch. This is wired to the ZombieVerter which reads the switch and responds accordingly — it will not send torque in Park or Neutral.

For bench testing without the full shifter assembly, specific pins on the Gear Shifter switch can be shorted to simulate a Drive signal and allow torque commands. Refer to the openinverter wiki for the specific pins.

VCU Parameter Settings

Parameter	Value for GS450h	Notes
inverter	2 (GS450H)	
vehicle	Match your donor vehicle (Classic, BMW E46, etc.)	
transmission	auto or per your shifter setup	
gearLever	Match your gear selector type	

Parameter	Value for GS450h	Notes
MotActive	0 (Mg1and2)	Controls which motor(s) receive torque commands. 0=both, 1=MG1 only, 2=MG2 only, 3=blend MG2→MG1 above 50% potnom. Default (0) drives both motors with the same percentage request.
OilPump	50 (default)	Oil pump PWM duty cycle at run setpoint — start at 50%, adjust based on operating temps

Oil Pump

The GS450h transmission requires hydraulic oil pressure to engage its internal clutch plates and enable traction. Without oil pressure the transmission cannot transmit torque — the inverter can be running perfectly and the wheels will not turn. An electric oil pump replaces the mechanical pump that the OEM engine would have driven.

The ZombieVerter controls this pump via:

VCU function	Type	Notes
GS450Hpump	PWM OUTPUT	Controls oil pump speed proportionally
OilPump	Parameter	PWM duty cycle — default 50%

Assign GS450Hpump to the Pump PWM pin in the configurable output assignments. The pump must run from precharge through to full de-energisation. Start at the default 50% duty cycle.

WARNING: ⚠ Wire the oil pump directly from the 12V bus with its own dedicated fuse (~25A) — do not power it from a VCU output pin.

Bench Testing the Sync Serial Connection

A basic connectivity check without HV:

Connect sync serial wires (8 wires, pins 16–23) from VCU to inverter connector

Connect 12V power to inverter connector (green wires + ground)

Apply permanent 12V to VCU (Pins 55, 56)

Connect to web interface

Go to Spot Values — check temphs (heat sink temperature)

If temphs shows a non-zero, plausible temperature value (e.g., 25–35°C at room temperature), the sync serial connection is working and the inverter is communicating. If it shows 0 or negative, the sync serial is not connected correctly.

Note: Note: The inverter will draw over 1A from the 12V supply at startup. If using a bench power supply, ensure it is set for at least 2A output.

Common GS450h Integration Issues

Issue	Cause	Fix
No temp reading from inverter	Sync serial not connected or wrong pins	Re-verify pin assignments against current wiki pinout
Ground loop / noise interference	Laptop USB connected to logic analyser during same test	Disconnect USB/logic analyser, check chassis grounds
System won't stay in run after restart	ISA shunt on ignition-switched power	Move shunt to permanent power supply
Resolver errors	Swapped MG1/MG2 connectors	Swap resolver plugs

What's Next

- **I03-X** — GS450h Wiring & Oil Pump Deep Dive (HV bypass procedure, brass pillar installation, sync serial detail)
- **I04** — Mitsubishi Outlander PHEV Drive Unit
- **A01** — Regen Tuning (GS450h has specific regen considerations)

Source: "ZombieVerter VCU Prius Inverter Connection" series — Good Enuff Garage (May 2024) | "2013 Lexus GS450h Transmission Wire Harness" — Good Enuff Garage (Jun 2025) | [openinverter.org](https://openinverter.org/wiki) wiki

Last verified against firmware: V2.30A (August 2025)

GS450h Wiring and Oil Pump Deep Dive

Track: Integration (Deep Dive Extension)

Prerequisites: I03 — GS450h Transaxle

Audience: Intermediate

Estimated reading time: 20 minutes

Overview

This module covers the three most critical details of a GS450h integration that are most often done wrong: the HV connection bypass, the sync serial wiring, and the oil pump controller setup.

HV Connection — The Critical Bypass

The GS450h inverter has two places you can connect HV. Only one is correct.

The OEM Terminals — DO NOT USE

Connecting HV to the OEM positive/negative terminals routes current through the boost inductor and boost IGBT. These components are rated for approximately 100A maximum. Everything works fine on the bench and at low speed. Under sustained load at highway speed, the inverter destroys itself. At least two confirmed inverter failures from this mistake are documented in the community.

The Brass Pillars — Correct Connection

The brass pillars are **custom spacers** — they are not supplied with the inverter. You will need to fabricate or purchase them. They are brass standoffs (typically M6, approx 10–15mm length) that allow your HV cables to reach the internal busbars of the inverter once the boost converter cover is removed.

How to access and connect:

Locate the black plastic cover on the inverter body

Gently prize off the cover — Toyota uses adhesive

Remove the two M6 bolts (10mm head) to expose the internal busbars

Install brass standoffs/pillars onto the busbar bolts to provide a connection point that extends out of the inverter body

Connect HV+ to one pillar, HV– to the other

WARNING: For bench testing: A 60V 5A power supply into the brass pillars is sufficient to verify sync serial communication and low-speed rotation. You do not need full pack voltage for initial bench testing.

Sync Serial Interface — 8 Wires, 4 Differential Pairs

Unlike the Leaf (CAN) and Outlander (CAN), the GS450h uses a synchronous serial bus.

Pin Assignments (ZombieVerter VCU)

Signal	Direction	VCU Pin (-)	VCU Pin (+)
REQ	VCU → Inverter	16	17
CLK	VCU → Inverter	18	19
MTH	Inverter → VCU	20	21
HTM	VCU → Inverter	22	23

Warning: **Always verify against the current wiki pinout.** Pin assignments may vary between board revisions and firmware versions.

Cable Requirements

- Use shielded twisted pair for each differential pair
- Toyota's OEM harness uses shielded twisted pairs — replicate this for reliable operation
- Building from scratch is achievable but error-prone; using the OEM harness is strongly recommended

Connectivity Check (No HV Required)

With sync serial correctly wired and 12V applied to both VCU and inverter:

- Check temps in Spot Values
- A reading of approximately 25-35°C at room temperature confirms the sync serial link is working
- A reading of 0 or negative means the sync serial is not communicating

12V Power to Inverter

From the OEM harness:

OEM wire colour	Function
Green (×2)	12V+ supply to inverter — join both, add fuse
White/black (×2)	Ground — join both

Start with a 1A fuse and work upwards if it blows (2A → 3A → 5A). The GS450h inverter draws more than the VCU at startup.

The 12V supply to the inverter is switched by the VCU via the inverter enable relay (Pin 32). The inverter should not power up until the VCU is ready.

Resolver Cable Extraction

The OEM "engine wire" harness contains the sync serial cables, resolver cables, and other signals. You use approximately 12% of the total harness.

What to Extract

Cable	Used for
MG1 resolver cable	Motor position sensing
MG2 resolver cable	Motor position sensing

Cable	Used for
Neutral safety switch harness (9-wire gray plug)	P/R/N/D detection
Shift solenoid harness (SL1, SL2)	Gear shift control (if used)

Key Tips

- **Blade technique:** Cut electrical tape and loom lengthwise with a sharp blade. A dull blade cuts wire insulation. Replace the blade before starting.
- **MG2 plug is at the rear of the transmission, MG1 is at the front.** Label them immediately on extraction — swapped resolver plugs cause resolver errors and the system won't run correctly.
- **Temperature sensor wires must be cut.** MG1 and MG2 temperature sensor wires (gray/green on MG2, red/blue on MG1) run inside the resolver cable bundle but terminate at the "harmonica" junction block and cannot be used. Cut them as far back as possible.
- **Gen 3 vs Gen 4:** Gen 3 (2007–2011) is the primary supported conversion platform. Gen 4 (2013+) harnesses may be physically compatible but test first.

Buying a Harness

Search eBay for "GS450h engine wire" or "GS450h transmission harness." Expect to pay \$125–200. Make lowball offers — sellers often accept. Always confirm from photos that the resolver plugs at the far end are not cut — this is the most commonly damaged part of a used harness.

Oil Pump System

The GS450h transmission uses hydraulic oil pressure to engage its internal clutch plates — this is what transfers torque from the motors to the output shaft. In an OEM vehicle this pressure is supplied by a mechanical pump driven by the engine. In an EV conversion there is no engine, so an electric oil pump is required.

Without oil pressure the transmission cannot transmit torque. The inverter and motors can be running perfectly, but if the oil pump is not running the wheels will not turn. The pump must run from precharge and throughout run mode.

Why It Matters

The oil pump is not about cooling the inverter — it is about enabling traction. No pump = no clutch engagement = no drive. Start the pump before commanding any torque.

Oil Pump Specifications

Parameter	Value
Current draw	~25A from 12V at full speed
Power source	Dedicated 12V — NOT from VCU output pin
Control method	External controller with PWM input from VCU
Starting point	30% PWM duty cycle

Oil Pump Controller Wiring

Wire colour	Function
Blue	12V+ supply to controller — 30A fuse minimum
Brown	Ground to controller casing
Black	PWM signal from VCU (GS450Hpump function)

The controller has two connectors: power (to the pump motor) and resolver (pump position feedback).

Warning: ⚠ **The ~25A draw cannot come from a VCU output pin.** These are low-current signal outputs. Wire the controller power directly from the 12V battery or auxiliary bus with its own dedicated fuse.

VCU Configuration

In Parameters → Dout, assign the GS450Hpump function to the Pump PWM output pin. Start at 30% duty cycle — this is sufficient for low-load bench operation. Increase for sustained high-load or elevated ambient temperatures.

Warning: From V2.20A, the PumpPWM output function is freely assignable — not limited to GS450h builds. This allows coolant pump PWM control on other drivetrains using the same pin assignment method.

Input Shaft Lock

Without locking the input shaft, MG1 can only spin the engine input shaft — it cannot send torque to the output. The lock forces MG1 torque through the power split device to the output shaft.

Without the lock, you lose approximately half your available torque and power. MG1 and MG2 combined with a properly locked shaft give a torque multiplication through the planetary gearset of approximately 2.6–2.7:1 for MG1's contribution.

The lock is a mechanical plate or bar with a spline fitting that bolts to the bellhousing flange and engages the engine input shaft splines. For bench testing only, Damien uses a vice grip and cable tie. For a real vehicle, a purpose-made billet lock plate is required.

Common Integration Issues

Issue	Cause	Fix
No temphs reading (0 or negative)	Sync serial not connected or wrong pins	Re-verify pin assignments against current wiki pinout
Ground loop / noise interference	Laptop USB connected to logic analyser during test	Disconnect USB/logic analyser, check chassis grounds
System won't stay in run after restart	ISA shunt on ignition-switched power	Move shunt to permanent power supply
Resolver errors	Swapped MG1/MG2 resolver connectors	Swap the resolver plugs
Inverter destroyed under load	HV connected to OEM terminals instead of brass pillars	Replace inverter, connect to brass pillars only

Source: Damien Maguire @Evbmw · Good Enuff Garage · openinverter.org/wiki

Video sources: "GS450h EV Drivetrain Deep Dive" (2020) · "GS450h and GS300h VCU Connection" (2022) · "GS450h Transmission Wire Harness" (2025) Last verified against firmware: V2.30A (August 2025)

Mitsubishi Outlander Drive Unit

Track: Integration

Prerequisites: C03 — Essential Parameters: First Start, W03 — Contactor & Precharge

Audience: Intermediate

Estimated reading time: 12 minutes

Overview

The Mitsubishi Outlander PHEV contains separate front and rear drive units — each with its own motor, inverter, and three-phase connection. Either or both can be used in a conversion. The interface is CAN-based, similar to the Nissan Leaf but with Mitsubishi-specific message IDs.

Front drive unit VCU parameter: inverter = (check current wiki — Outlander front) **Rear drive unit VCU parameter:** inverter = 8 (RearOutlander)

Note: Note on versions: Early firmware used inverter = 6 for Outlander, which has since been deprecated. Use inverter = 8 for the rear unit. Check the current wiki for the front unit value as this has changed across firmware versions.

Front vs Rear Drive Unit

The Outlander PHEV has distinct front and rear units:

	Front	Rear
Motor type	PMAC	PMAC
Peak power	~60kW	~60kW
Orientation	Transverse FWD	Longitudinal RWD
Interface	CAN	CAN
ZV support	Partial — check wiki	Full — V2.15A+
Common use	Front-wheel drive conversions	Rear-wheel drive conversions, longitudinal builds

The rear unit is the more commonly used in ZombieVerter conversions and has more complete firmware support.

Minimum Voltage

Warning: The Outlander inverter requires a minimum of approximately 200V to operate. Below this threshold the inverter will not produce torque. This is hardcoded in the inverter and cannot be changed from the VCU.

For bench testing: use a bench power supply at a low voltage to verify CAN communication. The inverter will communicate below 200V — it just won't spin. You need pack voltage to verify full

operation.

Wiring

The Outlander inverter needs:

Signal	Notes
12V+ permanent	To inverter logic power
Ground	Chassis ground
CAN H / CAN L	500 kbps, VCU CAN1 or CAN2
HV+ / HV−	Via contactors and precharge as normal
Motor three-phase	Three-phase AC to motor (integral in most units)
Resolver	Motor position sensor — OEM harness or equivalent

The motor and inverter are typically inseparable in the Outlander unit — the motor is integrated into the transaxle housing and the inverter bolts directly on. You take them together from the donor vehicle.

The Outlander OBC — Two 12V Feeds

The Outlander OBC (onboard charger) is a separate unit that provides both AC charging and DC-DC conversion. It is covered in Module A02.

The most common Outlander build failure: wiring only one of the two 12V power inputs to the OBC. Both must be connected. See A02 for full OBC wiring.

OEM Harness

For the Outlander drive unit, obtain as much of the OEM harness as possible. The resolver connector and cooling system connectors have non-standard pinouts that are significantly easier to work with if you have the original plugs.

Mitsubishi forum resources (as distinct from openinverter) can be valuable for Outlander-specific pinout information — Damien noted he sourced the initial CAN log data that enabled Outlander support from a Mitsubishi enthusiast forum.

Cooling

The Outlander drive unit requires active liquid cooling. The OEM system uses a dedicated inverter coolant circuit separate from the engine coolant. In a conversion:

- Electric coolant pump required (12V, controllable)
- Assign CoolantPump to a VCU digital output pin
- The pump should run whenever the inverter is active — assign to activate in precharge and run mode

Cabin Heater — CAN Variant Only (2016+)

The Outlander PHEV uses two different water heater designs across model years:

Variant	Years	Interface	ZombieVerter support
Resistive (non-CAN)	Pre-2016	Direct 12V/HV switching — no CAN	Not supported
CAN-controlled	2016+	CAN bus commands	Supported

Warning: ⚠ **The ZombieVerter can only control the CAN-bus water heater from 2016+ Outlander PHEVs.** Pre-2016 units use a resistive heater with no CAN interface and cannot be controlled by the VCU. If your donor vehicle is pre-2016, you will need an alternative cabin heating solution.

For supported (2016+) units, the heater communicates via CAN and requires a separate VCU parameter assignment. Refer to the openinverter wiki for the current CAN message IDs and parameter settings.

CAN Communication Verification

With 12V applied and CAN connected:

- Check invstat in Spot Values — should show non-zero when inverter is communicating
- Check tempms — should show a plausible temperature
- If no communication: verify inverter parameter is set correctly, check CAN bus assignment, verify 500 kbps baud rate, check twisted pair CAN wiring with 120Ω termination at both ends

Known Issues

Outlander front unit CAN parameter value changed between firmware versions. Always verify against the current openinverter wiki before setting this for the front unit — the deprecated value 6 (used before V2.17A) will not work correctly on current firmware.

Rear unit bug fixed in V2.17A (May 2024). If using the rear unit, firmware must be V2.17A or later. Earlier firmware had a bug that prevented correct rear unit torque control.

What's Next

- **A01** — Regen Tuning
- **A02** — Charge Control & Popular Chargers (Outlander OBC detail)

Source: [openinverter.org wiki](https://openinverter.org/wiki) | Damien Maguire / @EVBmw video series

Last verified against firmware: V2.30A (August 2025)

Regen Tuning

Track: Advanced

Prerequisites: C03 — Essential Parameters: First Start, C04 — Throttle Calibration

Audience: Intermediate

Firmware minimum: V2.15A (April 2024) — regen was not reliably functional before this release

Estimated reading time: 12 minutes

Important Firmware Note

WARNING: Regen was not reliably implemented until V2.15A (April 2024). Any content — videos, forum posts, tutorials — about ZombieVerter regen from before April 2024 refers to a non-functional or experimental implementation. Do not follow pre-V2.15A regen instructions.

Warning:

Warning: V2.20A (December 2024) added revRegen — control of regen behaviour in the reverse direction.

How Regen Works on the ZombieVerter

Regen reverses the normal energy flow. Instead of the battery powering the motor, the motor acts as a generator — converting kinetic energy (vehicle momentum) back into electrical energy and pushing it into the battery.

The VCU triggers regen by sending a **negative torque command** to the inverter. From the inverter's perspective, it's the same as a forward torque command but in the opposite direction. The inverter's control loops handle the actual energy reversal.

There are two distinct regen mechanisms:

Throttle-lift regen — triggered by lifting the throttle. When throtmin is set to a negative value (e.g. -5), lifting the throttle produces a light braking effect rather than pure coast.

Brake regen — triggered by the brake pedal signal. Additional regen on top of (or instead of) throttle-lift regen when the brake is pressed.

Understanding throtmin vs regenmax

These two parameters are frequently confused — they control different things:

Warning: throtmin is the **floor on potnom at all times** — the minimum normalised throttle demand the VCU will allow, even with your foot fully off the pedal. Setting it negative creates throttle-lift regen.

Warning:

Warning: regenmax is the **maximum regen torque level**, expressed as a potnom percentage, **scaled by motor RPM**. It caps how hard regen can bite at any given speed, tapering to zero at low RPM.

In practice: `throtmin` is what you feel when you lift the throttle at speed. `regenmax` governs how that regen level is tapered away as the vehicle slows, preventing a stall jerk at low speed.

The Key Parameters

Complete Regen Parameter Reference

Parameter	Default	Range	Description
throtmin	0%	-100 to 0	Floor on potnom at all times — sets throttle-lift regen level. 0 = no regen (pure coast).
regenmax	-10%	-35 to 0	Maximum regen as a potnom percentage, scaled by motor RPM
regenrpm	1500 rpm	100-10000	RPM at which regenmax is fully applied. Below this, regen tapers to 0 at 100 rpm
regenendrpm	100 rpm	100-10000	Below this RPM, regen is zero — prevents regen stall
regenBrake	-10%	-35 to 0	Brake pedal regen — negative potnom applied when brake pressed, scaled by RPM
regenramp	1 %/10ms	0.1-100	Ramp speed entering regen. At 1, reaching -30% braking takes 300ms
RegenBrakeLight	-15%	-100 to 0	Potnom threshold below which the brake light output activates
revRegen	Off	On/Off	Whether regen is active in reverse direction (V2.20A+)

throtmin — Throttle-Lift Regen

throtmin value	Effect
0 (default)	No regen on throttle lift — pure coast
-5	Light one-pedal style regen
-10	Noticeable engine-brake feel

throtmin value	Effect
-20	Moderate regen
-30	Strong regen — aggressive deceleration on throttle lift
-50+	Very strong — likely unnatural, unsettles passengers

Start conservative (e.g. -5) and increase gradually. Aggressive regen values feel unnatural and can be unsettling for passengers who aren't expecting strong deceleration on every throttle lift.

Warning: Damien's personal preference for his E39 GS450h build: "I'm not a huge fan of regen myself" — but notes that many builders want it for practical range extension and one-pedal driving feel.

regenmax and RPM tapering

regenmax is not applied flat — the VCU tapers it based on motor RPM. At regenrpm (default 1500) the full value applies. Below regenendrpm (default 100 rpm) regen is zero. This prevents regen stall at low speed and jerky stops.

Brake pedal regen — regenBrake

Separate from throttle-lift regen. Applies additional negative potnom when the brake pedal input (Pin 49) is active. Useful for stronger blended braking feel without making throttle-lift too aggressive. Default is -10% — same as regenmax.

RegenBrakeLight

Sets the potnom threshold below which the brake light output activates during regen. Default -15% means the brake lights come on when regen exceeds 15% — ensures following drivers see brake lights during meaningful deceleration, even with no mechanical braking.

brakelight output

Assign the BrakeLight output function to a digital output pin in the Dout section. Without this, regen deceleration happens without brake lights — a safety concern. **Do not skip this step.**

revRegen (V2.20A+)

Controls whether regen is active in reverse direction. Some builds need regen in reverse (regenerative downhill reverse), others don't. Toggle on or off.

Drivetrain-Specific Regen Behaviour

Regen behaviour varies by inverter type. This is why pre-V2.15A regen was unreliable — each drivetrain handles negative torque commands differently, and the VCU firmware needed per-drivetrain tuning.

Nissan Leaf inverter: Regen works well in V2.15A+. The Leaf motor and inverter were designed for aggressive regen in the OEM vehicle (e-Pedal). Supports strong negative torque values.

GS450h transaxle: Regen via MG2 works in V2.15A+. The automatic gear shifting (added V2.15A) accounts for regen: it downshifts before regen engages at low speeds to avoid drivetrain shock. Damien notes the downshift during regen deceleration on his E39. **This produces a brief jerk when regen activates — this is correct VCU behaviour, not a fault.** The downshift is required for the GS450h planetary gearset to transmit negative torque through the drivetrain.

Prius Gen3: Regen supported.

Outlander: Regen supported from V2.15A.

Tuning Process

Step 1 — Start with throttle-lift only

Set `throtmin` = -5. Drive at moderate speed and lift the throttle gently. You should feel a light braking effect. Watch `idc` in Spot Values (via OBD or web interface) — it should go negative (current flowing back into battery) when regen is active.

If `idc` stays at zero or positive: regen is not working. Check firmware version is V2.15A+, check that `throtmin` is saved to flash, verify that `throtmax` is not zero.

Step 2 — Increase gradually

Increase `throtmin` by -5 increments. At each step, test the feel:

- Does the deceleration feel proportional to throttle lift?
- Does it come on smoothly or with a jerk?
- Does it feel predictable to a passenger in the back seat?

Stop when it feels natural. Most builders settle between -10 and -30.

Step 3 — Verify brake lights

Plug in a brake light tester or have someone observe while you trigger regen. Brake lights must illuminate whenever regen is actively decelerating the vehicle.

Step 4 — Check BMS limits

The BMS charge current limit caps how much regen current can flow back into the battery. If `idc` during regen hits the BMS charge limit and flattens, the inverter is limiting regen to protect the battery — this is correct behaviour. You cannot tune past this limit without raising the BMS charge current setting.

Common Regen Problems

Symptom	Likely cause	Fix
No regen at all	Firmware < V2.15A	Update firmware
No regen at all	throtmin = 0	Set to negative value and save to flash
No regen at all	throtmax = 0	Set throtmax to 100 — zero throtmax blocks all torque including regen
Jerky/harsh regen onset	throtmin too negative	Reduce magnitude (closer to 0)
Regen cuts out unexpectedly	BMS charge limit hit	Raise BMS max charge current if battery allows
Brake lights not on during regen	BrakeLight output not assigned	Assign BrakeLight function to a Dout pin

Symptom	Likely cause	Fix
Regen feels inconsistent	Dual-pot mismatch triggering plausibility fault	Recalibrate both throttle channels
GS450h jerk when regen engages	Automatic downshift before regen entry	Correct VCU behaviour — downshift required for GS450h drivetrain, not a fault

One-Pedal Driving

With `throtmin` set to an appropriate negative value, the vehicle can be driven almost entirely with the throttle pedal — lift for regen deceleration, press for acceleration. Friction brakes are used only for final stops.

This style of driving:

- Increases range by recovering kinetic energy
- Reduces brake wear significantly
- Requires a learning period — new drivers should be warned about the strong deceleration on throttle lift
- Is most effective on routes with repeated stop-start patterns (urban driving)

Source: Damien Maguire @Evbmw — "ZombieVerter VCU Firmware 2.15" (April 2024) · "ZombieVerter VCU Software V2.04" (January 2024)

Regen first working: V2.15A (April 2024) · `revRegen` added: V2.20A (December 2024) Last verified against firmware: V2.30A (August 2025)

Charge Control and Popular Chargers

Track: Advanced

Prerequisites: C03 — Essential Parameters: First Start, W03 — Contactor & Precharge Circuit

Audience: All levels

Estimated reading time: 20 minutes

Overview

The ZombieVerter supports AC onboard charging, CHAdeMO DC fast charging, and CCS via the BMW i3 LIM or Foccci open-source controller. One VCU can manage any of these, and combinations are possible (e.g. Leaf PDM for AC plus CHAdeMO for fast charging in the same vehicle).

Supported Chargers

The **Community Experience** column reflects how well-documented and battle-tested each option is in real ZombieVerter builds as of V2.30A. Protocol support in firmware does not automatically mean easy integration — maturity varies significantly.

Charger	AC Power	Community Experience	Notes
Nissan Leaf PDM Gen1	3.3 kW	☐ Extensive	Simplest setup — no CP/PP wiring to VCU
Nissan Leaf PDM Gen2	6.6 kW	☐ Extensive	Recommended first build
Nissan Leaf PDM Gen3	6.6 kW	☐ Good	2018+ Leaf. Improved in V2.22A
Mitsubishi Outlander OBC	3.3 kW	☐ Good	AC + DC-DC. Two 12V feeds required — most common failure point
VW Golf PHEV OBC	3.3 kW	☐ Limited	VAG protocol. Two CAN buses required. Few documented builds.
Audi e-tron OBC	7.2–11 kW	☐ Limited	Most capable VW option. Working but complex — community experience thinner than Leaf/Outlander.
MG ZS Gen1 OBC	6.6 kW	☐ Limited	VAG protocol — same two-CAN-bus requirement as VW

Charger	AC Power	Community Experience	Notes
MG ZS Gen2 OBC	6.6–11 kW	□ Limited	3-phase capable. Protocol supported; limited real-world build reports as of V2.30A.
Tesla Gen2 charger	10 kW	□ Moderate	Requires open-source controller board from evbmw.com
Tesla Gen3 charger	18.5 kW	□ Moderate	Requires Gen3 v2 controller board
Elcon	1.5–6.6 kW	□ Moderate	Universal J1772 charger
CHAdemo	DC 50+ kW	□ Good	Native VCU support. 6 PCB jumpers required.
CCS (BMW i3 LIM)	DC fast	□ Good	Salvaged from i3. Type 1 and 2.
CCS (Foccci)	DC fast	□ Good	Open-source controller. V2.20A one-click setup.

WARNING: □ VW and MG chargers: The VAG protocol is implemented in firmware and has been demonstrated working. However, the two-CAN-bus requirement is a common integration stumbling block, and real-world documented builds using VW or MG OBCs are significantly fewer than Leaf or Outlander builds. Treat these as working-but-complex options that require more troubleshooting and community forum research. Do not assume the same level of step-by-step documentation exists as for Leaf PDM or Outlander.

Understanding CP and PP Signals

Every J1772 / Type 2 AC charging setup uses two signals from the charge port to the VCU. Understanding these is essential before wiring any charger except the Leaf PDM.

Control Pilot (CP)

The EVSE (charge station) generates the CP signal. It tells the car how much current is available and detects when the car is ready to charge. The car signals readiness by pulling down the pilot voltage through a resistor.

State	CP voltage	Meaning
A	+12V DC	No vehicle connected
B	+9V DC	Vehicle present, not ready
C	+6V PWM	Vehicle ready — charging permitted
D	+3V PWM	Ready + ventilation required

The PWM duty cycle encodes available current: $10\% = 6A \cdot 25\% = 15A \cdot 50\% = 30A$.

CpSpooF (V2.20A+): The VCU can generate a fake State C signal via a PWM output pin, so chargers that need CP confirmation (Outlander OBC, Foccci, some VW OBCs) start charging without a full J1772 charge port. Assign the CpSpooF function to a high-side PWM output pin in the Dout section.

Proximity Pilot (PP)

PP detects whether a cable is physically plugged in. A resistor inside the cable connector pulls down a 5V sense line. Different cable current ratings use different resistor values.

PP resistor in cable	Cable rating
Open circuit	No cable / disconnected
1.5 k Ω	13A cable
680 Ω	20A cable
220 Ω	32A cable

PP wiring — voltage divider circuit:

The PP sense line requires a two-resistor voltage divider:

- **Upper resistor (pull-up):** 330 Ω from VCU 5V supply (Pin 48) to the PP sense line
- **Lower resistor (to ground):** This is the resistor built into the charge port or cable connector:
- **Type 1 (J1772) connectors:** 2.7k Ω to ground
- **Type 2 (IEC 62196) connectors:** 4.7k Ω to ground
- Note: many charge ports already have this lower resistor installed — check before adding one

Setup: Assign the ProxPilot function to an analogue input pin. Watch PPVal in Spot Values while plugging in a cable. Set PPthreshold just above the reading when the cable is fully inserted. Note that different cables (13A vs 32A) will give different PPVal readings — set your threshold for the lowest-rated cable you intend to use.

WARNING: Leaf PDM exception: The Leaf PDM detects cable insertion internally via its own CP circuit. You do not need to wire PP to the VCU for a Leaf PDM build.

Charge Mode Operation

Charge mode is triggered automatically when the VCU detects cable insertion via PP/ProxPilot, or a CHAdeMO/CCS handshake begins.

What Changes in Charge Mode

Aspect	In charge mode
opmode	Shows "charge"
HV contactors	Closed — HV bus live for charging
Throttle commands	Ignored — zero torque to inverter
DC-DC converter	Active — charges 12V battery from HV
Charge current	VCU sends power/current command to OBC via CAN

Key Charge Parameters

Parameter	What it does
chargerType	Selects your OBC (LeafPDM, Outlander, VAG, Tesla...)
chargePower	Watts requested from OBC (Leaf PDM)
chargeCurrent	Max AC current limit (A)
chargeVoltage	Target / maximum pack voltage
PPthreshold	PP ADC value that triggers charge mode
chargeStart / chargeStop	Scheduled charge timer (hour of day)

Setup Checklist

Set chargerType — save to flash

Wire PP if required (not Leaf PDM): 330Ω pull-up to 5V → analogue input → assign ProxPilot → calibrate PPthreshold

Wire CpSpoofer if required: assign to PWM output. Needed for Outlander, Foccci, most VW OBCs

Set chargeVoltage and chargeCurrent limits

Test: plug in cable → watch opmode in Spot Values → should transition to "charge" → watch idc for current flow

Nissan Leaf PDM

The simplest charging setup. The PDM combines the AC charger and DC-DC converter in one unit.

Minimum requirements: 12V power · HV bus · CAN. No CP/PP wiring to VCU. No PWMs, no HVIL, no special relay logic. Set chargerType = LeafPDM and the VCU handles all CAN messaging.

Power control:

```
chargerType = LeafPDM
```

```
chargePower = 1000 → start at 1kW, ramp up
```

```
chargePower = 3300 → Gen1 max
```

```
chargePower = 6600 → Gen2/3 max
```

```
chargePower = 0 → stop charger
```

Warning: **DC-DC note:** The PDM's DC-DC activates in charge mode. Ensure permanent 12V on VCU — a 12V dropout reboots the VCU mid-session.

Mitsubishi Outlander OBC

3.3 kW AC charger + DC-DC converter. Supported since V2.01A.

Warning: ⚠ **Two 12V power inputs — both must be fed.** The Outlander OBC has two separate 12V connectors. Both must be powered. Wiring only one: opmode shows "charge" but idc = 0 and no current flows. This is the most common Outlander charging failure.

CP/PP wiring required. From V2.20A, use CpSpoofer PWM output for CP. Wire PP with 330Ω pull-up to 5V.

V2.20A also added a required CAN heartbeat message — without it the OBC stops after a timeout. Current firmware handles this automatically.

VW / Audi OBC and MG ZS

All use VAG protocol — set chargerType = VAG.

Warning: ⚠ **Two separate CAN buses required.** VW chargers use "Hybrid CAN" and "Powertrain CAN" internally. Both must connect to different VCU CAN ports. Wire only one bus and the charger ignores you.

Audi e-tron OBC (7.2–11 kW): The VCU sends startup CAN commands that disable OEM requirements for charge port lock feedback and LIN features — no OEM Audi charge port needed. Community credit: Mitch Elliott's reverse engineering.

MG ZS Gen2 (11 kW): 3-phase capable. Also VAG protocol. Support is present in firmware but documented builds are limited — check the openinverter.org forum for current community experience before committing to this option.

All VW/MG chargers need PP wiring (330Ω pull-up) and CpSpoofer or physical CP circuit.

Tesla Gen2 and Gen3 Chargers

Both require dedicated open-source controller boards — available from evbmw.com or build from GitHub.

Gen2 (10 kW, Model S pre-2017): Requires HV first, then CAN frames, then AC mains applied at a specific moment (~frame 31,645 in the message sequence). The controller board handles this automatically. Do not attempt to drive with raw CAN.

Gen3 (18.5 kW, Model S/X 2017+): Three-phase AC. Requires Gen3 v2 controller board. US and EU variants have different internal junction box wiring.

Both need a standard Type 2 charge port. PP sense → VCU ProxPilot analogue input (330Ω pull-up). CP is generated by the controller board.

Tesla Gen2 DCDC converter also supported as of V2.04A.

CHAdemo DC Fast Charging

Native VCU support — no extra controller board needed, but the PCB must be configured first.

PCB Configuration — 6 Solder Jumpers

On the PCB rear silk screen, solder:

- **SJ6, SJ7, SJ8, SJ9** — enables HS CAN3
- **2 jumpers above R31** — adds fault-tolerant termination

Without all 6, CHAdemo handshake will not complete.

12V Signal Wiring

CHAdemo socket pin	Connect to
Pin 1	12V chassis GND
Pin 2	Contactor coil + (both coils paralleled)

CHAdEMO socket pin	Connect to
Pin 4	VCU GP Out 3 (charge enable/disable)
Pin 10	Contactor coil – (both coils)
CAN H/L	CAN3 on VCU

CHAdEMO does not need PP — handshake is fully CAN + 12V signal based.

CCS — BMW i3 LIM or Foccci

Method	Notes
BMW i3 LIM	Salvaged from i3. Provides CCS Type 1 and 2. Connect via ZV Comms CAN parameters.
Foccci	Open-source CCS controller. V2.20A one-click setup. Default CAN Node ID = 22. Uses CpSpooF PWM from VCU.

Charger Selection Guide

Charger	Power	Complexity	CP/PP to VCU	Best for
Leaf PDM Gen2	6.6 kW	Low	Not needed	First build, Leaf stack
Outlander OBC	3.3 kW	Medium	PP + CpSpooF	Outlander builds
VW Golf PHEV OBC	3.3 kW	High	PP + CP + dual CAN	European builds — expect integration effort
Audi e-tron OBC	7.2-11 kW	High	PP + CpSpooF + dual CAN	High-power AC — expect integration effort
MG ZS Gen2 OBC	11 kW	High	PP + CpSpooF + dual CAN	3-phase 11kW — limited community reference
Tesla Gen2	10 kW	High — controller board	Full charge port	High AC power
Tesla Gen3	18.5 kW	High — controller board	Full charge port	Maximum AC speed
CHAdEMO	DC 50+ kW	High — PCB jumpers + wiring	Not needed	Public fast charging

Warning: Recommended first build: Leaf PDM Gen2. Most documented, no PP wiring to VCU, fewest gotchas.

Source: Damien Maguire @Evmw · Good Enuff Garage · openinverter.org · Mitch Elliott (Audi e-tron OBC reverse engineering)

Outlander OBC: V2.01A · Tesla Gen2 DCDC: V2.04A · Foccci one-click setup: V2.20A Last verified against firmware: V2.30A (August 2025)

BMS Integration

Track: Advanced

Prerequisites: W03 — Contactor & Precharge Circuit, C03 — Essential Parameters: First Start

Audience: All levels

Estimated reading time: 12 minutes

Overview

The ZombieVerter does **not** implement cell-level BMS functions. It reads pack-level data from a separate BMS via CAN and enforces current limits from that BMS. This is an intentional division of labour:

- **ISA shunt** — pack-level monitoring (voltage, current, SoC). Feeds precharge logic and contactor control.
- **Cell BMS (SimpBMS, Orion, etc.)** — cell-level monitoring, balancing, per-cell protection. Sends charge/discharge limits to VCU over CAN.
- **VCU** — enforces BMS limits, manages contactors, controls inverter and charger.

ISA IVT Shunt

The ISA IVT-S shunt is the recommended pack-level monitor for all ZombieVerter builds. It provides continuous measurement of voltage, current, power, and state of charge over CAN.

What It Measures

Measurement	Spot value
Pack voltage (U1)	udc
DC current	idc
State of charge	SOC
Instantaneous power	power
Amp-hours (coulomb count)	AMPh
Kilowatt-hours	KWh

For the complete CAN message ID list, spot value mapping, and init procedure, see **W08 — ISA IVT-S Shunt: Complete Reference**.

Permanent 12V — The Critical Rule

Warning: ☐ **The ISA shunt must have always-on 12V — the same permanent feed as the VCU itself.**

If powered from an ignition-switched supply, the shunt loses its SoC state and UDC reading when the key is off. On the next startup, UDC reads zero → precharge fails every time.

Rule: ISA shunt power = always on. Same feed as VCU. Never ignition-switched.

U1 Placement — Inverter Side of Main Contactor

The ISA shunt's U1 voltage sense wire must connect to the HV bus on the **inverter side** of the main contactor — not the battery side.

During precharge, U1 rises from 0V toward pack voltage as the inverter capacitors charge through the precharge resistor. When U1 crosses `udcsw`, the VCU closes the main contactor.

If U1 is connected to the battery side, it reads full pack voltage immediately, and the VCU tries to close the main contactor before the capacitors are charged — potentially welding the contacts on first use.

CAN Bus

The ZombieVerter-specific IVT-S flash runs at **500 kbps** — the same as CAN1 and CAN2 on the VCU. The shunt can share either bus with other 500 kbps devices. Set `ShuntCan` to match your wiring.

Warning: ⚠ Stock IVT-S units default to 1 Mbps. Reflashing with the ZV-specific firmware is required before use. See W08.

Initialisation (One-Time)

`ShuntType = ISA`

`ShuntCan = CAN1 or CAN2 (match your wiring)`

→ Set `IsaInit = 1` → save to flash → power cycle VCU and shunt simultaneously

→ Verify `udc` shows a reading in Spot Values

→ Set `IsaInit = 0` → save to flash → reboot VCU

Warning: ⚠ `IsaInit` is not cleared automatically. Leaving it at 1 re-runs the full init — and resets the coulomb counter — on every boot.

If initialisation fails, power the shunt 2–3 seconds before the VCU. This allows the shunt to fully boot before the VCU attempts the init sequence.

BMW S-BOX Alternative

Set `ShuntType = SB0X`. The BMW Safety Box provides pack voltage monitoring and HVIL (High Voltage Interlock Loop) in one unit. Popular in BMW-based builds.

Cell-Level BMS Options

BMS	Notes
SimpBMS	Most popular. Open-source. Designed for EV conversions. Connects to Leaf cell modules via UART (120-pin Leaf BMS connector), sends charge/discharge limits to VCU over CAN.
Orion BMS	Commercial, professional-grade. Plug-and-play CAN integration with VCU. High cost but very reliable.
Nissan Leaf ZE1 BMS	Salvaged from 62kWh Leaf. Native Leaf cell integration. Supported in V2.30A.
Renault Kangoo BMS	36-module BMS. Added V2.20A.

BMS	Notes
BMW S-BOX	Combined pack monitor + HVIL. Set ShuntType = SBOX .
VW E-BOX	Volkswagen equivalent to S-BOX. Set ShuntType = VAG .

How BMS Limits Work

The BMS sends maximum charge and discharge current limits to the VCU via CAN. The VCU derates inverter torque commands and charge current to stay within those limits.

BMS sends via CAN → VCU enforces:

Max discharge current → caps motor torque demand

Max charge current → caps regen / charge current

BMS fault / over-volt → VCU opens contactors

BMS Parameters

Parameter	What it does
BMSCan	CAN bus the BMS is connected to
BMSMode	BMS protocol (SimpBMS, Orion, ZE1, etc.)
maxDischargeCurrent	VCU hard limit (A) — overrides BMS if lower
maxChargeCurrent	VCU hard limit for regen and charging

Warning: ⚠ **Set BMSCan correctly.** If the BMS shares a bus with the inverter, check for CAN ID conflicts between devices. The ISA shunt and cell BMS can be on the same bus — both run at 500 kbps with the ZV IVT-S flash — or on separate buses if preferred.

ISA vs Cell BMS — Both Needed

The ISA shunt and a cell BMS are complementary, not interchangeable:

- The **ISA shunt** provides UDC for precharge logic and idc for power monitoring. Without it (or ShuntType = SBOX/VAG), precharge sequencing cannot work.
- The **cell BMS** provides per-cell voltage monitoring, temperature, balancing, and current limits. Without it, you have no protection against individual cell over-voltage or thermal runaway.

From V2.20A, it is possible to run with ShuntType = None and use BMS data from the inverter and BMS directly for pack-level monitoring — demonstrated in Damien's Mitsubishi L200 truck build using a Leaf ZE1 battery pack with integrated BMS.

Source: openinverter.org/wiki · Damien Maguire @Evmw · ZombieVerter firmware source

Last verified against firmware: V2.40A

Gear Selectors and Shifters

Track: Advanced

Prerequisites: C03 — Essential Parameters: First Start, W06 — 12V Auxiliary & Ignition Wiring

Audience: All levels

Estimated reading time: 10 minutes

Overview

The VCU supports multiple methods of direction selection — from simple toggle switches to OEM electronic shifters communicating over CAN. The method is set via two parameters: `dirmode` (controls the 12V input logic) and `gearLever` (selects OEM CAN shifter protocol if used).

Warning: GS450h exception: The GS450h has a physical P/R/N/D linkage connected to the transmission neutral safety switch. Note that with the exception of Park, which physically engages the parking pawl, the other positions are a simple electrical switch. The VCU reads switch state directly — it does not use the `gearLever` CAN parameter. See Module I03.

dirmode — Simple Direction Control

For builds using 12V switches or buttons rather than OEM CAN shifters.

dirmode	Wiring	Behaviour
defaultFwd	Pin 53 only	Auto-forward in run mode. Momentary 12V to Pin 53 = reverse. Release = back to forward.
switch	Pin 54 (Fwd) + Pin 53 (Rev)	3-position. Neither active = neutral. Best for PRND selector.
button	Single momentary	Press to toggle between forward and reverse.
switchReversed	Same as switch	Motor direction inverted — use when drivetrain spins backwards.
buttonReversed	Same as button	Toggle + inverted direction.

3-Position Switch (switch mode) — Recommended

D position → 12V to Pin 54 (`din_forward` = on)

N position → neither pin (no torque output)

R position → 12V to Pin 53 (`din_reverse` = on)

Warning: ⚠ **Verify with Spot Values before road testing.** Sit with the web interface open. Move through each selector position. Confirm `dinforward` and `dinreverse` toggle correctly. A

wiring error means unintended reverse at startup.

defaultFwd — Simplest Possible Build

Only needs one wire for reverse. Auto-Drive when in run mode. Single momentary button on Pin 53 for reverse. When released, returns to Drive. No PRND lever needed — ideal for minimal builds.

OEM CAN-Based Electronic Shifters

Modern vehicles use electronic shifters that send gear position over CAN. The gear shifter class (added V2.04A) reads these messages and maps them to the VCU's internal direction and Park/Neutral states.

Set the gearLever parameter to match your shifter:

Shifter	Added	Notes
12V inputs (default)	Always	Standard switch wiring — Pins 53/54
BMW F30 CAN e-shifter	V2.04A	Sends PRND via CAN. Clean OEM look. Reverse engineering by "project Gus."
JLR rotary shifter	V2.05A	Jaguar/Land Rover CAN rotary. Popular for custom builds.
BMW E65 7-series	V2.20A	Now uses shifter class (was hardcoded before V2.20A).

Check the openinverter wiki for the specific CAN ID required for your shifter model.

Park Gear (V2.05A+)

When Park is selected via an OEM shifter or a dedicated 12V input:

- **Torque inhibited** — no motor movement regardless of throttle position
- **HV bus maintained** — contactors stay closed, DC-DC converter active
- **Charge mode enabled** — some OBCs require Park to be selected before accepting a charge command

E91 Build Example

Jamie Jones (Ohm-Grown EVs) used the OEM BMW E91 automatic shifter assembly for his conversion. The shifter physically clicks into P/R/N/D positions but sends simple 12V signals — wired directly to VCU Pins 53/54. This is the simplest way to retain an OEM shifter feel without CAN complexity.

Source: Damien Maguire @Evbmw · Jamie Jones / Ohm-Grown EVs · openinverter.org

Gear shifter class added V2.04A · JLR added V2.05A · Park gear added V2.05A Last verified against firmware: V2.30A (August 2025)

CAN Gateway and Multi-Node Systems

Track: Advanced

Prerequisites: H01 — VCU Hardware Walkthrough, C02 — Web Interface Walkthrough

Audience: Intermediate

Estimated reading time: 15 minutes

Overview

The ZombieVerter has three independent CAN buses. Understanding how to assign devices, terminate correctly, and integrate OEM instrument clusters is what separates a functional conversion from a polished one.

CAN Bus Physical Wiring

Termination — The Most Skipped Step

A CAN bus must have exactly **two 120Ω termination resistors** — one at each physical end of the bus. Without correct termination you get signal reflections, corrupted frames, and intermittent communication failures that are very hard to diagnose.

Warning: ⚠ **Exactly two 120Ω resistors — one at each end of each bus.**

One resistor at the VCU end (often built into the VCU board). One at the furthest device on that bus. Between CAN H and CAN L.

Verifying termination: With power off, measure resistance across CAN H and CAN L. With two 120Ω resistors in parallel you should read approximately 60Ω. More than 60Ω = too few resistors. Less than 60Ω = too many.

Warning: ⚠ **Check device datasheets.** Some inverters and OBCs have built-in 120Ω termination. Adding an external resistor to a device that already has one results in too many terminators and bus instability.

Topology — Bus Not Star

CAN is a bus topology. All devices connect to the same two wires (CAN H and CAN L) running from one end to the other. Avoid star topologies and long stub branches.

Correct – bus topology:

[120Ω] — VCU — ISA Shunt — Inverter — [120Ω]

Wrong – star topology:

VCU

├— ISA Shunt ← signal reflections at each stub

├— Inverter

└— BMS

Use twisted pair for all CAN runs. Shielded twisted pair for runs near HV cables or in noisy environments.

Baud Rates by Device

Device	Baud rate
Nissan Leaf inverter	500 kbps
Mitsubishi Outlander	500 kbps
ISA IVT shunt	1000 kbps (1 Mbps)
SimpBMS / Orion BMS	500 kbps
BMW i3 LIM	500 kbps
CHAdemo socket	500 kbps (fault-tolerant CAN3)
OBD2 / ELM327	500 kbps

All devices on the same CAN bus must run at the same baud rate. The ISA shunt at 1 Mbps typically needs its own bus separate from 500 kbps devices.

Assigning Devices to CAN Buses

The Three Buses

Bus	Notes
CAN1	High-speed CAN. Primary bus for most inverters.
CAN2	High-speed CAN. Secondary bus — BMS, shunt, second devices.
CAN3	Configurable via PCB jumpers — high-speed, fault-tolerant, or single-wire. Used for CHAdemo.

Assignment Parameters

Parameter	What it assigns
inverterCan	Which bus the traction inverter is on
shuntCan	Which bus the ISA shunt is on
bmsCan	Which bus the cell BMS is on
chargerCan	Which bus the OBC is on

Typical Configuration

CAN1 @ 500 kbps:

Traction inverter (inverterCan = CAN1)

Leaf PDM (same bus as inverter)

BMW i3 LIM (CCS — same bus)

CAN2 @ 1000 kbps:

ISA shunt (shuntCan = CAN2)

SimpBMS / Orion (bmsCan = CAN2, if 500k: separate bus from shunt)

CAN3 @ 500 kbps fault-tolerant:

CHAdEMO socket (6 PCB jumpers required)

Warning: ⚠ **CAN1/CAN2 label swap:** There is a known documentation issue where CAN1 and CAN2 are labelled in reverse in some older wiring diagrams. If a device won't communicate despite correct wiring, try swapping the bus assignment parameter value before rewiring.

Vehicle CAN Integration

The vehicle Parameter

Set vehicle to tell the VCU which OEM CAN profile to use. The VCU then sends spoofed CAN messages to keep the instrument cluster functional.

vehicle setting	Covers	What it drives
BMW E46	1998–2005 3-series	Speedo, tacho, temp, fuel level, CEL suppression
BMW E6x+	E60/E90/E9x/E91	Speedo, tacho, lock state, warning lights
BMW E39	1996–2003 5-series	Speedo, tacho, instrument functions
BMW E31	8-series	Added V2.04A
VAG	Mid-2000s VW/Audi	Instrument cluster CAN
Subaru	Various	CAN integration
Classic	Any vehicle	No vehicle CAN — uses analog outputs only
None	—	No vehicle CAN output

BMW E9x — Added V2.20A

The E90/E91/E92/E93 CAN integration added in V2.20A includes:

- Lock state over CAN
- Warning lights control (suppresses CEL and ICE-specific warnings)
- Vehicle speed driven from motor RPM

If Your Vehicle Isn't Listed

Use Classic mode and drive gauges with analog PWM outputs. You can also submit a CAN log from your OEM vehicle as a GitHub feature request — see Module A06.

Classic Mode & Analog Gauges

Set vehicle = Classic for any vehicle without CAN-based instruments — pre-2000 vehicles, classic cars, trucks, kit cars.

Tachometer — TachoPPR Parameter (V2.20A+)

The VCU outputs a pulse signal from a PWM output pin that can drive an OEM analog tachometer. TachoPPR sets pulses per revolution:

TachoPPR	Equivalent cylinder count
1	1-cylinder
2	4-cylinder (typical default)
3	6-cylinder
4	8-cylinder

Tune until the needle reads actual motor RPM. Assign the tacho output function in the Dout section.

Speedometer

A speedo pulse output is also available via the GS450h pump pin in V2.20A+ when set to Classic mode. Configure PPR to match your OEM speedo sender expectation.

Fuel Gauge — Digipot Method

For vehicles without CAN fuel gauge control, a digital potentiometer (digipot) IC wired between the VCU and the OEM fuel gauge sender wire can drive the fuel needle to show state of charge.

The VCU controls the digipot resistance in steps. Calibrate by setting step values and observing needle position:

- 0 steps = low fuel warning triggered
- Tune mid-scale and full-scale steps to match actual SoC at those points

The temperature gauge can also be driven from temphs (inverter heatsink temperature) or tempm (motor temperature) in some vehicle classes.

OBD2 & Live Telemetry

OBD2 via CAN (V2.04A+)

The VCU can respond to OBD2 requests over CAN, allowing standard Bluetooth OBD dongles and apps like Torque Pro to read live VCU data — motor speed, pack voltage, current, SoC, temperatures.

Warning: ⚠ **Dongle quality matters.** Cheap AliExpress ELM327 clones have inconsistent results. Higher-quality Bluetooth or WiFi OBD adapters work reliably. If Torque Pro connects but shows no data, try a different dongle before assuming a configuration problem.

Web Interface Graphing

The VCU web interface includes built-in real-time graphing. Select spot values to plot, set the update interval. Useful for:

- Throttle linearity during calibration (plot potnom vs pot)
- Precharge voltage rise (plot udc over time)
- Temperature monitoring under load (temphs, tempm)
- Regen tuning (idc going negative during deceleration)

Parameter Files on GitHub

Damien publishes parameter files for his fleet vehicles (E39, E46, E31, L200 truck) on the ZombieVerter GitHub repo. These are real working configurations for tested vehicles — a useful starting point for matching drivetrain/vehicle combinations before tuning your own build.

Diagnosing CAN Communication Failures

Work through in order:

Is the correct bus assignment parameter set?

Is the baud rate consistent across all devices on that bus?

Is termination correct? Measure $\sim 60\Omega$ across H/L with power off.

Try swapping inverterCan or shuntCan value (CAN1/CAN2 label swap issue).

Are CAN H and CAN L wires the right way around? Try swapping them.

Is the cable twisted pair? Long untwisted runs in noisy environments cause intermittent failures.

Source: Damien Maguire @EVBmw · openinverter.org/wiki · V2.04A changelog · V2.20A changelog

TachoPPR added V2.20A · E90 CAN added V2.20A · OBD2 added V2.04A · E31 added V2.04A Last verified against firmware: V2.30A (August 2025)

Firmware Customisation and Contributing

Track: *Advanced*

Prerequisites: *None — standalone reference module*

Audience: *Intermediate to advanced*

Estimated reading time: 8 minutes

Overview

The ZombieVerter firmware is fully open source, hosted at [\[github.com/damienmaguire/Stm32-vcu\]\(https://github.com/damienmaguire/Stm32-vcu\)](https://github.com/damienmaguire/Stm32-vcu). When you need something the firmware doesn't yet support, you have three paths — and one of them requires no coding at all.

Path 1 — GitHub Feature Request (No Coding Required)

The simplest path. File an issue on GitHub describing what you need. The maintainers and community evaluate it, and if it fits the project goals and has sufficient demand, it gets implemented.

How to File a Good Request

Search existing issues first. Your feature may already have been requested, discussed, or even implemented in a recent release that you haven't updated to yet. Check both open and closed issues.

Describe the use case, not the implementation. "I need to control a Renault Zoe OBC" is a better issue than "add parameter X with value Y to CAN message Z." The maintainers understand the codebase far better than most requesters — explain the problem you're trying to solve.

Provide CAN logs if relevant. For new inverter or charger support, a CAN log captured from the OEM vehicle is the single most valuable thing you can provide. Without it, reverse engineering has to start from zero. This is why the forum is full of "does anyone have a CAN log for X?" threads — they unlock new hardware support.

Post on the forum first. openinverter.org/forum is the right first stop before GitHub. If others need the same feature, the forum thread becomes evidence of demand. It also filters out requests that are already supported with a simple parameter change.

What Gets Prioritised

Priority	Type of request
High	Bug fixes · Safety issues · Widely-used hardware with CAN log data available

Priority	Type of request
Medium	New inverter/charger support with community demand · Features partially implemented
Lower	Niche hardware with single user · Features requiring significant reverse engineering from scratch

Path 2 — Fork and Self-Implement

Clone the repo, make your changes, compile, flash to your VCU, and run it. You're entirely in control. No obligation to contribute back to the main project — it's your fork.

```
# Clone with all submodules – this step is critical
```

```
git clone --recurse-submodules git@github.com:damienmaguire/Stm32-vcu.git
```

Warning: ⚠ **--recurse-submodules is mandatory.** The firmware depends on libopeninv (Johannes Huebner's openinverter framework) and other submodules. Without this flag the build fails with missing includes.

Build Environment

The firmware uses an ARM GCC toolchain. Build documentation is on the openinverter.org wiki. Johannes Huebner's GitHub (github.com/jsphuebner) is the reference for the underlying parameter system and CAN class architecture — understanding his framework is essential for meaningful customisation.

GD32 Boards

Warning: ⚠ **If you have a GD32F107 board, the GD_Zombie branch was abandoned and has substantially diverged from main.** Self-building for a GD board means starting from that old branch — you will not have features from V2.00A onwards. See Module C01 for chip identification.

Path 3 — Pull Request to Main Branch

If you implement something genuinely useful for the wider community, you can submit a pull request. This is how most new hardware support, vehicle classes, and firmware features have landed — contributions from community members working on their own builds who then share the work.

Examples of community contributions that made it into main: BMW E31 vehicle class, OBD2 support, analog IO matrix, JLR gear shifter, Kangoo BMS, MG ZS OBC.

The Testing Requirement

Warning: ⚠ **Changes must be tested in real vehicles before merging. This is non-negotiable.**

The firmware runs high-voltage systems in moving vehicles. A bug that gets through to a release can cause a precharge failure at speed, unintended motor torque, a contactor fault, or worse. Damien personally tests new features across multiple vehicles — typically 3–4 different builds covering different drivetrain and charger combinations — before any release goes out.

This is why release cycles are measured in months, not weeks. V2.20A was tested across four running vehicles (BMW E39, E46, E31, and the Mitsubishi L200 truck) before release. The testing

isn't just "does it compile and power up" — it's sustained real-world driving with the full complement of features active.

What a Good Pull Request Includes

- Working, clean code
- Evidence of testing in at least one real vehicle (ideally more)
- CAN log data if it's a new device integration
- A description of what was tested, what edge cases were considered, and any known limitations
- No breaking changes to existing functionality without discussion first

The community is welcoming to well-tested contributions. The project's growth from a single-vehicle GS450h controller to a universal VCU supporting dozens of drivetrains and chargers was built on exactly this kind of contribution.

The Openinverter Framework

The ZombieVerter firmware is built on Johannes Huebner's openinverter firmware framework (libopeninv). Understanding the framework is important for anyone doing serious firmware work:

- **Parameter system** — all configurable parameters are defined in a structured way that automatically exposes them in the web interface
- **CAN class architecture** — each device (inverter, charger, BMS) is implemented as a CAN class that can be selected at runtime via the `inverter / chargerType` parameters
- **IO Matrix** — the flexible input/output assignment system introduced in V2.00A

Johannes's GitHub and the openinverter forum are the best resources for understanding the framework internals.

Source: github.com/damienmaguire/Stm32-vcu · openinverter.org/forum · Damien Maguire @Evbmw

Last verified against firmware: V2.30A (August 2025)